Mathematical Stories

Lecture Notes

July 19, 2026

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Introduction

Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

The course begins with geometric objects and questions, introduces notation only when it is needed, derives formulas from those objects, and then uses examples to test the resulting language. The reader should always distinguish the object, its coordinate description, the argument connecting them, and the final result.

How the Course Works

Each chapter supports the live lecture and later independent study. During the lecture it provides the mathematical route and the core results to discuss. After the lecture it remains a complete narrative containing definitions, examples, arguments, and interpretations that the student can revisit without the lecturer being present.

This book is part of a weekly learning process rather than material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. In a fifteen-week semester, the remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

The Weekly Learning Cycle

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant objects and methods should be read.

2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF contains the full argument, not merely the material shown or discussed in detail live.

3. **Ask questions and clarify the material.** Mark every unclear definition, transition, calculation, or conclusion and resolve it using the text, discussion, examples, and appropriate tools.

4. **Solve the complete assigned problem set.** Move from following explanations to constructing arguments independently.

5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion.

6. **Present and discuss your work.** Be prepared to explain a solution, answer questions about its steps, and describe what was learned while preparing it.

The lecture prepares the reading, the reading prepares the problem solving, the written solution prepares the presentation, and feedback leads back to revision. No stage is intended to replace the others.

Exercise Classes: Presentation, Feedback, and Revision

A student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether it can be communicated clearly. Feedback given during exercise classes is intended for the whole group, not only for the student whose work is being presented. By comparing different solutions, students learn which explanations are complete, which steps need justification, and which weaknesses require revision.

Feedback concerns the mathematical work and its results, not the personal worth of the student. It should identify what already works and what must be changed to produce a stronger argument.

The first submitted version need not be the final version. Receiving comments, reconsidering choices, correcting errors, and adding missing explanations are normal parts of serious academic and professional work, not evidence of failure.

Progress and Assessment

By the end of a weekly cycle, the student should be able to identify the main objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Evidence of serious engagement includes:

- regular preparation and completion of assigned problem sets;
- documented attempts rather than bare answers;
- active participation in presentations and discussion;
- separate, self-contained notes for individual problems;
- careful revision after feedback;
- increasing clarity, independence, and mathematical precision.

A weak first solution or presentation does not determine the final exercise component grade. Sustained work, revision, and demonstrated development can lead to a very good final result. This does not lower the standard; it recognises the process required to reach it.

The examination is assessed separately and tests mathematical knowledge and the ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills needed to complete it successfully.

Attend, reread, question, solve, write, present, and revise. The lecture provides orientation; the chapter preserves the complete argument; independent work turns that argument into understanding.

How to Prepare Mathematical Notes

A mathematical note does not merely record an answer. It shows how the problem is understood, which information matters, why a method is appropriate, how each important step is justified, and how the conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain objects, operations, conditions, and logical relationships. Reading a formula therefore means being able to explain what each object represents, why the expression is meaningful, and what information it provides.

A sequence of formulas without explanatory sentences rarely communicates a complete argument. Words connecting the formulas are part of the mathematics because they state the logical relationship between successive steps.

Make the Note Self-Contained

Prepare one separate note for each assigned problem, including all subparts. Begin with the problem or a complete and faithful formulation of it. Record what is given, what must be found or proved, and what notation will be used.

Before calculating, identify the mathematical objects involved and decide what would count as a satisfactory answer. After several weeks, opening the note should still reveal both the question and the full solution without searching elsewhere.

Make the Reasoning Visible

Sentences that reveal mathematical reasoning include:

- Because the determinant is nonzero, the matrix is invertible.

- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is nonzero, it determines a line.
- Substituting into the original equation verifies the solution.

The amount of explanation should be proportional to the problem. A simple exercise may need only a few sentences; a substantial argument may need many. The goal is not maximal length but complete visibility of every essential step.

Write as briefly as possible, but as completely as necessary.

A Complete Solution

1. **Understand the problem.** Record the question, the given information, and the required conclusion.
2. **Choose a method.** Select definitions, theorems, or techniques connecting the data with the goal.
3. **Construct the argument.** Carry out the necessary steps and explain why each important transition is valid.
4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning.
5. **State the conclusion.** Answer the original question clearly and in context.

A bare final answer, an unexplained sequence of formulas, or disconnected calculations constitute incomplete work. Quality cannot be replaced by quantity, but a single polished solution does not replace completing the rest of the assigned problem set.

Working with AI

AI may clarify notation, generate simpler examples, check reasoning, identify missing steps, and help organise mathematical prose. The student remains responsible for every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Receiving a polished answer is not the same as understanding it. A note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently.

Work on one problem at a time and prepare one complete note for it. Do not paste an entire problem set into one conversation and request all answers at once. Repeat the focused process separately for every problem in the assigned set.

Useful questions to ask while working with AI include:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler explanation of this transition?
- How can the result be checked?

- What should I be able to explain during class?

Preparing notes in this way develops the ability to read notation, recognise objects, organise information, construct arguments, and communicate ideas precisely. Repeating the process for the complete problem set builds habits of systematic academic and professional work.

Student GitHub Workspace

The current instructions, problem templates, and technical workflow are maintained in the public repository `dchorazkiewicz/Math_Problems_Repo`. Students should create a public fork and commit notes to their own fork; the common course repository is a template, not the submission location.

- `README.md` explains the workspace and weekly index;
- `SOURCE_MATERIAL.md` maps weeks to the current PDF and LaTeX problem sources;
- `NOTE_GUIDELINES.md` defines a complete note;
- `AI_WORKFLOW.md` describes focused work, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives connected assistants operational rules;
- `docs/` contains weekly folders and one Markdown file per problem.

The workspace uses Markdown, Git and GitHub, MathJax, MkDocs, and GitHub Actions. Because the technical instructions may change independently of this PDF, students should consult the repository for the current workflow.

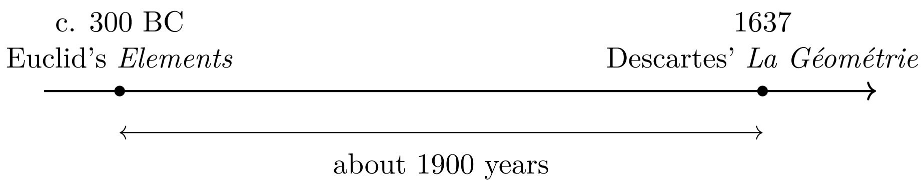
Chapter 1

Cartesian Geometry and Coordinates

1.1 Introduction

The Cartesian coordinate system is something students have already encountered many times, both in primary school and in secondary school. They have learned how to draw the axes and how to use coordinates, but usually without a constructive explanation of where this system comes from.

The main purpose of this chapter is to introduce the Cartesian coordinate system constructively. We will begin with the basic objects and constructions provided by Euclidean geometry and show how they can be used to build a coordinate system.



We will focus mainly on the plane and on low-dimensional spaces, because they are easy to draw, study, and grasp as a whole. This makes them the natural setting in which to build geometric intuition and understand the central ideas. At the same time, many of the constructions and descriptions considered here extend easily to higher-dimensional structures once they are expressed in Cartesian coordinates and, later, in vector notation. We will nevertheless remain close to drawable and directly accessible situations so that the underlying ideas stay visible.

We will then look at elementary ways of using Cartesian coordinates to describe geometric objects and their relations. This will only be the beginning of a language and a collection of methods that will be developed further in later material.

1.2 Brief Euclidean Background

Before introducing coordinates, we recall only the geometric ingredients that the Cartesian construction requires.

What is available before coordinates. Points, straight lines, right angles, lengths, and perpendicular projection are already meaningful geometrically. A Cartesian system does not create these objects; it assigns numerical descriptions to them after axes, an origin, directions, and a unit have been chosen.

The detailed postulates and construction rules are collected later as a self-study reference. The main narrative now proceeds directly to the Cartesian plane.

1.3 Constructing and Using Cartesian Coordinates

The preceding section established the Euclidean objects and constructions available before coordinates. We now make the additional choices that turn that plane into a Cartesian coordinate plane: axes, an origin, directions, and a common numerical scale.

We now carry out the construction of a Cartesian coordinate system.

1.3.1 The Plane Before Coordinates

We begin with an ordinary Euclidean plane: a set of points in which we may construct lines and circles.

a Euclidean plane

1.3.2 First Construction: A Straight Line

First choice. Choose two distinct points of the plane and construct the unique straight line through them.

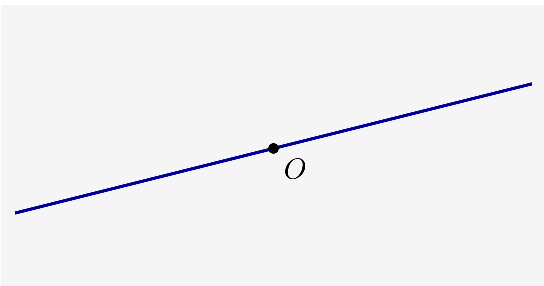
What has and has not been created. At this stage we have only a chosen Euclidean line. It is not yet a coordinate axis: there is no second line, no origin, no positive direction, and no numerical scale.

the first constructed line

1.3.3 Second Construction: A Point on the Line

Second choice. Choose an arbitrary point O on the constructed line.

A chosen point is not yet an origin. The point O will later become the common zero of both coordinate axes, but it becomes an origin only after the second axis, orientations, and numerical scale have been constructed.

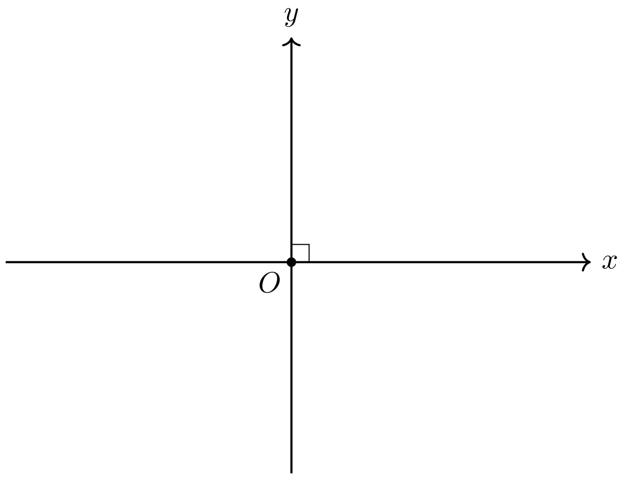


a point chosen on the first line

1.3.4 Third Construction: The Perpendicular Line

Third construction. Through O, construct the unique line perpendicular to the first line. The two constructed lines now intersect at a right angle.

> We call the first line the x-axis, the perpendicular line the y-axis, and their intersection O the origin.



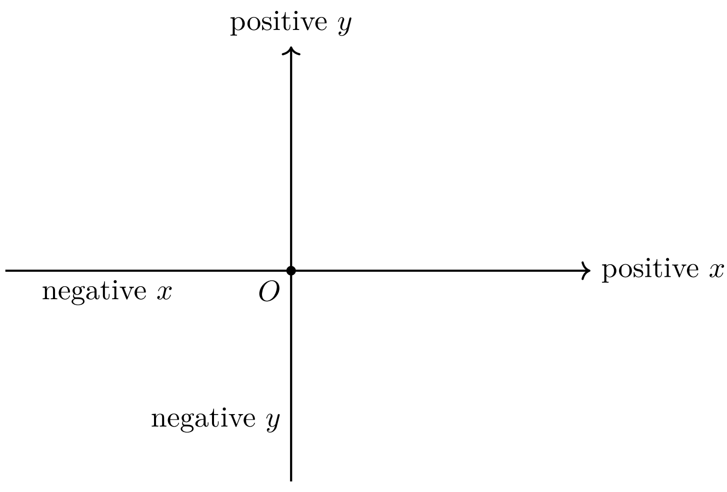
What has now been created. Only after the perpendicular line has been constructed do we have the geometric framework of two coordinate axes with a common origin.

Remark. The axes are perpendicular because this was required in their Euclidean construction. No coordinate formula and no numerical angle measure was used to create that right angle.

1.3.5 Choosing Positive and Negative Directions

Choosing orientations. Each axis is divided by O into two opposite rays. Choose one ray on the x-axis as positive and one ray on the y-axis as positive. The opposite rays are then called negative.

Orientation is a choice. The Euclidean plane does not force which ray should be positive. Reversing one orientation produces a different coordinate system on the same geometric plane.

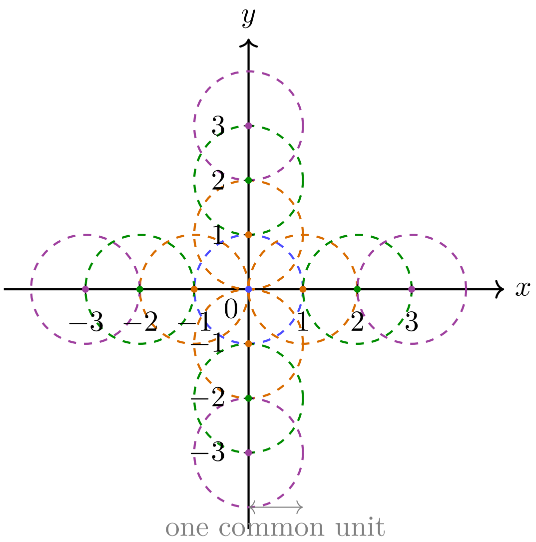


What signs will mean. A positive or negative coordinate will record on which oriented side of the origin the corresponding projection lies. The sign is not a geometric length; it is directional information supplied by the chosen orientation.

1.3.6 Constructing a Common Unit Length

Place the point of the compass at the origin O and draw a circle with any chosen radius. This radius becomes the unit length. The circle meets each axis in two points. On the positive parts of the axes these points are labelled 1, and on the negative parts they are labelled -1.

To construct the next marks, keep the same compass opening. Place the compass at the point 1 on the x-axis and draw another circle. Its next intersection with the positive part of the axis gives the point 2. Repeating the same step gives $3, 4, \ldots$. Moving in the opposite direction constructs $-2, -3, -4, \ldots$. The same construction is carried out on the y-axis.

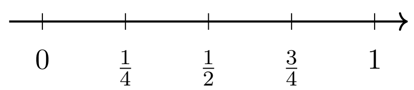


1.3.7 Refining the Scale by Bisection

Constructing finer marks. The compass-and-straightedge construction does not stop at integer marks. Bisecting each unit segment constructs halves. Bisecting again constructs quarters, then eighths, and so on. Repeating this process on every integer interval and on both sides of the origin gives all dyadic marks

$$\frac{k}{2^n}, \qquad k \in \mathbb{Z}, \quad n \in \mathbb{N}.$$

successive bisections of one unit



> The dyadic marks are dense on the axis: every nonempty interval contains a number of the form $k/2^n$. They provide arbitrarily fine numerical approximations to positions on the line.

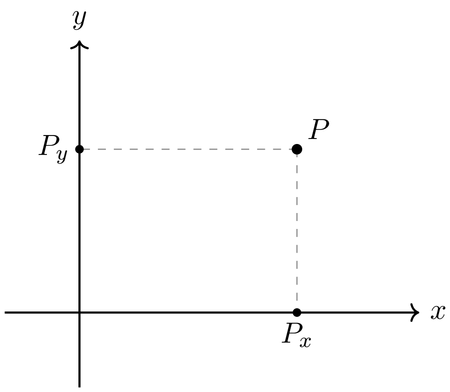
> **Continuity assumption.** Once an origin, a positive direction, and a unit length are fixed on a Euclidean line, every point of the line corresponds to exactly one real number, and every real number corresponds to exactly one point of the line.

What continuity contributes. Finite straightedge-and-compass operations construct many particular positions, including every dyadic mark, but not every real position. Continuity supplies the complete real scale. After this additional assumption, every point of each axis has a unique signed real label.

1.3.8 Assigning Coordinates by Orthogonal Projection

When coordinates can be assigned. Only after the axes, orientations, common unit, refined scale, and real labels have been constructed can an arbitrary point of the plane receive a coordinate pair.

Constructing the two coordinate values. Let P be any point of the plane. Through P, construct the line perpendicular to the x-axis; it meets that axis at a unique point P_x. Through P, construct the line perpendicular to the y-axis; it meets that axis at a unique point P_y.



> If the signed real labels of P_x and P_y are x and y, respectively, the coordinates of P are the ordered pair
> $$P \longleftrightarrow (x, y).$$

Why the correspondence is one-to-one. The construction is unambiguous because the perpendicular through a given point to a given line is unique. Conversely, given real numbers x and y, locate the corresponding points on the axes and construct through them lines perpendicular to the respective axes. The two lines meet in exactly one point. □

> Once the two perpendicular axes, their positive directions, and a common unit are fixed, orthogonal projection gives a one-to-one correspondence
> $$\text{points of the Euclidean plane} \longleftrightarrow \text{pairs } (x, y) \in \mathbb{R}^2.$$

Coordinates are addresses. The ordered pair is the numerical address of a geometric point in the chosen coordinate system. It is not the point itself and it is not the source of the geometry.

1.3.9 The Coordinate Formula for Euclidean Distance

The Euclidean distance between two points is the length of the straight segment joining them. Coordinates now allow this geometric length to be calculated.

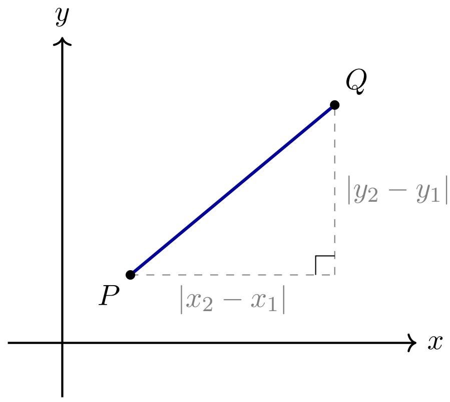
The question. Given the Cartesian coordinates of two points, how can the already existing Euclidean length of the segment between them be recovered algebraically?

For points
$$P = (x_1, y_1), \qquad Q = (x_2, y_2),$$

their projections show that the horizontal and vertical separations have lengths

$$|x_2 - x_1| \qquad \text{and} \qquad |y_2 - y_1|.$$

Together with the segment PQ, these two perpendicular segments form a right triangle. Here "right" refers to the Euclidean right angle already used in the construction; no numerical measure of that angle is required.



From the right triangle to the formula. By the Pythagorean theorem,

$$d_E(P,Q)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

Distance is nonnegative, so taking the nonnegative square root gives the desired coordinate expression.

> For points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ in the Cartesian plane,
>
> $$d_E(P,Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

What the formula means. The formula does not create a new notion of distance. It is the numerical coordinate expression of the Euclidean segment length already present in the plane. This is the first major payoff of the Cartesian construction: a geometric magnitude has become computable by algebra.

From Euclidean plane to Cartesian distance. The construction proceeds in the following order:

1. begin with a Euclidean plane containing no chosen coordinates;
2. construct one straight line;
3. choose a point O on that line;
4. construct through O the perpendicular line, and only then name the two lines as coordinate axes and O as their origin;
5. choose positive directions on both axes;
6. choose one common unit length and transfer it with a compass in both positive and negative directions to construct the integer marks;
7. repeatedly bisect intervals to construct the dense family of dyadic marks;

8. use Euclidean continuity to identify every axis position with a real number;
9. assign to every point its unique pair of orthogonal projections;
10. derive the coordinate distance formula from the resulting right triangle and the Pythagorean theorem.

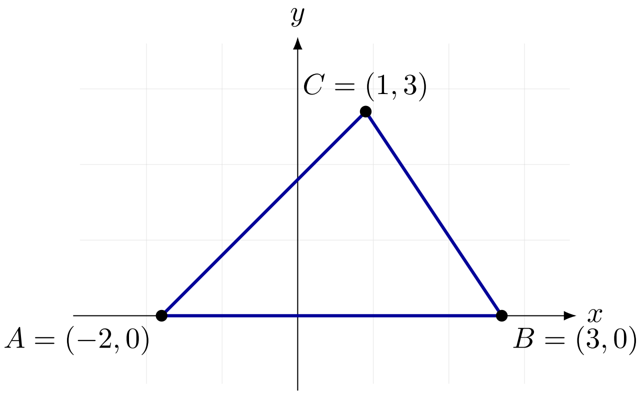
1.3.10 Elementary Graphs and Their Intersections

Once coordinates are available, equations and inequalities can be read as conditions selecting points of the plane. The following examples show how familiar geometric objects can be described in the Cartesian language.

A triangle described by its vertices. Consider the three points

$$A = (-2, 0), \qquad B = (3, 0), \qquad C = (1, 3).$$

These three ordered pairs determine the vertices of a triangle. Joining A to B, B to C, and C to A gives its three sides.



Coordinates therefore allow a geometric figure to be specified by listing a finite collection of points and declaring how they are connected.

A circle and a disk. The equation

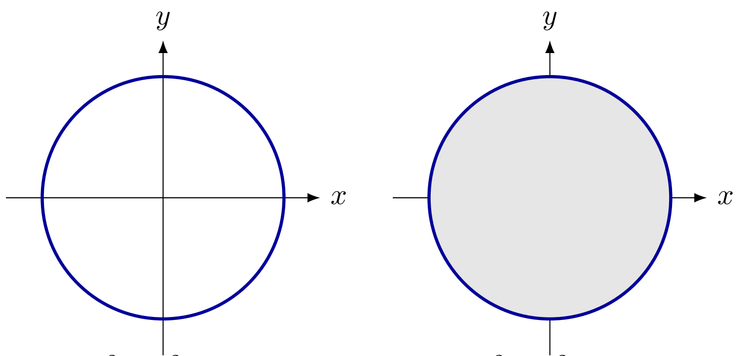
$$x^2 + y^2 = 4$$

selects exactly the points whose distance from the origin is 2. It describes the *circle*: only the boundary.

The inequality

$$x^2 + y^2 \leq 4$$

selects the boundary together with every point inside it. It describes the *disk*.



$x^2 + y^2 = 4$

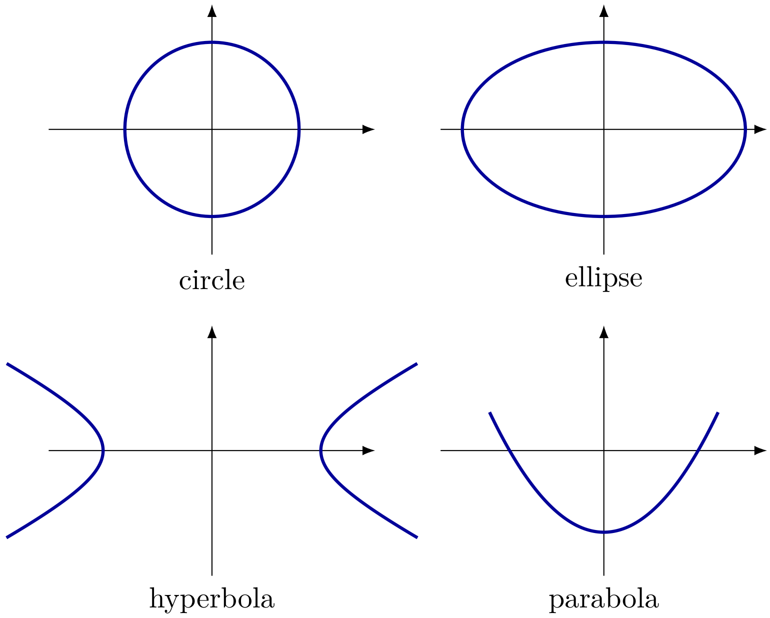
circle

$x^2 + y^2 \leq 4$

disk

Several curves described by equations. Different equations select different shapes:

$$\begin{aligned} x^2+y^2 &= 4, && \text{circle},\\ \frac{x^2}{9}+\frac{y^2}{4} &= 1, && \text{ellipse},\\ \frac{x^2}{4}-\frac{y^2}{1} &= 1, && \text{hyperbola},\\ y &= \frac{x^2}{2}, && \text{parabola}. \end{aligned}$$

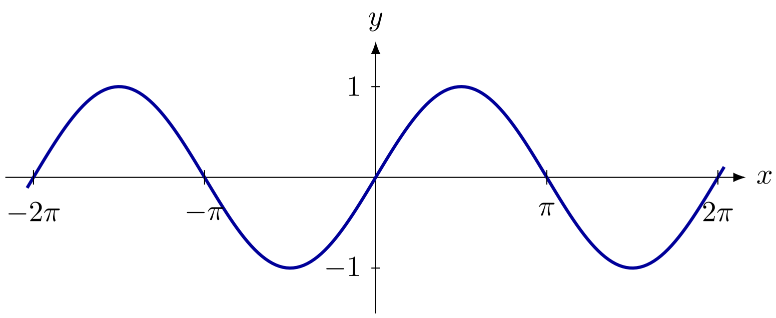


At this point these formulas are examples of how equations encode shapes. Their systematic study will come later.

The graph of the sine function. The equation

$$y = \sin x$$

selects a periodic curve. The same pattern repeats every 2π units along the x-axis.



A line from a value table. For

$$y = x + 2,$$

choosing several values of x gives

x	-2	-1	0	1	2
y	0	1	2	3	4

Plotting these points reveals a straight line. The equation is a rule for generating all points of the graph.

A parabola and a line. For

$$y = x^2 \qquad \text{and} \qquad y = x + 2,$$

a common point must satisfy both conditions. Hence

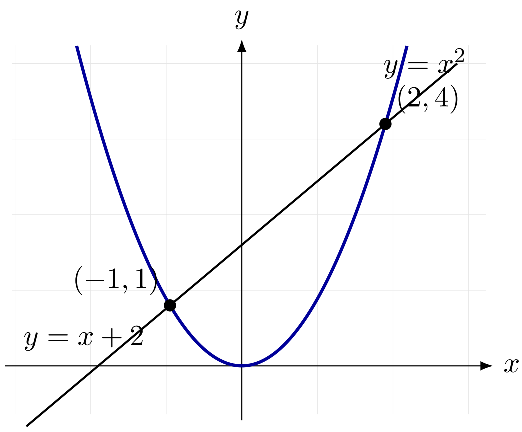
$$x^2 = x + 2,$$

so

$$(x - 2)(x + 1) = 0.$$

The graphs meet at

$$(-1, 1) \qquad \text{and} \qquad (2, 4).$$



A point belongs to a graph when its coordinates satisfy the equation or inequality that defines it. An intersection point belongs to both graphs, so its coordinates satisfy both conditions simultaneously.

Cartesian space. The Cartesian plane is produced by explicit choices: two perpendicular axes, their intersection as origin, positive orientations, and a common unit. Orthogonal projection assigns an ordered pair (x, y) to every point. Equations then select geometric sets of points, intersections impose several conditions simultaneously, and the distance formula translates Euclidean length into arithmetic.

1.4 Euclidean Reference for Self-Study

The following material preserves the geometric background behind the Cartesian construction. It is included so that the chapter remains self-contained, but it is not intended to occupy the same lecture time as coordinates, graphs, and intersections.

1.4.1 Three Different Levels Must Not Be Confused

The logical hierarchy.

1. *Euclid's postulates* describe the geometric world being assumed.
2. *Straightedge-and-compass rules* describe which elementary drawings and intersections may be used in a construction.

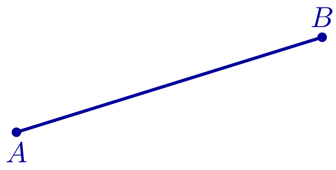
3. *Derived constructions* must be built from those elementary operations and justified by a proof.

Remark. A perpendicular bisector, a perpendicular through a point, an angle bisector, or a parallel through a point is not treated as a primitive permission. Each must be constructed from the elementary toolkit and then justified.

1.4.2 Euclid's Five Postulates

The following are the five postulates used in the first book of Euclid's *Elements*. They specify the elementary geometric operations and relations from which the later constructions begin. The diagrams illustrate their meaning; they are not proofs.

Postulate I: joining two points. A straight segment can be drawn from any point to any other point.

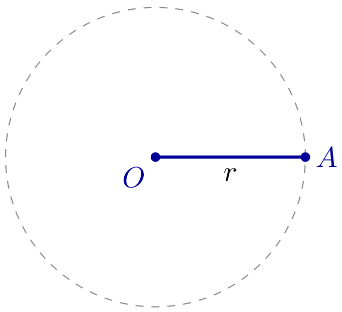


a segment is drawn from A to B

Postulate II: extending a segment. A finite straight segment can be extended continuously in the same straight line.

AB is extended beyond B

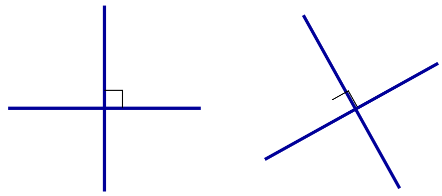
Postulate III: drawing a circle. A circle can be drawn with any chosen centre and any chosen radius.



centre O and radius r determine a circle

Why the circle postulate matters. This postulate permits a length to be copied geometrically without first assigning a number to it. The compass transfers a geometric magnitude directly.

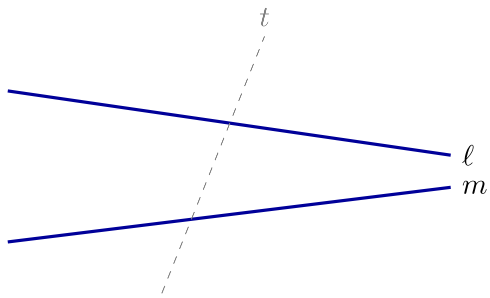
Postulate IV: equality of right angles. All right angles are congruent.



every right angle represents the same geometric magnitude

Remark. This postulate does not say that a perpendicular may simply be drawn without construction. It says that once right angles have been constructed, they are all congruent.

Postulate V: the parallel postulate. If a transversal meets two straight lines and the interior angles on one side sum to less than two right angles, then the two lines, when extended, meet on that side.



the angle condition determines on which side ℓ and m meet

Modern working form. A familiar equivalent is Playfair's formulation: through a point outside a given line there passes exactly one parallel to that line. We shall use that form later, after the relation between the two formulations has been made clear.

What the postulates provide. The first three postulates provide elementary constructions, the fourth makes right angles comparable, and the fifth controls parallelism. Together they specify the geometric world available before coordinates and algebra are introduced.

1.4.3 Elementary Rules for Straightedge-and-Compass Work

For explicit constructions we use the following operational rules.

The elementary construction toolkit.

1. Through two already constructed distinct points, draw the straight line.
2. With an already constructed point as centre and an already constructed segment as radius, draw a circle.
3. Retain intersection points of two constructed lines, a line and a circle, or two circles whenever such intersections exist.

How new objects are produced. The first two rules are operational forms of Euclid's first three postulates. The third rule explains how new constructible points arise: they are intersections of objects already obtained. Every later construction in this chapter must be reducible to this toolkit.

1.4.4 Constructions Derived from the Toolkit

The following objects are not added as new primitive permissions. Each is constructed from lines, circles, and their intersections.

The Perpendicular Bisector and the Midpoint

Construction. Let A and B be distinct points. Draw two circles with the same radius, one centred at A and one centred at B, choosing the radius larger than half of $|AB|$. Let their intersection points be P and Q.

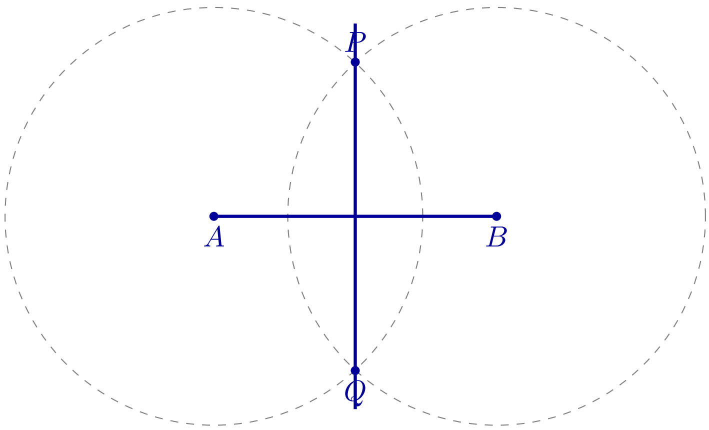
Because P lies on both circles,

$$|PA| = |PB|,$$

and similarly,

$$|QA| = |QB|.$$

Thus both P and Q are equidistant from A and B.



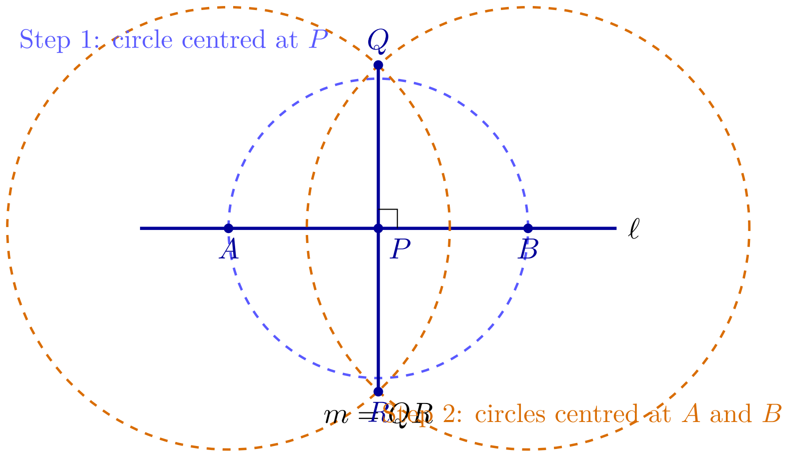
> The line PQ is the perpendicular bisector of AB. Its intersection with AB is the midpoint of AB. Thus both midpoint and perpendicularity are constructed from equal radii and intersections.

A Perpendicular Through a Point on a Line

Construction. Let P lie on a line ℓ.

Step 1. Draw a circle centred at P. It meets ℓ at two points A and B. Since $|PA| = |PB|$, the point P is the midpoint of AB.

Step 2. Apply the perpendicular-bisector construction to AB. The resulting perpendicular bisector passes through its midpoint P.



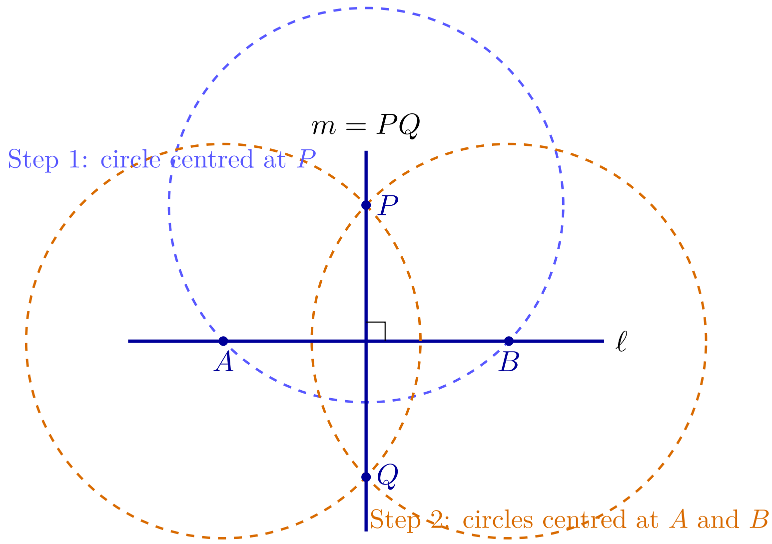
> The constructed line m passes through P and is perpendicular to ℓ.

A Perpendicular Through a Point Outside a Line

Construction. Let P lie outside a line ℓ.

Step 1. Draw a circle centred at P with radius large enough to meet ℓ at two points A and B. Then $|PA| = |PB|$, so P lies on the perpendicular bisector of AB.

Step 2. Draw equal-radius circles centred at A and B, chosen so that they meet again at a second point Q. The points P and Q are both equidistant from A and B.



> The line PQ is the perpendicular bisector of AB and therefore is perpendicular to ℓ through the prescribed external point P.

Bisecting an Angle

Construction and justification. Let two rays with common endpoint O form an angle. Draw a circle centred at O meeting the rays at A and B. Then draw equal-radius circles centred at A and B, and let one of their intersections inside the angle be P. Since

$$|OA| = |OB|, \qquad |AP| = |BP|, \qquad |OP| = |OP|,$$

the triangles OAP and OBP are congruent.

The congruence of OAP and OBP gives

$$\angle AOP = \angle POB.$$

Therefore the ray OP bisects the angle.

Constructing a Parallel Through a Point

Construction. Let P lie outside a line ℓ. First construct a line m through P perpendicular to ℓ. Then construct a line n through P perpendicular to m.

The transversal m forms equal corresponding right angles with n and ℓ. By the converse of the corresponding-angles theorem, the two lines are parallel.

The constructed line satisfies

$$n \parallel \ell.$$

The fifth postulate, equivalently Playfair's postulate, guarantees that this is the unique parallel to ℓ through P.

1.4.5 Parallel Lines and Proportional Segments

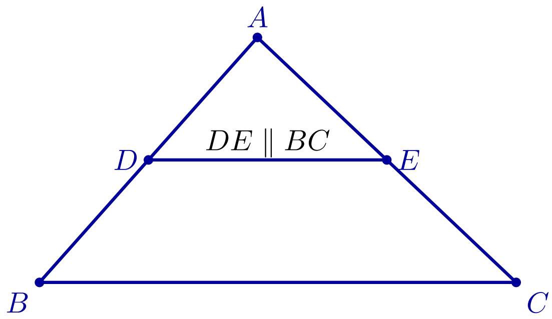
A central fact of Euclidean geometry is that parallel lines preserve ratios. This is the content of Thales' theorem in the form needed here.

Let D lie on AB and E lie on AC in a triangle ABC. If

$$DE \parallel BC,$$

then

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}.$$



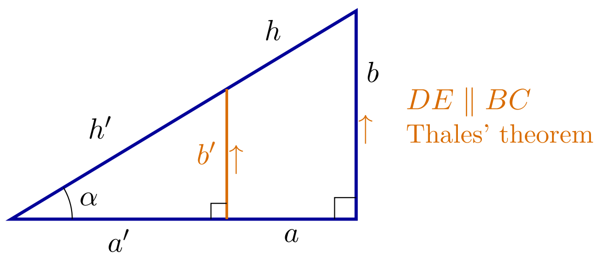
Why the ratios are equal. Because $DE \parallel BC$, corresponding angles in triangles ADE and ABC are equal. The triangles therefore have the same three angles and differ only by one common scale factor. Every corresponding side is multiplied by that same factor, which gives the three equal ratios. □

Two triangles are *similar* when their corresponding angles are equal. In that case their corresponding side lengths are proportional.

What similarity preserves. Similarity changes the size of a figure while preserving its shape. It is the reason ratios of corresponding lengths can depend only on angles and later makes trigonometric ratios well defined.

1.4.6 Sine and Cosine Before Coordinates

Take a right triangle containing an acute angle α. Let h denote the hypotenuse, let a be the leg adjacent to α, and let b be the leg opposite α.



$$\frac{a'}{a} = \frac{b'}{b} = \frac{h'}{h}, \qquad \frac{a'}{h'} = \frac{a}{h}, \qquad \frac{b'}{h'} = \frac{b}{h}.$$

Why the ratios depend only on the angle. All right triangles containing the same acute angle are similar. Corresponding side lengths therefore differ by one common scale factor, which cancels in the ratios a/h and b/h. These ratios depend on the angle, not on the size of the triangle chosen to represent it. □

For an acute geometric angle α,

$$\cos\alpha = \frac{a}{h}, \qquad \sin\alpha = \frac{b}{h}.$$

For a right angle,

$$\cos\alpha = 0.$$

If α is obtuse and β is its acute supplement, define

$$\cos\alpha = -\cos\beta.$$

Thus cosine is positive for acute angles, zero for a right angle, and negative for obtuse angles.

Trigonometry is geometric before it is coordinate-based. Sine and cosine are ratios determined by similarity. Their definitions do not require Cartesian axes or ordered pairs. Coordinates will later provide a new way to calculate and use these same geometric quantities.

The fundamental identity. For an acute angle, the Pythagorean theorem gives

$$a^2 + b^2 = h^2.$$

Dividing by h^2 yields

$$\left(\frac{a}{h}\right)^2 + \left(\frac{b}{h}\right)^2 = 1.$$

Substituting the definitions of cosine and sine gives the identity below.

For every angle in the geometric range considered here,

$$\cos^2\alpha + \sin^2\alpha = 1.$$

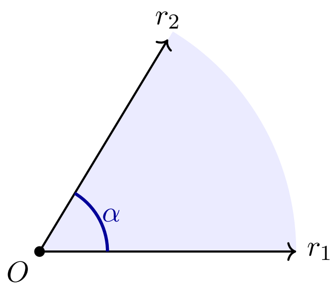
For acute angles this is a direct reformulation of the Pythagorean theorem.

1.4.7 Angles and Their Numerical Measure

Angle and angle measure are different objects. An angle exists geometrically before it is assigned a number. We can compare, copy, add, and bisect angles without degrees or radians. A numerical angle measure requires an additional choice of unit.

> Let r_1 and r_2 be two rays with the same initial point O. The rays, together with a choice of one of the two regions between them, determine an angle α with vertex O.

Remark. The angle is the chosen geometric region. It is neither the small arc drawn near the vertex nor a number. The arc is only a graphical mark indicating which region has been selected.



Why circles give a consistent measure

Draw a circle centred at O. The rays cut from the circle an arc corresponding to the chosen angle.

> For every Euclidean circle, the ratio of circumference C to diameter D is the same constant:
>
> $$\frac{C}{D} = \pi.$$
>
> Since $D = 2r$, every circle of radius r has circumference
>
> $$C = 2\pi r.$$

Why the quotient s/r is independent of the circle. Similarity implies that enlarging a circle by a factor λ multiplies its diameter, circumference, and corresponding arc lengths by λ. If the same rays cut arc lengths s_1, s_2 from circles of radii r_1, r_2, then

$$\frac{s_1}{s_2} = \frac{r_1}{r_2}, \qquad \text{hence} \qquad \frac{s_1}{r_1} = \frac{s_2}{r_2}.$$

The quotient therefore depends only on the angle. □

> Let s be the length of the selected arc on a circle of radius r centred at the vertex. The radian measure of α is
>
> $$m_{\text{rad}}(\alpha) = \frac{s}{r}.$$

Full turn, half-turn, and right angle. For a full turn, $s = 2\pi r$, so

$$m_{\text{rad}}(\text{full turn}) = 2\pi.$$

A half-turn therefore has measure π, and a quarter-turn has measure $\pi/2$.

Degrees provide a second numerical scale. A full turn is assigned $360°$, so

$$2\pi \text{ radians} = 360°, \qquad m_{\mathrm{deg}}(\alpha) = \frac{180°}{\pi}\, m_{\mathrm{rad}}(\alpha).$$

Example. A constructible right angle has the two numerical descriptions

$$m_{\mathrm{rad}}(\text{right angle}) = \frac{\pi}{2}, \qquad m_{\mathrm{deg}}(\text{right angle}) = 90°.$$

The geometric angle is the same object; only the chosen numerical scale changes.

Sine and cosine as functions of a number

The trigonometric ratios were first defined for acute geometric angles. We now extend them to a full turn. For $t \in [0, 2\pi)$, let $\beta \in [0, \pi/2]$ be the acute reference angle between the direction of t and the nearest horizontal ray. The magnitudes $\sin\beta$ and $\cos\beta$ come from the right-triangle definition; their signs are determined by the quadrant:

	$\cos t$	$\sin t$
$0 \le t \le \pi/2$	$+\cos\beta$	$+\sin\beta$
$\pi/2 \le t \le \pi$	$-\cos\beta$	$+\sin\beta$
$\pi \le t \le 3\pi/2$	$-\cos\beta$	$-\sin\beta$
$3\pi/2 \le t < 2\pi$	$+\cos\beta$	$-\sin\beta$

For arbitrary real t, define sine and cosine periodically:

$$\sin(t + 2\pi k) = \sin t, \qquad \cos(t + 2\pi k) = \cos t, \qquad k \in \mathbb{Z}.$$

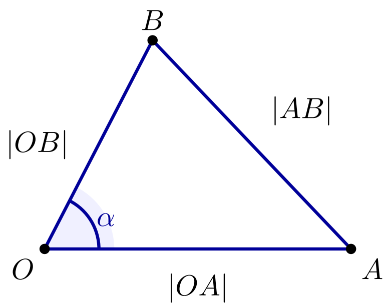
From a geometric angle to a numerical input. If $t = m_{\mathrm{rad}}(\alpha)$, then $\cos t$ is the same value as the geometric cosine $\cos\alpha$. The notation $\cos t$ therefore treats the numerical measure as an input while preserving the earlier geometric meaning.

The Euclidean law of cosines

Directly measuring an arc is often difficult. A triangle provides another way to recover the measure of an angle. Choose points

$$A \in r_1 \setminus \{O\}, \qquad B \in r_2 \setminus \{O\}.$$

The points form the Euclidean triangle OAB.



If $\alpha = \angle AOB$, then

$$|AB|^2 = |OA|^2 + |OB|^2 - 2|OA|\,|OB| \cos\alpha.$$

Equivalently, the last factor may be written as $\cos(m_{\mathrm{rad}}(\alpha))$.

Recovering the angle from side lengths. Solving the law of cosines for the angle measure gives

$$m_{\mathrm{rad}}(\alpha) = \arccos\left(\frac{|OA|^2 + |OB|^2 - |AB|^2}{2|OA|\,|OB|}\right).$$

Later, coordinates will provide an arithmetic method for calculating the three Euclidean lengths in this formula.

Remark. At this point no vector has been introduced. We have only Euclidean objects, postulates, constructions, similarity, trigonometric ratios, numerical angle measure, and the Euclidean law of cosines. Vectors will appear only after the Cartesian coordinate system has been constructed.

What numerical measurement adds. The angle is geometric; radians and degrees are numerical descriptions of it. Similarity makes the radian quotient s/r independent of scale, while the law of cosines allows the same angle to be recovered from three side lengths.

What the Cartesian construction uses. The coordinate plane needs perpendicular axes, an origin, chosen directions, a common unit, and orthogonal projection. The detailed Euclidean reference explains the geometric background of those ingredients; the Cartesian section shows how they become a numerical language.

1.5 Review Exercises

1.5.1 Elementary computational and graphical exercises

1. Plot the points

$$A = (3, 2), \qquad B = (-2, 4), \qquad C = (-3, -1), \qquad D = (2, -3).$$

   For each point, state its quadrant and draw its orthogonal projections onto both axes.

2. For $P = (-4, 3)$, write the two projection points on the coordinate axes and compute the distances from P to the x-axis, the y-axis, and the origin.

3. A point has horizontal coordinate -5 and vertical coordinate 2. Construct it from the axes by orthogonal projection. Repeat for a point with coordinates $(0, -4)$ and explain what is special about its position.

4. Compute and illustrate with right triangles the distances

$$d\big((1, 2), (4, 6)\big), \qquad d\big((-2, 1), (4, 1)\big), \qquad d\big((-3, -2), (1, 1)\big).$$

5. Find the midpoint of the segment joining $A = (-2, 1)$ and $B = (4, 5)$. Verify that the midpoint is equally distant from both endpoints.

6. A point has coordinates $P = (6, -4)$ when one unit on each axis has its original length. Find its coordinates when one new unit is equal to two old units and the origin and axis directions remain unchanged.

7. A point has coordinates $P = (5, 3)$. Move the origin to the old point $(2, 1)$ without changing the units or axis directions. Compute the coordinates of P in the new system.

8. A point has coordinates $P = (4, -2)$. Reverse only the positive direction of the x-axis, leaving the origin, unit, and y-axis unchanged. Find the new coordinates and draw both descriptions of the same point.

9. Prepare a value table for
$$y = -2x + 4$$
at five convenient integer values of x. Plot the points, draw the line, and find its intersections with both coordinate axes.

10. Plot on one coordinate system
$$y = x + 2, \qquad y = x^2.$$
Find their common points algebraically and verify each point in both equations.

1.5.2 Development and reasoning exercises

1. Describe how to construct a Cartesian coordinate system starting only from a plane, one chosen point, one chosen line through that point, and the ability to construct perpendicular lines. Identify every additional choice that must be made before points can receive numerical coordinates.

2. Explain why a geometric point is not the same object as an ordered pair. Choose one fixed point and describe it in two coordinate systems with different origins. State precisely what changes and what remains unchanged.

3. Derive
$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
from the horizontal and vertical displacements between $P = (x_1, y_1)$ and $Q = (x_2, y_2)$. Include a diagram and explain where perpendicularity is used.

4. A drawing uses twice the scale horizontally as vertically. Explain why the picture of
$$x^2 + y^2 = 4$$
may look like an ellipse even though the equation describes a circle. Explain what information belongs to the geometry and what information belongs only to the drawing convention.

5. Find all points on the x-axis whose distance from $P = (2, 3)$ is 5. Before calculating, predict geometrically how many solutions can occur. Then solve and explain why the algebra agrees with the picture.

6. A point is described by $(6, 2)$. A new coordinate system is obtained by moving the origin to $(2, -1)$, reversing the x-axis direction, and replacing the old unit by a unit twice as long. Compute the new coordinates by treating these three changes one at a time, and explain why the order of the reasoning matters.

7. Three vertices of an axis-aligned rectangle are
$$A = (-2, -1), \qquad B = (3, -1), \qquad C = (3, 2).$$
Find the fourth vertex and compute the side and diagonal lengths. Explain how the coordinate description makes the rectangle structure visible without measuring angles numerically.

8. The line $y = 2x - 3$ and the parabola $y = x^2 - 2$ are drawn in the same coordinate system. Find their common points and explain why solving

$$2x - 3 = x^2 - 2$$

is exactly the algebraic version of asking for points that belong to both curves.

9. Orthogonal projection assigns to every point of the plane one number on each axis. Explain why two different points cannot receive the same ordered pair. Then explain why every ordered pair of real numbers determines exactly one point of the plane.

10. The construction of a scale begins with integer marks and repeated bisection. Explain what points can be labelled after finitely many bisections and why an additional continuity assumption is needed to regard every point of the axis as having a real coordinate. Keep the argument geometric rather than set-theoretic.

Chapter 2

Vectors, Bases, and Coordinate Systems

2.1 Vectors in Cartesian Space

In the previous chapter, points of the plane were described by Cartesian coordinates. We now use that numerical description to construct vectors. The order is important: we begin with points and the arithmetic already available for their coordinates, then introduce an arrow determined by an ordered pair of points, decide geometrically when two such arrows are equal, and only then pass to free vectors and operations on them.

The coordinate formulas will not replace the geometry. They will provide practical criteria for recognising and calculating the geometric objects that have already been introduced.

2.1.1 From Cartesian Points to Free Vectors

A point of the Cartesian plane is an ordered pair of real numbers. Let

$$P = (p_1, p_2), \qquad Q = (q_1, q_2).$$

Before vectors are introduced, we first define several arithmetic operations on such points. These definitions are possible because a point of the Cartesian plane is represented by an ordered pair of numbers.

The point

$$O = (0, 0)$$

plays the role of zero. Addition of points is defined coordinatewise:

$$\boxed{P + Q = (p_1 + q_1,\ p_2 + q_2)}.$$

The point opposite to P is

$$\boxed{-P = (-p_1, -p_2)},$$

so that

$$P + (-P) = O.$$

Subtraction of points is defined by adding the opposite point:

$$\boxed{Q - P = Q + (-P) = (q_1 - p_1,\ q_2 - p_2)}.$$

For a real number λ, multiplication of a point by a number is also defined coordinatewise:

$$\boxed{\lambda P = (\lambda p_1,\ \lambda p_2)}.$$

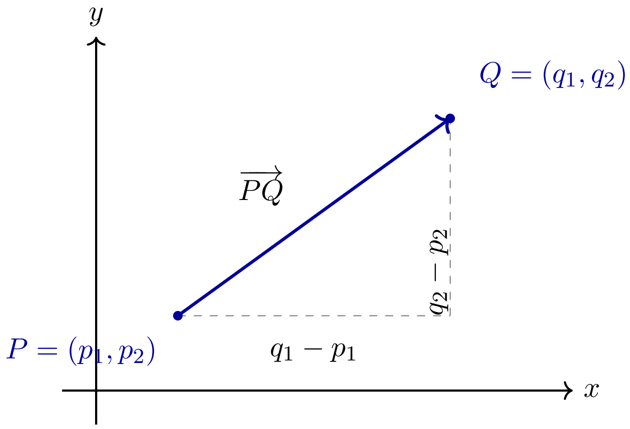
At this stage these are arithmetic operations on ordered pairs. They are useful computational rules, but their geometric meaning has not yet been established. The first expression to acquire such a meaning will be the difference $Q - P$.

An ordered pair of points determines an anchored vector

The ordered pair (P, Q) determines an arrow beginning at P and ending at Q. We denote it by

$$\overrightarrow{PQ}$$

and call it an *anchored vector*. The order matters: P is its initial point and Q is its endpoint.



The point difference is

$$Q - P = (q_1 - p_1,\ q_2 - p_2).$$

The two numbers $q_1 - p_1$ and $q_2 - p_2$ are, by definition, the coordinates of the anchored vector $\overrightarrow{PQ}$. We do not use round brackets for the vector: round brackets remain reserved for Cartesian points.

Its length is the distance between the two points:

$$\boxed{|\overrightarrow{PQ}| = \varrho(P, Q)}.$$

Reversing the order reverses the arrow and changes the signs of both coordinates. The vector $\overrightarrow{QP}$ therefore has coordinates

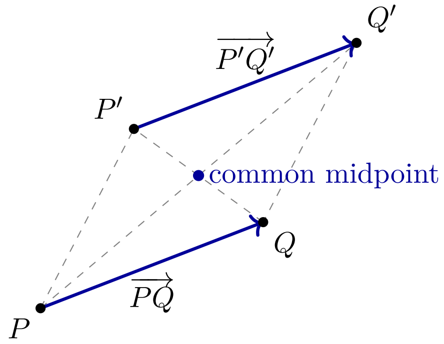
$$p_1 - q_1, \qquad p_2 - q_2.$$

When do arrows drawn in different places represent the same vector?

We first define equality geometrically. Consider two anchored vectors

$$\overrightarrow{PQ} \qquad \text{and} \qquad \overrightarrow{P'Q'}.$$

They are called equal when the midpoint of the pair P, Q' is also the midpoint of the pair Q, P'. Equivalently, the four points form a parallelogram in which the two arrows are opposite directed sides.



The midpoint condition is

$$\frac{1}{2}(P+Q') = \frac{1}{2}(Q+P').$$

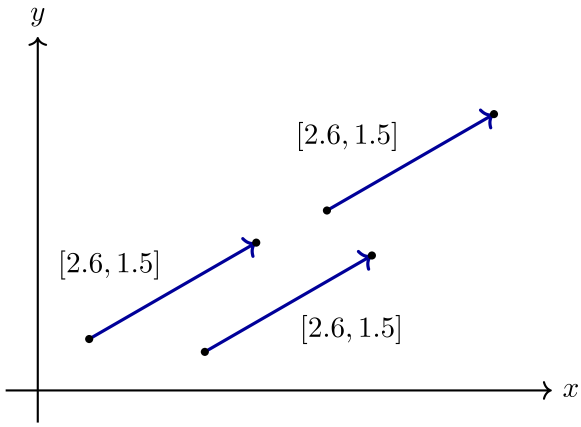
Rearranging this equality gives

$$Q - P = Q' - P'.$$

We have therefore obtained the coordinate criterion for the geometric relation:

$$\boxed{\overrightarrow{PQ} = \overrightarrow{P'Q'} \quad \Longleftrightarrow \quad Q - P = Q' - P'}.$$

Thus two anchored vectors are equal precisely when their corresponding coordinates are equal. Their initial points may be different.



Free vectors

Equality divides all anchored vectors into disjoint classes. Each class contains all arrows representing one and the same directed change. Such a class is called a *free vector.*

If $\overrightarrow{PQ}$ is one representative of a free vector $\mathbf{v}$, we write

$$\boxed{\mathbf{v} = [\overrightarrow{PQ}]}.$$

Here the square brackets denote the equivalence class of all anchored vectors equal to $\overrightarrow{PQ}$.

If the common coordinates of these representatives are v_1, v_2, then the same free vector is written

$$\boxed{\mathbf{v} = [v_1, v_2]}.$$

Thus round brackets distinguish points from vectors:

$$V = (v_1, v_2) \qquad \text{is a point, whereas} \qquad \mathbf{v} = [v_1, v_2] \qquad \text{is a free vector.}$$

The book's notation can now be used without ambiguity. Since

$$Q - P = (q_1 - p_1,\, q_2 - p_2)$$

is a Cartesian point, placing square brackets around this point means the free vector with the same coordinate entries:

$$\boxed{[\overrightarrow{PQ}] = [Q - P] = [q_1 - p_1,\, q_2 - p_2]}.$$

The outer square brackets change the type of object: $(q_1 - p_1, q_2 - p_2)$ is a point, while $[q_1 - p_1, q_2 - p_2]$ is a free vector.

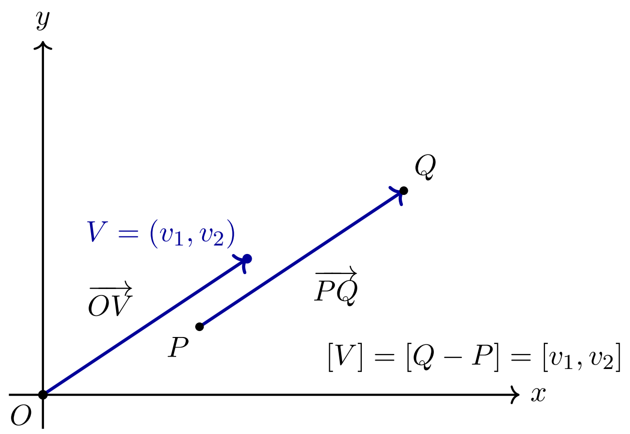
A particularly convenient representative begins at the origin. If

$$O = (0,0), \qquad V = (v_1, v_2),$$

then

$$\mathbf{v} = [\overrightarrow{OV}] = [V] = [v_1, v_2].$$

This does not identify the point V with the vector $\mathbf{v}$; it gives a canonical representative of the free vector.



Only now, after free vectors have been constructed, do we define operations on them. Each operation will first be described using representatives and then written in coordinates.

2.1.2 Addition, Opposite Vectors, and Subtraction

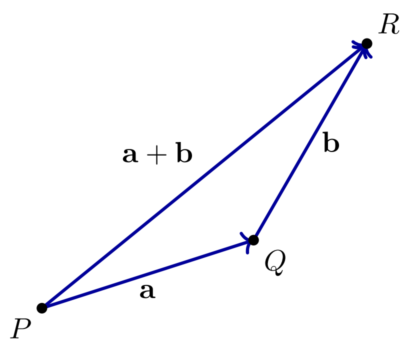
Let $\overrightarrow{PQ}$ represent the free vector

$$\mathbf{a} = [a_1, a_2],$$

and choose a representative $\overrightarrow{QR}$ of

$$\mathbf{b} = [b_1, b_2]$$

whose initial point is the endpoint of the first arrow. The arrow $\overrightarrow{PR}$ records the combined change.



Since

$$Q = P + (a_1, a_2)$$

and

$$R = Q + (b_1, b_2),$$

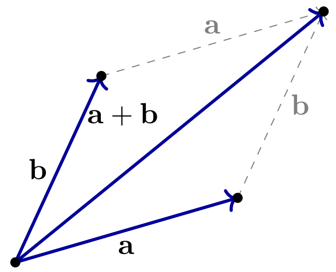
we have

$$R = P + (a_1 + b_1,\ a_2 + b_2).$$

Therefore the free vector represented by $\overrightarrow{PR}$ depends only on $\mathbf{a}$ and $\mathbf{b}$, not on the chosen point P. We define

$$\boxed{\mathbf{a}+\mathbf{b} = [\overrightarrow{PR}] = [a_1+b_1,\, a_2+b_2]}.$$

The same sum can be constructed in the opposite order. Placing a representative of $\mathbf{b}$ first and then a representative of $\mathbf{a}$ gives the same endpoint. This is the parallelogram rule.



Thus

$$\boxed{\mathbf{a}+\mathbf{b} = \mathbf{b}+\mathbf{a}}.$$

For every vector

$$\mathbf{a} = [a_1, a_2]$$

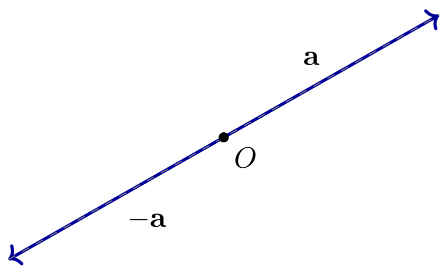
there is an *opposite vector*

$$\boxed{-\mathbf{a} = [-a_1, -a_2]},$$

characterised by

$$\mathbf{a} + (-\mathbf{a}) = \mathbf{0}.$$

Geometrically it has the same length and direction line as $\mathbf{a}$, but the opposite orientation.

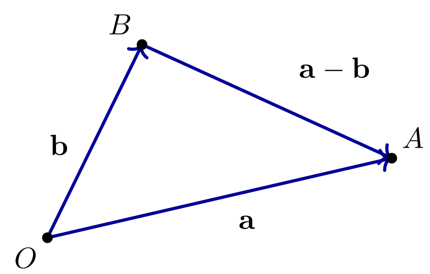


Subtraction is defined by adding the opposite vector:

$$\boxed{\mathbf{a}-\mathbf{b} = \mathbf{a}+(-\mathbf{b}) = [a_1-b_1,\, a_2-b_2]}.$$

It is the missing vector in the equation

$$(\mathbf{a}-\mathbf{b}) + \mathbf{b} = \mathbf{a}.$$



The last diagram also explains the relation between points and vectors. For any two points P and Q,

$$[\overrightarrow{PQ}] = Q - P.$$

The subtraction is performed on their coordinate descriptions and produces the free vector represented by the arrow from P to Q.

2.1.3 Parallel Vectors and Multiplication by a Number

Before multiplying a vector by a number, we identify the geometric relation that this operation is meant to preserve.

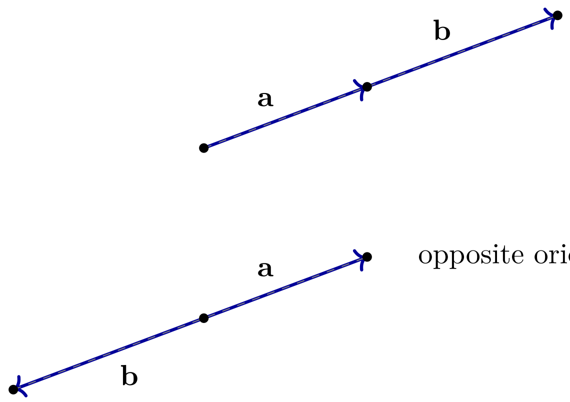
Two nonzero free vectors $\mathbf{a}$ and $\mathbf{b}$ have the same direction and the same orientation when placing them successively produces no shortening:

$$|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|.$$

They have the same direction and opposite orientations when

$$|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|.$$

In both cases the vectors are called *parallel*.



Let

$$\mathbf{a} = [a_1, a_2], \qquad \mathbf{b} = [b_1, b_2].$$

The equality condition in the triangle inequality implies that the two vectors are parallel exactly when their coordinate pairs are proportional. Thus

$$\boxed{\mathbf{a} \parallel \mathbf{b} \quad \Longleftrightarrow \quad \lambda \mathbf{a} = \mu \mathbf{b}}$$

for some real numbers λ and μ that are not both zero. In the plane, this condition may also be written as

$$\boxed{a_1 b_2 - a_2 b_1 = 0}.$$

If λ and μ have the same sign, the vectors have the same orientation; if their signs are opposite, the orientations are opposite.

We can now define multiplication of a vector by a real number t.

If $t > 0$, the vector $t\mathbf{a}$ is parallel to $\mathbf{a}$, has the same orientation, and has length

$$|t\mathbf{a}| = t|\mathbf{a}|.$$

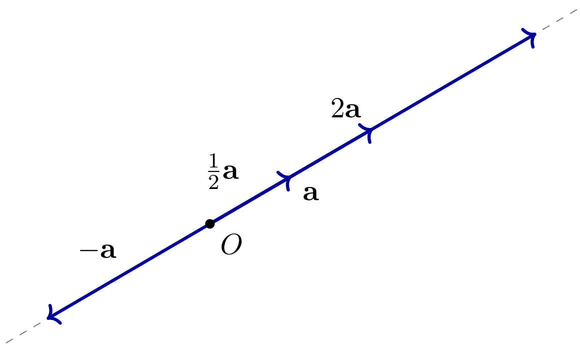
If $t < 0$, it is parallel to $\mathbf{a}$, has the opposite orientation, and has length

$$|t\mathbf{a}| = |t|\,|\mathbf{a}|.$$

For $t = 0$, the result is the zero vector.

In coordinates this geometric definition becomes

$$\boxed{t[a_1, a_2] = [ta_1, ta_2]}.$$



The diagram shows all cases at once:

t	length	orientation
$t > 1$	increased	unchanged
$0 < t < 1$	decreased	unchanged
$t = 0$	0	none
$t < 0$	$\lvert t\rvert\, \lvert\mathbf{a}\rvert$	reversed

Multiplication by a number is compatible with vector addition:

$$\boxed{t(\mathbf{a} + \mathbf{b}) = t\mathbf{a} + t\mathbf{b}}$$

and

$$\boxed{(s + t)\mathbf{a} = s\mathbf{a} + t\mathbf{a}}.$$

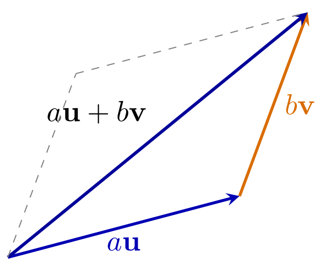
These identities follow directly from coordinatewise addition and multiplication, but their geometric content is that scaling commutes with combining directed changes.

2.1.4 Linear Combinations: What Addition and Scaling Can Build

Addition and scalar multiplication were introduced as two separate operations, but they are most useful when they are combined.

> Given vectors $\mathbf{u}$ and $\mathbf{v}$ and real numbers a and b, the vector
>
> $$a\mathbf{u} + b\mathbf{v}$$
>
> is called a *linear combination* of $\mathbf{u}$ and $\mathbf{v}$.

How a linear combination is constructed. First scale $\mathbf{u}$ by a and $\mathbf{v}$ by b; then add the resulting displacements. A linear combination is therefore not a new operation. It is a controlled use of addition and scalar multiplication.



One direction versus two directions. A single nonzero vector generates only the line

$$\{\lambda\mathbf{v} : \lambda \in \mathbb{R}\}.$$

Two parallel vectors still generate only one line. Two nonparallel vectors provide two independent directions and can generate the entire plane.

Why nonparallel directions reach the plane. If $\mathbf{u}$ and $\mathbf{v}$ are parallel, every scalar multiple of either vector lies on the same line, so every sum $a\mathbf{u} + b\mathbf{v}$ remains on that line. If they are not parallel, the parallelogram construction allows a target displacement to be decomposed into one component along each direction. The two coefficients record those two signed motions. □

> Two nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$ determine every vector in the plane uniquely in the form
>
> $$\mathbf{w} = a\mathbf{u} + b\mathbf{v}.$$
>
> The numbers a and b record how much motion occurs in each chosen direction.

Bridge to a basis. A basis is not an unrelated new object. It is a pair of independent directions whose linear combinations describe every vector uniquely. Basis coordinates are precisely the coefficients in that reconstruction.

2.1.5 How Much Motion and in Which Direction?

The coordinate pair of a vector records two changes, but it does not immediately answer two basic geometric questions.

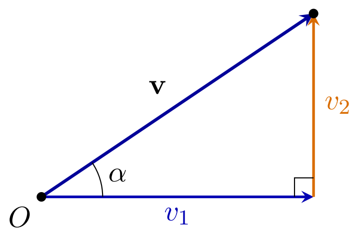
Two questions carried by every nonzero vector.

1. How long is the displacement?
2. In which direction does it point?

Let

$$\mathbf{v} = (v_1, v_2)$$

be drawn from the origin. Its endpoint, together with its horizontal and vertical coordinate changes, forms a right triangle. The legs have lengths $|v_1|$ and $|v_2|$, while the vector itself is the hypotenuse.



Length from the Pythagorean theorem. The right triangle gives

$$\|\mathbf{v}\|^2 = v_1^2 + v_2^2.$$

The length is nonnegative, so

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$

> The Euclidean length, or norm, of $\mathbf{v} = (v_1, v_2)$ is
>
> $$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$
>
> The double bars distinguish the vector from the number measuring its length.

$$\|\mathbf{0}\| = 0, \qquad \mathbf{v} \neq \mathbf{0} \implies \|\mathbf{v}\| > 0.$$

Moreover, for points A and B,

$$d_E(A, B) = \|B - A\|.$$

A displacement packages distance and direction. The vector $B - A$ records how to move from A to B, while its norm records the Euclidean distance travelled. The same object therefore carries both directional and metric information.

Now assume $\mathbf{v} \neq \mathbf{0}$ and let α be its angle of inclination, measured from the positive x-axis.

Separating magnitude from direction. The right triangle gives

$$\cos\alpha = \frac{v_1}{\|\mathbf{v}\|}, \qquad \sin\alpha = \frac{v_2}{\|\mathbf{v}\|}.$$

Therefore

$$v_1 = \|\mathbf{v}\| \cos\alpha, \qquad v_2 = \|\mathbf{v}\| \sin\alpha.$$

Every nonzero vector admits the decomposition

$$\mathbf{v} = \|\mathbf{v}\|(\cos\alpha, \sin\alpha).$$

The vector $(\cos\alpha, \sin\alpha)$ has length one.

The two pieces of a vector.

- $\|\mathbf{v}\|$ says how much displacement occurs;
- $(\cos\alpha, \sin\alpha)$ says in which direction it occurs.

Thus every nonzero vector is a positive magnitude multiplied by a unit direction.

2.1.6 One Angle Calculation, Two Geometric Tests

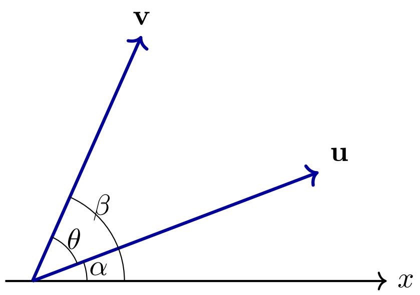
Take two arbitrary nonzero vectors

$$\mathbf{u} = (u_1, u_2), \qquad \mathbf{v} = (v_1, v_2).$$

Let their angles of inclination be α and β. Then

$$\mathbf{u} = \|\mathbf{u}\|(\cos\alpha, \sin\alpha), \qquad \mathbf{v} = \|\mathbf{v}\|(\cos\beta, \sin\beta).$$

One comparison controls both tests. The directed turn from $\mathbf{u}$ to $\mathbf{v}$ is $\beta - \alpha$. Its sine detects parallel directions, while its cosine detects perpendicular directions. The coordinate tests will be derived from these geometric facts.



Parallel directions arise from the sine of the angle difference

Deriving the determinant test. Two nonzero vectors determine the same unoriented direction precisely when

$$\beta - \alpha \equiv 0 \pmod{\pi},$$

equivalently when $\sin(\beta - \alpha) = 0$. Using

$$\sin(\beta - \alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha$$

and substituting

$$\cos\alpha = \frac{u_1}{\|\mathbf{u}\|}, \quad \sin\alpha = \frac{u_2}{\|\mathbf{u}\|}, \quad \cos\beta = \frac{v_1}{\|\mathbf{v}\|}, \quad \sin\beta = \frac{v_2}{\|\mathbf{v}\|},$$

we obtain

$$\sin(\beta - \alpha) = \frac{u_1 v_2 - u_2 v_1}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

The denominator is nonzero, so the sine vanishes exactly when the numerator vanishes.

> For nonzero vectors in the plane,
>
> $$\mathbf{u} \parallel \mathbf{v} \iff u_1 v_2 - u_2 v_1 = 0.$$
>
> Equivalently,
>
> $$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} = \lambda\mathbf{v} \text{ for some } \lambda \neq 0.$$

Why the determinant condition gives a scalar multiple. If $v_1 \neq 0$, set $\lambda = u_1/v_1$. The equality $u_1 v_2 - u_2 v_1 = 0$ gives $u_2 = \lambda v_2$, hence $\mathbf{u} = \lambda\mathbf{v}$. If $v_1 = 0$, then $v_2 \neq 0$ and the same argument uses $\lambda = u_2/v_2$. □

Meaning of the determinant expression. The quantity $u_1 v_2 - u_2 v_1$ measures whether the two coordinate directions span genuine area. It is zero exactly when both vectors lie on one line.

The cosine of the same difference reveals perpendicularity

Deriving the dot-product test. The vectors are perpendicular precisely when

$$\beta - \alpha \equiv \frac{\pi}{2} \pmod{\pi},$$

equivalently when $\cos(\beta - \alpha) = 0$. The identity

$$\cos(\beta - \alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha$$

gives

$$\cos(\beta - \alpha) = \frac{u_1 v_1 + u_2 v_2}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

Since the denominator is nonzero, perpendicularity is equivalent to the vanishing of the numerator.

> For vectors in the plane, the *dot product* is
>
> $$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

In three-dimensional space,

$$(u_1, u_2, u_3) \cdot (v_1, v_2, v_3) = u_1v_1 + u_2v_2 + u_3v_3.$$

For nonzero vectors,

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0.$$

More generally, if θ is the smaller angle between them, then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta.$$

What the sign of the dot product says. The dot product is positive for an acute angle, zero for a right angle, and negative for an obtuse angle. It converts angular information into one scalar.

Example. Take

$$\mathbf{u} = (3, 2), \qquad \mathbf{v} = (-2, 3).$$

Then

$$\mathbf{u} \cdot \mathbf{v} = 3(-2) + 2(3) = 0.$$

Therefore $\cos\theta = 0$, so the vectors are perpendicular.

Two coordinate tests from one geometric comparison. The sine of the angle difference produces the parallelism test $u_1v_2 - u_2v_1 = 0$. The cosine of the same difference produces the perpendicularity test $\mathbf{u} \cdot \mathbf{v} = 0$. Neither rule is imposed arbitrarily; both are consequences of the geometry of angle.

2.1.7 Why the Dot Product Can Be Expanded

The dot product was discovered from the cosine of the angle between two vectors. Before using it in longer calculations, we must verify how it behaves under addition and scalar multiplication.

Verification from coordinates. Let

$$\mathbf{u} = (u_1, u_2), \qquad \mathbf{v} = (v_1, v_2), \qquad \mathbf{w} = (w_1, w_2).$$

Then

$$\begin{aligned}(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} &= (u_1 + w_1)v_1 + (u_2 + w_2)v_2 \\ &= (u_1v_1 + u_2v_2) + (w_1v_1 + w_2v_2) \\ &= \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v}.\end{aligned}$$

Thus the dot product distributes over vector addition. Similarly,

$$(\lambda\mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v}).$$

The symmetry of multiplication of real numbers also gives

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}.$$

Finally,

$$\mathbf{u} \cdot \mathbf{u} = u_1^2 + u_2^2 = \|\mathbf{u}\|^2.$$

□

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and a real number λ,

$$(\mathbf{u}+\mathbf{w}) \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v},$$
$$(\lambda \mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v}),$$
$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}.$$

Moreover,

$$\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2.$$

Remark. These properties justify every later expansion involving expressions such as $(\mathbf{u}-\mathbf{v}) \cdot (\mathbf{u}-\mathbf{v})$. They are not formal rules borrowed from elsewhere; they follow from the coordinate definition of the dot product.

2.1.8 The Angle Between Two Vectors

Directed turn versus ordinary angle. The difference of inclination angles $\beta - \alpha$ is a directed turn modulo a full turn. The ordinary angle between two nonzero vectors is the smaller, unoriented separation between their directions.

For nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, their angle is the unique number

$$\theta \in [0, \pi]$$

satisfying

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

Why the interval is $[0, \pi]$. The interval chooses the smaller unoriented angle. Directions with inclinations 10° and 350° are separated by 20°, not 340°.

The angle distinguishes the principal relative positions:

$$\theta = 0 \iff \text{same direction}, \qquad \theta = \frac{\pi}{2} \iff \text{perpendicular}, \qquad \theta = \pi \iff \text{opposite directions}.$$

Remark. The Cauchy–Schwarz inequality guarantees that

$$-1 \le \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \le 1,$$

so the angle is well defined. The inequality will be justified from the vector geometry developed later.

2.1.9 The Cosine Theorem Emerges Immediately

Draw $\mathbf{u}$ and $\mathbf{v}$ from a common point. The displacement from the endpoint of $\mathbf{v}$ to the endpoint of $\mathbf{u}$ is

$$\mathbf{u} - \mathbf{v}.$$

Thus these three vectors determine a triangle with side lengths

$$\|\mathbf{u}\|, \qquad \|\mathbf{v}\|, \qquad \|\mathbf{u}-\mathbf{v}\|.$$

Expanding the third side.

$$\begin{aligned}\|\mathbf{u}-\mathbf{v}\|^2 &= (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v})\\ &= \mathbf{u}\cdot\mathbf{u}-\mathbf{u}\cdot\mathbf{v}-\mathbf{v}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v}\\ &= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2-2\mathbf{u}\cdot\mathbf{v}\\ &= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2-2\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta.\end{aligned}$$

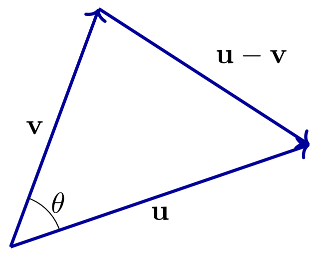
If

$$a=\|\mathbf{u}\|,\qquad b=\|\mathbf{v}\|,\qquad c=\|\mathbf{u}-\mathbf{v}\|,$$

then the familiar triangle formula follows.

> For a triangle with side lengths a, b, and c, where θ is the angle between the sides of lengths a and b,
> $$c^2=a^2+b^2-2ab\cos\theta.$$

Why the theorem appears here. The cosine theorem is not inserted as an external formula. It is the expansion of the squared displacement between the endpoints of two arbitrary vectors. The triangle law is therefore encoded in vector subtraction and the dot product.



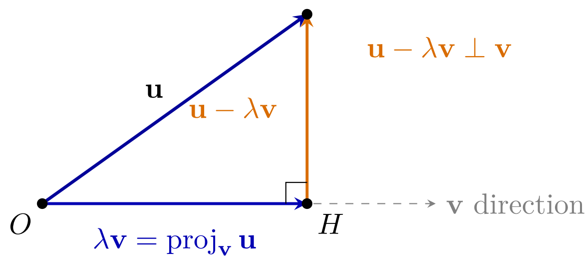
2.1.10 Projection Is Forced by the Right-Angle Condition

Let $\mathbf{v}\neq\mathbf{0}$. We seek a multiple $\lambda\mathbf{v}$ such that the remainder

$$\mathbf{u}-\lambda\mathbf{v}$$

is perpendicular to $\mathbf{v}$. Geometrically, this means decomposing $\mathbf{u}$ into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$:

$$\mathbf{u}=\lambda\mathbf{v}+(\mathbf{u}-\lambda\mathbf{v}).$$



Geometric requirement. The point H is chosen on the line spanned by $\mathbf{v}$. The vector from O to H is the parallel component $\lambda\mathbf{v}$, while the vector from H to the endpoint of $\mathbf{u}$ is the perpendicular remainder. The right-angle condition determines the value of λ.

Derivation. The perpendicularity criterion requires

$$(\mathbf{u}-\lambda\mathbf{v})\cdot\mathbf{v}=0.$$

Expanding gives

$$\mathbf{u} \cdot \mathbf{v} - \lambda(\mathbf{v} \cdot \mathbf{v}) = 0,$$

so

$$\lambda = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}.$$

For $\mathbf{v} \neq \mathbf{0}$, the projection of $\mathbf{u}$ onto the direction of $\mathbf{v}$ is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Remark. The formula is the solution of a geometric requirement, not a rule to be memorised without explanation.

2.1.11 Orthogonal Decomposition and Its Consequences

For $\mathbf{v} \neq \mathbf{0}$, the orthogonal decomposition of $\mathbf{u}$ relative to the direction of $\mathbf{v}$ is

$$\mathbf{u} = \operatorname{proj}_{\mathbf{v}} \mathbf{u} + \left(\mathbf{u} - \operatorname{proj}_{\mathbf{v}} \mathbf{u}\right).$$

The first term is parallel to $\mathbf{v}$ and the second is perpendicular to $\mathbf{v}$.

Because the two components are perpendicular,

$$\|\mathbf{u}\|^2 = \left\|\operatorname{proj}_{\mathbf{v}} \mathbf{u}\right\|^2 + \left\|\mathbf{u} - \operatorname{proj}_{\mathbf{v}} \mathbf{u}\right\|^2.$$

This is the Pythagorean theorem applied to vector components.

Scalar component and vector projection

The signed scalar component of $\mathbf{u}$ in the direction of nonzero $\mathbf{v}$ is

$$\operatorname{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \operatorname{comp}_{\mathbf{v}} \mathbf{u} \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Scalar amount and vector displacement. The scalar component measures signed length along the chosen direction. The vector projection turns that signed amount back into an actual displacement by multiplying it by the unit vector $\mathbf{v}/\|\mathbf{v}\|$.

The Cauchy–Schwarz inequality

Projection cannot be longer than the whole vector. The length of the parallel component cannot exceed the length of $\mathbf{u}$:

$$|\operatorname{comp}_{\mathbf{v}} \mathbf{u}| \leq \|\mathbf{u}\|.$$

Substituting the scalar-component formula gives

$$\frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{v}\|} \leq \|\mathbf{u}\|.$$

Multiplying by $\|\mathbf{v}\| > 0$ yields the inequality below. □

For all vectors $\mathbf{u}$ and $\mathbf{v}$,

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|.$$

Consequently,

$$-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|} \leq 1$$

for nonzero vectors, so the angle formula using arccos is meaningful.

The triangle inequality

The direct route is no longer than a broken route. Using Cauchy–Schwarz,

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\| \, \|\mathbf{v}\| + \|\mathbf{v}\|^2 \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2. \end{aligned}$$

Both sides are nonnegative, so taking square roots gives the desired result. □

For all vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Geometric meaning. The triangle inequality is the algebraic form of the Euclidean fact that the straight segment between two points is no longer than a route that changes direction at an intermediate point.

2.2 One Vector in a Different Basis

A vector does not change when we describe it in another basis. Only the numbers used to reconstruct it from the chosen basis vectors change. One numerical example is enough to see the mechanism.

Let

$$\mathbf{b}_1 = [1, 1], \qquad \mathbf{b}_2 = [1, -1]$$

and consider the vector

$$\mathbf{v} = [4, 2].$$

We seek numbers α and β such that

$$\mathbf{v} = \alpha \mathbf{b}_1 + \beta \mathbf{b}_2.$$

Using the Cartesian coordinates of the basis vectors gives

$$[4, 2] = \alpha[1, 1] + \beta[1, -1] = [\alpha + \beta, \alpha - \beta].$$

Therefore

$$\alpha + \beta = 4, \qquad \alpha - \beta = 2.$$

Solving this small system gives

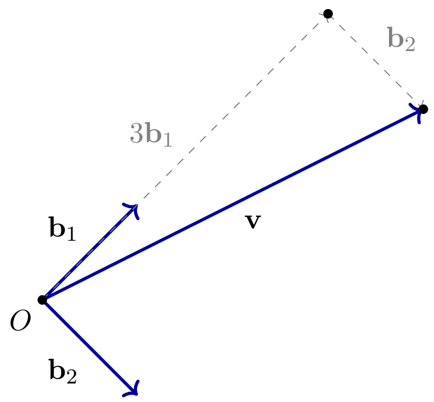
$$\alpha = 3, \qquad \beta = 1.$$

Hence

$$\boxed{\mathbf{v} = 3\mathbf{b}_1 + \mathbf{b}_2}$$

and the coordinates of the same vector in the basis $B = (\mathbf{b}_1, \mathbf{b}_2)$ are

$$\boxed{[\mathbf{v}]_B = [3, 1].}$$



The Cartesian coordinates $[4, 2]$ and the basis coordinates $[3, 1]_B$ are two numerical descriptions of the same geometric vector. Changing the basis means solving for the coefficients in one concrete linear combination; no additional general machinery is needed here.

2.3 Review Exercises

2.3.1 Elementary computational and graphical exercises

1. Let $A = (1, 2)$ and $B = (5, -1)$. Compute the free vector $[\overrightarrow{AB}]$ and draw equal anchored representatives with initial points $O = (0, 0)$ and $P = (-2, 3)$.

2. For
$$\mathbf{u} = [3, -2], \qquad \mathbf{v} = [-1, 4],$$
compute $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $2\mathbf{u}$, and $-3\mathbf{v}$. Draw $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} + \mathbf{v}$ using the parallelogram rule.

3. Compute $2\mathbf{u} - 3\mathbf{v}$ for $\mathbf{u} = [1, 2]$ and $\mathbf{v} = [2, -1]$. Reproduce the same result by joining anchored representatives head to tail.

4. Compute the lengths and unit vectors for
$$\mathbf{u} = [3, 4], \qquad \mathbf{v} = [-5, 12], \qquad \mathbf{w} = [2, -2].$$

5. For
$$\mathbf{u} = [2, 1], \qquad \mathbf{v} = [1, -2], \qquad \mathbf{w} = [4, 2],$$
compute the three dot products and decide which pairs are perpendicular or parallel.

6. Find the angle between $\mathbf{u} = [1, 0]$ and $\mathbf{v} = [1, 1]$, and between $\mathbf{v} = [1, 1]$ and $\mathbf{w} = [-1, 1]$.

7. Compute the projection of $\mathbf{u} = [4, 3]$ onto $\mathbf{v} = [1, 0]$ and onto $\mathbf{w} = [1, 1]$. In each case write $\mathbf{u}$ as a parallel component plus a perpendicular remainder.

8. Let
$$\mathbf{b}_1 = [1, 1], \qquad \mathbf{b}_2 = [1, -1], \qquad \mathbf{u} = [4, 2].$$
Find α, β such that
$$\mathbf{u} = \alpha\mathbf{b}_1 + \beta\mathbf{b}_2,$$
and verify the result by reconstruction. State the coordinates of $\mathbf{u}$ in the basis $B = (\mathbf{b}_1, \mathbf{b}_2)$.

9. Convert between Cartesian and polar coordinates:
   (a) $(\sqrt{3}, 1)$ and $(-1, \sqrt{3})$ to polar form;
   (b) $(r, \theta) = (4, \pi/6)$ and $(3, \pi)$ to Cartesian form.

2.3.2 Development and reasoning exercises

1. For
$$\mathbf{u} = [5, 1], \qquad \mathbf{v} = [2, 1],$$
compute the projection of $\mathbf{u}$ onto $\mathbf{v}$, the perpendicular remainder, and verify by a dot product that the remainder is perpendicular to $\mathbf{v}$. Draw the decomposition.

2. Choose two nonzero vectors $\mathbf{u}, \mathbf{v}$ with the same length but with $\mathbf{u} + \mathbf{v}$ shorter than either vector. Compute the relevant lengths and explain the geometric cancellation.

3. Find all values of t for which
$$\mathbf{u} = [1, t], \qquad \mathbf{v} = [2, -1]$$
are perpendicular. Then find all values for which they are parallel. Interpret both conditions geometrically.

4. Prove from the construction with anchored representatives, rather than only from coordinates, that vector addition is associative. Illustrate the argument with three concrete vectors.

5. Show that two nonzero vectors in the plane are parallel exactly when one is a scalar multiple of the other. Connect the geometric statement with the proportionality of their coordinates.

6. Explain how
$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos\theta$$
contains both length and angular information. Use one acute, one right, and one obtuse example.

7. Compare Cartesian coordinates, coordinates in the basis $B = (\mathbf{b}_1, \mathbf{b}_2)$ from the numerical example, and polar coordinates. Give one concrete situation in which each description is useful.

Chapter 3

Lines and Planes in the Plane and in Space

3.1 Lines in the Cartesian Plane

A line should not first appear as a formula to be recognised. We already have all the ingredients needed to construct it geometrically: a point tells us where to begin, and a nonzero vector tells us in which direction motion is allowed. The question is therefore:

Which points can be reached from one fixed point by moving only in one fixed direction?

Data that determine a line. A line in the Cartesian plane is determined by one point P_0 on the line and one nonzero direction vector $\mathbf{v}$. Every point of the line is reached by adding a scalar multiple of $\mathbf{v}$ to P_0.

Let

$$P_0 = (x_0, y_0)$$

be fixed and let

$$\mathbf{v} = (u, v) \neq \mathbf{0}$$

be a fixed direction vector. Every permitted displacement is a scalar multiple $t\mathbf{v}$. Starting from P_0 gives

$$P = P_0 + t\mathbf{v}, \qquad t \in \mathbb{R}.$$

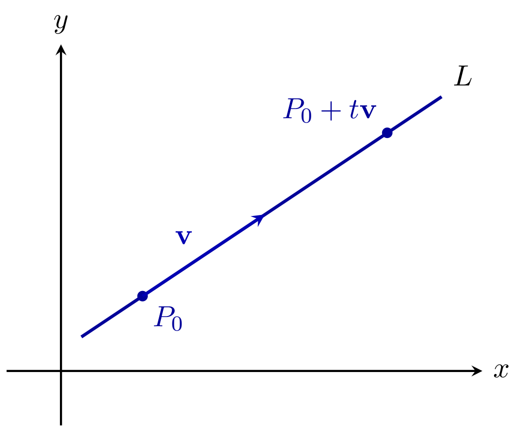
The line through P_0 with direction vector $\mathbf{v} \neq \mathbf{0}$ is

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{aligned} x &= x_0 + tu, \\ y &= y_0 + tv, \end{aligned} \qquad t \in \mathbb{R}.$$

Meaning of the parameter. At $t = 0$ we are at P_0; positive and negative values move in the two opposite orientations along the same direction. Changing the length of $\mathbf{v}$ only changes how quickly the parameter runs through the points, not the geometric line itself.



3.1.1 The same line described by a perpendicular direction

A direction vector tells us how to move along the line. There is a second, equally important description: choose a nonzero vector $\mathbf{n} = (a, b)$ perpendicular to $\mathbf{v}$. A point $P = (x, y)$ lies on the line precisely when the displacement $P - P_0$ is perpendicular to $\mathbf{n}$.

From direction to normal form. The dot-product test from the previous chapter gives

$$\mathbf{n} \cdot (P - P_0) = 0.$$

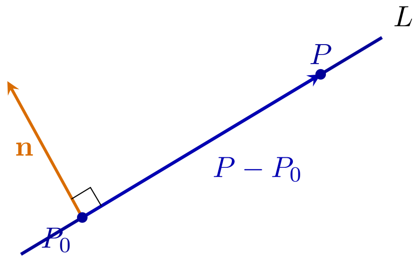
Written in coordinates,

$$a(x - x_0) + b(y - y_0) = 0.$$

After collecting the constant terms,

$$ax + by = c, \qquad c = ax_0 + by_0.$$

Interpretation. The general linear equation is not a second, unrelated definition of a line. It records the same object by specifying a direction that no displacement along the line may contain.



3.1.2 Slope form as a special coordinate description

Eliminating the parameter. If the horizontal component u of the direction vector is nonzero, eliminate t from

$$x = x_0 + tu, \qquad y = y_0 + tv.$$

Since

$$t = \frac{x - x_0}{u},$$

we obtain

$$y - y_0 = \frac{v}{u}(x - x_0).$$

Therefore the slope is

$$m = \frac{v}{u},$$

and after collecting constants the equation takes the familiar form

$$y = mx + b.$$

Remark. If $u = 0$, the line is vertical and is described by $x = x_0$. The vector and normal descriptions treat this case without any exception; only slope notation fails.

Summary. The three forms answer different questions about the same set:

$$\begin{array}{c|c} P_0 + t\mathbf{v} & \text{how to move along the line,} \\ \mathbf{n} \cdot (P - P_0) = 0 & \text{which displacements are allowed,} \\ y = mx + b & \text{how vertical position depends on horizontal position.} \end{array}$$

No representation is the line itself. The line is the geometric set of points that all these descriptions identify.

3.1.3 What the Familiar Cartesian Forms Let Us Read

Choose the representation adapted to the question. The parametric, normal, and slope forms are not competing definitions. They expose different information about the same geometric line, so the useful form depends on what must be calculated or interpreted.

Intersections with the coordinate axes

Impose both defining conditions. For a nonvertical line

$$y = mx + b,$$

setting $x = 0$ imposes the condition for the y-axis and gives $(0, b)$. If $m \neq 0$, setting $y = 0$ imposes the condition for the x-axis and gives

$$x = -\frac{b}{m}.$$

> For $y = mx + b$,
> $$L \cap \{x = 0\} = \{(0, b)\},$$
> and, when $m \neq 0$,
> $$L \cap \{y = 0\} = \left\{\left(-\frac{b}{m}, 0\right)\right\}.$$

General intersection principle. To find an intersection of two sets, impose both defining conditions at once. Axis intercepts are the simplest example of this general method.

Example. For

$$y = -\frac{1}{2}x + 3,$$

the y-axis intersection is $(0, 3)$. Setting $y = 0$ gives $x = 6$, so the x-axis intersection is $(6, 0)$.

The line through two points

Extracting a direction from two points. Let

$$P = (x_1, y_1), \qquad Q = (x_2, y_2), \qquad P \neq Q.$$

The displacement

$$Q - P = (x_2 - x_1, y_2 - y_1)$$

is nonzero and therefore provides a direction vector of the unique line through the points.

> The unique line through distinct points P and Q is
> $$L = \{P + t(Q - P) : t \in \mathbb{R}\}.$$
> This formula handles vertical and nonvertical lines uniformly.

Recovering slope notation. If $x_1 \neq x_2$, the horizontal component of $Q - P$ is nonzero, so

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

and

$$y - y_1 = m(x - x_1).$$

If $x_1 = x_2$, the line is vertical and is described by

$$x = x_1.$$

Remark. The exceptional vertical case belongs only to slope notation. The geometric line and its parametric description require no exception.

Parallel lines in slope notation

Why equal slopes mean equal directions. The nonvertical lines

$$y = m_1 x + b_1, \qquad y = m_2 x + b_2$$

have direction vectors $(1, m_1)$ and $(1, m_2)$. These directions are parallel exactly when $m_1 = m_2$. $\square$

> For distinct nonvertical lines,
>
> $$L_1 \parallel L_2 \iff m_1 = m_2 \text{ and } b_1 \neq b_2.$$
>
> If also $b_1 = b_2$, the two equations describe the same line.

Remark. Slope formulas are efficient coordinate consequences of the vector construction. They should be used when they simplify a question, but not at the cost of hiding the more general descriptions that handle vertical lines and extend directly to space.

3.1.4 Vector Tests Behind the Slope Rules

> Let two lines have nonzero direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Their directions are parallel precisely when
>
> $$\mathbf{v}_1 \parallel \mathbf{v}_2 \iff \mathbf{v}_2 = \lambda \mathbf{v}_1 \text{ for some } \lambda \neq 0.$$
>
> They are perpendicular precisely when
>
> $$\mathbf{v}_1 \perp \mathbf{v}_2 \iff \mathbf{v}_1 \cdot \mathbf{v}_2 = 0.$$

Why vector tests are more general. These criteria do not require a slope and therefore include vertical directions without any special convention. In three dimensions, parallel directions will also be detected by a zero cross product.

Recovering the familiar slope rule. For nonvertical lines with slopes m_1, m_2, choose direction vectors

$$\mathbf{v}_1 = (1, m_1), \qquad \mathbf{v}_2 = (1, m_2).$$

Then

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 1 + m_1 m_2.$$

Therefore

$$\mathbf{v}_1 \perp \mathbf{v}_2 \iff 1 + m_1 m_2 = 0.$$

> For two nonvertical lines,
>
> $$L_1 \perp L_2 \iff m_1 m_2 = -1.$$

This is the slope form of the dot-product criterion, not an independent rule.

3.1.5 The Nearest Point on a Line

Let

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}, \qquad \mathbf{v} \neq \mathbf{0},$$

and let P be any point.

Projecting onto the allowed direction. Decompose the displacement $P - P_0$ into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$. The parallel component is

$$\mathrm{proj}_{\mathbf{v}}(P - P_0) = \frac{(P - P_0) \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

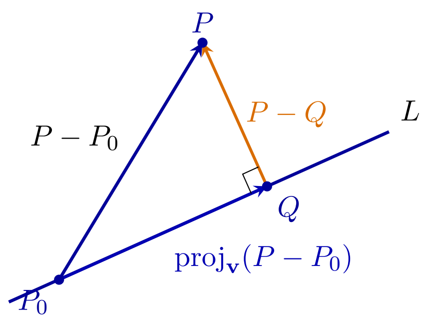
Starting from P_0 and moving by this component reaches

$$Q = P_0 + \mathrm{proj}_{\mathbf{v}}(P - P_0).$$

The nearest point on L to P is

$$Q = P_0 + \mathrm{proj}_{\mathbf{v}}(P - P_0).$$

The remainder $P - Q$ is perpendicular to the line.



The distance from P to the line L is

$$d(P, L) = \|(P - P_0) - \mathrm{proj}_{\mathbf{v}}(P - P_0)\|.$$

Why the perpendicular is shortest. Every other route from P to a point of the line contains the same perpendicular component together with an additional component along the line. The orthogonal component alone is therefore the shortest possible displacement.

3.1.6 Distance from the Normal Equation

Suppose the line is written as

$$ax + by = c, \qquad (a, b) \neq (0, 0),$$

with normal vector $\mathbf{n} = (a, b)$.

Reading the normal component. For $P = (x_P, y_P)$ and any fixed P_0 on the line,

$$\mathbf{n} \cdot (P - P_0) = ax_P + by_P - c.$$

The signed scalar component in the normal direction is

$$\frac{ax_P + by_P - c}{\sqrt{a^2 + b^2}}.$$

Distance is its absolute value.

For the line $ax + by = c$ and the point $P = (x_P, y_P)$,

$$d(P, L) = \frac{|ax_P + by_P - c|}{\sqrt{a^2 + b^2}}.$$

The formula is a projection. The numerator measures signed displacement in the normal direction, and the denominator normalises the normal vector. The formula is therefore the length of a projection, not a detached rule to memorise.

3.2 Planes in Three-Dimensional Space

A line was produced by allowing one real parameter:

$$P = P_0 + t\mathbf{v}.$$

Only one independent direction of motion was available.

The geometric question. What set is obtained when motion from one fixed point is allowed in two independent directions?

Let $P_0 \in \mathbb{R}^3$ be fixed and let $\mathbf{u}, \mathbf{v}$ be nonparallel vectors.

Generating the plane. Every permitted displacement is a linear combination

$$s\mathbf{u} + t\mathbf{v}.$$

Starting at P_0 gives

$$P = P_0 + s\mathbf{u} + t\mathbf{v}.$$

One parameter produces a line; two independent parameters produce a plane. If the two directions were parallel, both parameters would still move only along one line.

The plane through P_0 generated by nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$ is

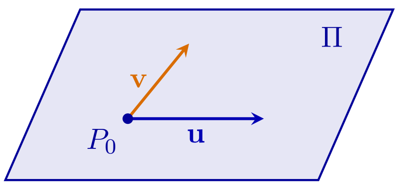
$$\Pi = \{P_0 + s\mathbf{u} + t\mathbf{v} : s, t \in \mathbb{R}\}.$$

If

$$P_0 = (x_0, y_0, z_0), \quad \mathbf{u} = (u_1, u_2, u_3), \quad \mathbf{v} = (v_1, v_2, v_3),$$

then

$$\begin{cases} x = x_0 + su_1 + tv_1, \\ y = y_0 + su_2 + tv_2, \\ z = z_0 + su_3 + tv_3. \end{cases}$$



3.2.1 A Normal Vector from Two Directions

The plane is generated by two directions. To write one equation for it, we need a vector perpendicular to both of them.

For

$$\mathbf{u} = (u_1, u_2, u_3), \qquad \mathbf{v} = (v_1, v_2, v_3),$$

their *cross product* is

$$\mathbf{u} \times \mathbf{v} = \big(u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1\big).$$

Direct substitution shows that

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0, \qquad (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0.$$

Therefore $\mathbf{u} \times \mathbf{v}$ is perpendicular to both generating directions and may be used as a normal vector of the plane. Reversing the order only reverses the normal:

$$\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v}).$$

Both choices describe the same plane.

From the normal direction to an equation. Let

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}.$$

A point P belongs to the plane exactly when $P - P_0$ is an allowed combination. Hence

$$\mathbf{n} \cdot (P - P_0) = 0.$$

For $\mathbf{n} = (a, b, c)$ this becomes

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

or, after collecting constants,

$$ax + by + cz = d.$$

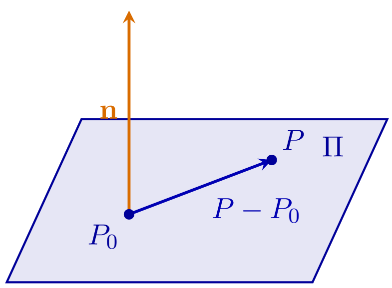
The parametric and normal descriptions of the same plane are

$$P = P_0 + s\mathbf{u} + t\mathbf{v}$$

and

$$\mathbf{n} \cdot (P - P_0) = 0 \quad \Longleftrightarrow \quad ax + by + cz = d.$$

Two descriptions, one plane. The parametric equation lists all permitted motions in the plane. The normal equation states that the forbidden normal component of every in-plane displacement must vanish.



3.2.2 A Plane Through Three Points

Extracting two independent directions. Let P, Q, R be three noncollinear points. Then

$$\mathbf{u} = Q - P, \qquad \mathbf{v} = R - P$$

are nonparallel and provide the two directions required to generate a plane. Their cross product

$$\mathbf{n} = (Q - P) \times (R - P)$$

is nonzero and perpendicular to the plane.

> The unique plane through three noncollinear points P, Q, R is
>
> $$\Pi = \{P + s(Q - P) + t(R - P) : s, t \in \mathbb{R}\}.$$
>
> A normal vector is
> $$\mathbf{n} = (Q - P) \times (R - P).$$

Example. For

$$P = (1, 0, 0), \qquad Q = (0, 1, 0), \qquad R = (0, 0, 1),$$

we obtain

$$Q - P = (-1, 1, 0), \qquad R - P = (-1, 0, 1),$$

and

$$(Q - P) \times (R - P) = (1, 1, 1).$$

Using the point P gives

$$(1, 1, 1) \cdot \big((x, y, z) - (1, 0, 0)\big) = 0,$$

so

$$x + y + z = 1.$$

3.2.3 Relative Positions of Planes

Let

$$\Pi_1 : \mathbf{n}_1 \cdot P = d_1, \qquad \Pi_2 : \mathbf{n}_2 \cdot P = d_2.$$

Normals control relative position. If the normals are parallel, the planes have the same orientation and are either identical or distinct and parallel. If the normals are not parallel, the planes intersect in a line.

> The angle between planes is measured through their normals. In particular,
>
> $$\Pi_1 \perp \Pi_2 \iff \mathbf{n}_1 \cdot \mathbf{n}_2 = 0.$$
>
> If the planes are nonparallel, a direction vector of their intersection line is
>
> $$\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2.$$

Finding the intersection line. The cross product gives the direction because the line must be perpendicular to both normals. A point on the line is then found by solving the two plane equations simultaneously.

3.2.4 Distance from a Point to a Plane

Distance as a normal projection. Let P_0 be a fixed point on the plane. The shortest displacement from P to the plane follows the normal direction. Hence the distance is the magnitude of the scalar component of $P - P_0$ along $\mathbf{n}$:

$$d(P, \Pi) = \frac{|\mathbf{n} \cdot (P - P_0)|}{\|\mathbf{n}\|}.$$

> For a plane $ax + by + cz = d$ and a point $P = (x_P, y_P, z_P)$,
>
> $$d(P, \Pi) = \frac{|ax_P + by_P + cz_P - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

What the formula measures. The numerator is the signed normal displacement encoded by the plane equation; the denominator converts the chosen normal vector to unit length. The formula is an orthogonal projection, not an isolated computational rule.

Line and plane from permitted directions. A line is generated by one independent direction,

$$P = P_0 + t\mathbf{v},$$

whereas a plane is generated by two,

$$P = P_0 + s\mathbf{u} + t\mathbf{v}.$$

The cross product of the two generating directions produces a normal, and the dot product turns perpendicularity into the equation

$$\mathbf{n} \cdot (P - P_0) = 0.$$

The parametric and normal forms therefore arise from the same geometric construction.

3.3 Review Exercises

3.3.1 Elementary computational and graphical exercises

1. Write a parametric equation of the line through $P = (2, -1)$ with direction $v = (3, 4)$. Find the points corresponding to $t = -1, 0, 2$ and plot them.

2. Find an implicit equation of the line through $P = (1, 3)$ with normal vector $n = (2, -1)$. Check that P satisfies the equation.

3. Convert

   $$2x - 3y = 6$$

   to slope-intercept form, find both axis intercepts, and give one direction vector and one normal vector.

4. Find the line through $A = (-1, 2)$ and $B = (3, -4)$ in parametric, implicit, and slope form.

5. Find and plot the intersection of

   $$L_1 : y = 2x - 1, \qquad L_2 : x + y = 5.$$

   Verify the point in both equations.

6. Determine whether each pair is parallel, perpendicular, identical, or neither:
   (a) $2x - y = 1$ and $4x - 2y = 7$;
   (b) $x + 2y = 3$ and $2x - y = 4$;
   (c) $3x + y = 5$ and $6x + 2y = 10$.
7. Compute the distance from $P = (3, 1)$ to the line $2x - y - 4 = 0$. Find the foot of the perpendicular and verify the distance from coordinates.
8. Find the line through $P = (1, -2)$ perpendicular to $3x + 2y = 5$. Find their intersection.
9. Find the line through $P = (-1, 3)$ parallel to $2x-y = 4$. Write it in implicit and parametric form.
10. A line has direction vector $(2, -3)$ and passes through $(4, 1)$. Find its intersections with both coordinate axes and draw the line.

3.3.2 Development and reasoning exercises

1. Write a parametric equation of the plane through $P = (1, 2, 0)$ with directions $u = (1, 0, 1)$ and $v = (0, 2, -1)$. Produce four points of the plane and explain why the nonparallel directions generate a two-dimensional set.
2. For the same plane, compute a normal vector using $u \times v$ and derive an implicit equation. Verify that the points found previously satisfy it.
3. Find the plane through
$$P = (1, 0, 0), \qquad Q = (0, 1, 0), \qquad R = (0, 0, 1).$$
Give both parametric and normal forms and explain how the three-point problem reduces to two direction vectors.
4. Determine the relative position of
$$\Pi_1 : x + y + z = 1, \qquad \Pi_2 : x - y + z = 3.$$
Find a direction vector of their intersection line and one point on it, then write the line parametrically.
5. Compare the line formula
$$P = P_0 + tv$$
with the plane formula
$$P = P_0 + su + tv.$$
Use one concrete line and one concrete plane to show that the number of independent directions determines the geometric dimension.
6. Explain why a normal equation describes a line in the plane and a plane in space by the same idea: the displacement from a fixed point has zero normal component. Give both formulas side by side.
7. A student converts every line immediately to slope form. Give two problems in which parametric or normal form is more natural, solve them briefly, and explain what slope form hides.
8. Give a complete strategy for choosing a representation of a line or plane. Your answer must discuss constructing the object, testing membership, finding intersections, checking parallelism or perpendicularity, and measuring distance.

Chapter 4

Conic Sections Through Cartesian and Polar Eyes

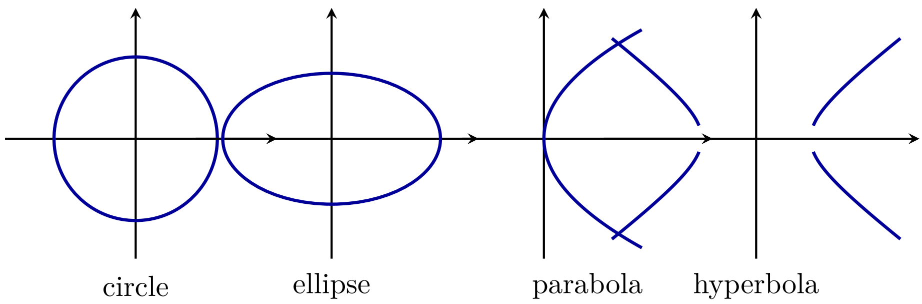
4.1 The Classical Conics Before Polar Coordinates

4.1.1 Why curves of second degree appear

The previous chapter used first-degree equations to describe lines and planes. A new phenomenon appears when the defining condition contains Euclidean distance: squares enter automatically.

The question of this chapter. Can one choose coordinates so that the geometry of circles, ellipses, parabolas, and hyperbolas becomes visible rather than hidden inside algebra?

We begin with the familiar Cartesian pictures. They provide a reference point, not the final language. The real goal is to place the origin at a focus and ask for the distance to the curve in each direction.



A map of the argument. First we identify the four shapes in Cartesian coordinates. Then we construct polar coordinates. Finally one focus–directrix condition produces one polar formula whose parameter decides which conic appears.

4.1.2 Circle and ellipse: from one scale to two

A circle centred at the origin is described by

$$\boxed{x^2 + y^2 = R^2}.$$

Its equation treats every direction equally. The parameterisation

$$\boxed{x = R\cos t, \qquad y = R\sin t}$$

makes this symmetry visible: one angle determines one point on the circle.

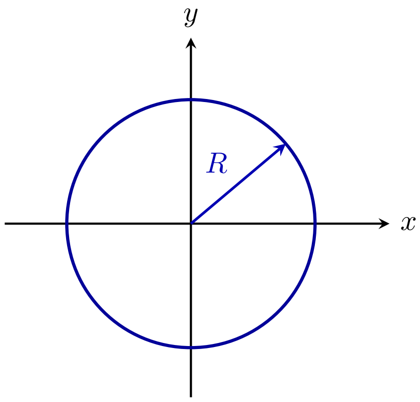
An ellipse introduces two perpendicular scales:

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}, \qquad a \geq b > 0.$$

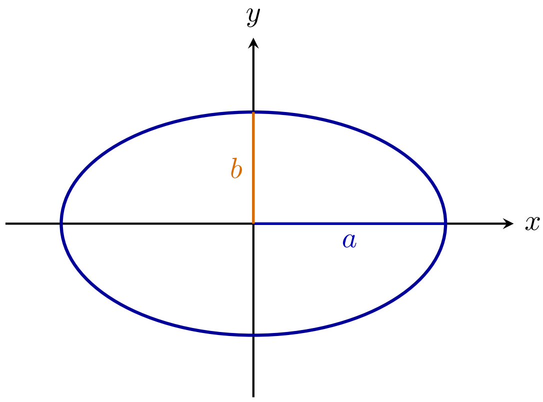
It is parameterised by

$$\boxed{x = a\cos t, \qquad y = b\sin t}.$$

The same circular parameter is stretched by different factors in the two coordinate directions.



$x^2 + y^2 = R^2$



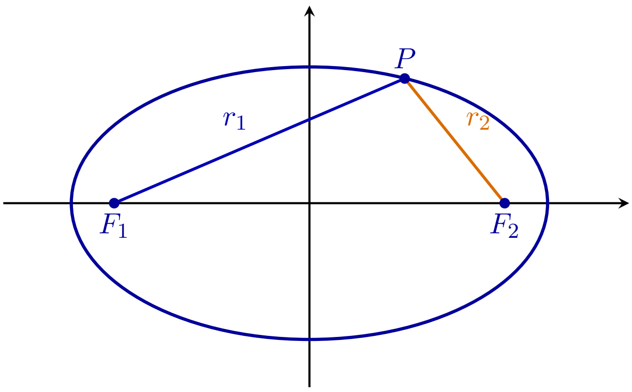
$x^2/a^2 + y^2/b^2 = 1$

The ellipse also has a second geometric description. Define

$$c = \sqrt{a^2 - b^2}.$$

Its foci are $F_1 = (-c, 0)$ and $F_2 = (c, 0)$, and every point P on the ellipse satisfies

$$\boxed{d(P, F_1) + d(P, F_2) = 2a}.$$



$r_1 + r_2 = 2a$

Two useful descriptions. The centre-based equation exposes symmetry and semiaxes. The focal description exposes distances from the points that later become natural locations of a central force source.

4.1.3 Parabola: equal distance from a focus and a line

Fix a focus $F = (p, 0)$ and a directrix $D : x = -p$, where $p > 0$. The parabola is the set of points $P = (x, y)$ satisfying

$$\boxed{d(P, F) = d(P, D)}.$$

Since

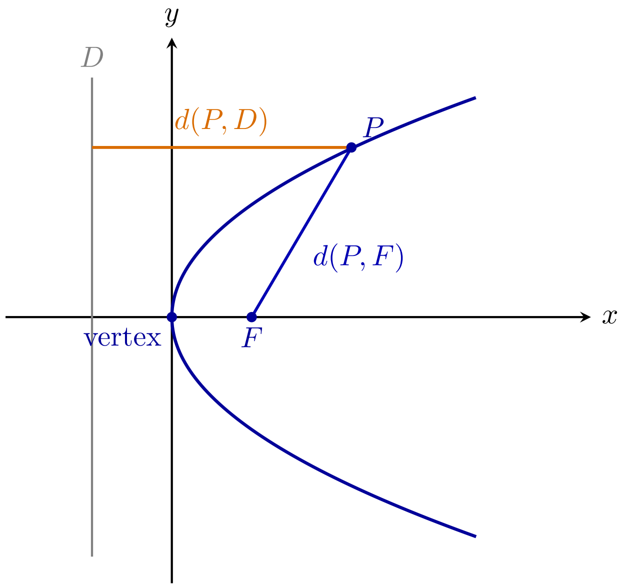
$$d(P, F) = \sqrt{(x-p)^2 + y^2}, \qquad d(P, D) = x + p,$$

squaring the equality gives

$$(x-p)^2 + y^2 = (x+p)^2,$$

and therefore

$$\boxed{y^2 = 4px}.$$



A convenient parameterisation is

$$\boxed{x = pt^2, \qquad y = 2pt}.$$

Substitution gives $y^2 = 4p^2t^2 = 4px$, so every parameter value produces a point on the curve.

Why the parabola is the boundary case. The parabola is neither closed like an ellipse nor split into two branches like a hyperbola. It will later appear exactly at the transition between bounded and unbounded focal motion.

4.1.4 Hyperbola: two complete branches and two asymptotes

For $a > 0$ and $b > 0$, the standard hyperbola is

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}.$$

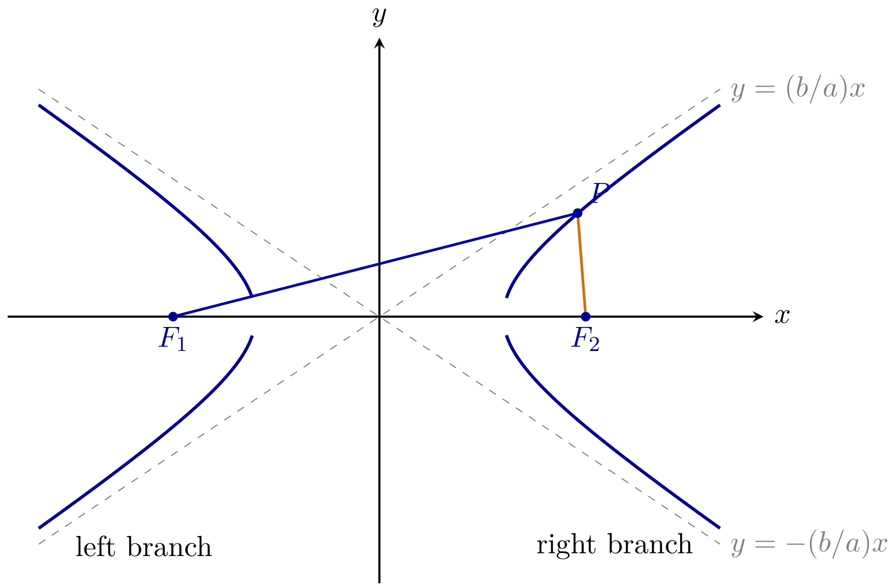
Define

$$c = \sqrt{a^2 + b^2}.$$

The foci are $F_1 = (-c, 0)$ and $F_2 = (c, 0)$, and the geometric condition is

$$\boxed{|d(P, F_1) - d(P, F_2)| = 2a}.$$

The absolute value is essential: one sign describes the right branch and the opposite sign describes the left branch.



The asymptotes

$$\boxed{y = \pm\frac{b}{a}x}$$

are not parts of the hyperbola. They record the directions approached by both branches far from the centre.

Do not confuse a branch with the whole hyperbola. A focus-centred polar equation will naturally describe the branch visible from that chosen focus and directrix. The Cartesian equation above remains the cleanest single picture of the complete two-branch curve. Later we will state explicitly which branch a polar formula traces and how the second branch is recovered.

4.1.5 Four familiar shapes, one geometric problem

The four curves differ in their Cartesian equations, but the decisive geometry is already visible:

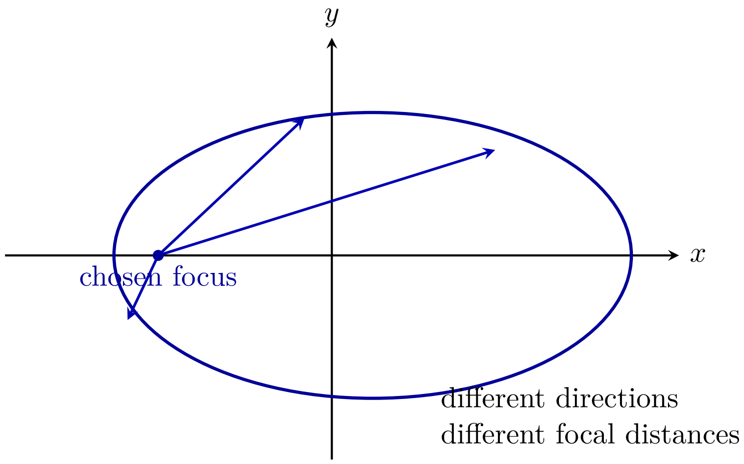
curve	distinguished data	large-scale behaviour
circle	one centre and one radius	closed
ellipse	two foci, fixed sum	closed
parabola	one focus and one directrix	one open branch
hyperbola	two foci, fixed difference	two open branches

The centre is the natural origin for a circle. It is also convenient for a Cartesian ellipse or hyperbola. But it is not the natural point from which to study focal distance. For that purpose one should move the origin to a focus.

The need for a new coordinate system. From a focus, a point on a conic is determined by two questions:

1. In which direction do we look?
2. How far must we travel in that direction before reaching the curve?

These are exactly the questions answered by polar coordinates.



The next section constructs this coordinate language before using it on conics.

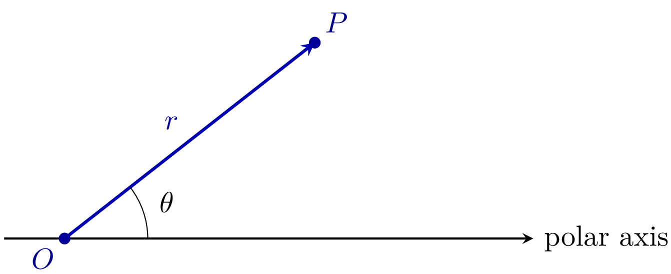
4.2 Polar Coordinates as Coordinates Adapted to a Focus

4.2.1 A point described by direction and distance

Choose a point O, called the pole, and a directed ray from O, called the polar axis. A point P is described by

$$\boxed{(r, \theta)},$$

where r is the distance from O to P and θ is the directed angle from the polar axis to the ray OP.



For this chapter we use $r \geq 0$ and $0 \leq \theta < 2\pi$. The pole is represented by $r = 0$; its angle is geometrically irrelevant.

Why this is adapted to a focus. If the pole is placed at a focus, then the first coordinate r is already the focal distance. No square root is needed to recover it from x and y.

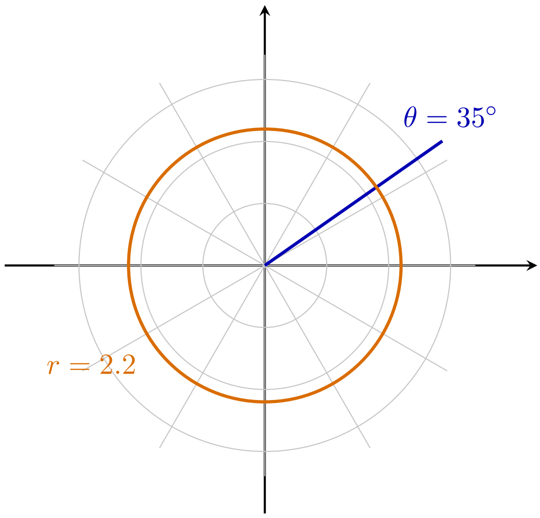
The coordinate curves also change character. In Cartesian coordinates, $x =$ constant and $y =$ constant are lines. In polar coordinates,

$$\theta = \text{constant}$$

is a ray, while

$$r = \text{constant}$$

is a circle centred at the pole.



4.2.2 Converting between Cartesian and polar descriptions

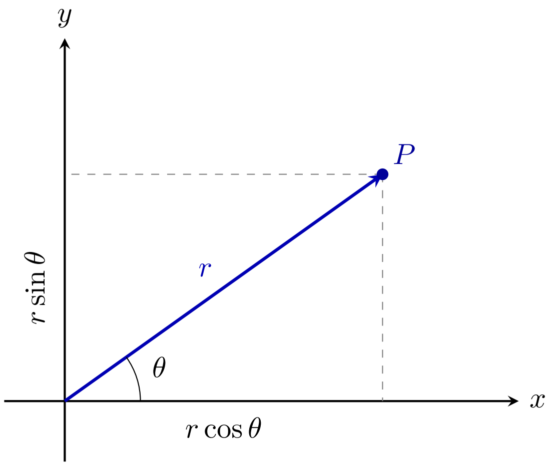
The right triangle formed by the ray OP gives

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta}.$$

Conversely,

$$\boxed{r = \sqrt{x^2 + y^2}}.$$

The angle is determined by the direction of (x, y); in computation this is the role of the two-argument function $\operatorname{atan2}(y, x)$.



Example. For $(r, \theta) = (4, \pi/3)$,

$$x = 4\cos\frac{\pi}{3} = 2, \qquad y = 4\sin\frac{\pi}{3} = 2\sqrt{3}.$$

Thus the same point has Cartesian coordinates $(2, 2\sqrt{3})$.

Coordinates are descriptions, not objects. Changing from (x, y) to (r, θ) does not move the point. It changes which questions are easy to answer. Cartesian coordinates expose horizontal and vertical displacement; polar coordinates expose direction and distance from a chosen point.

4.2.3 Reading a polar equation as a radial instruction

A polar equation

$$r = f(\theta)$$

should be read operationally: choose a direction θ, then travel the prescribed distance r from the pole.

The simplest example is

$$\boxed{r = R},$$

which gives the circle of radius R centred at the pole.

A more revealing example is

$$\boxed{r = 2R\cos\theta}.$$

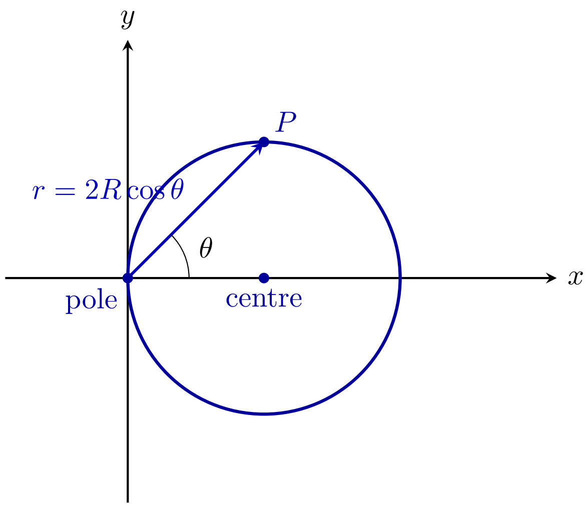
Using $x = r\cos\theta$ and $r^2 = x^2 + y^2$,

$$r^2 = 2Rr\cos\theta \quad\Longrightarrow\quad x^2 + y^2 = 2Rx,$$

so

$$\boxed{(x - R)^2 + y^2 = R^2}.$$

The same curve is therefore a circle centred at $(R, 0)$ and passing through the pole.



What the equation is doing. The factor $\cos\theta$ shortens the permitted radial distance as the ray turns away from the positive axis. At $\theta = \pi/2$ the distance becomes zero, so the curve returns to the pole. A Cartesian equation hides this directional story; the polar equation displays it directly.

This way of reading a formula prepares the conic equation. The next question is whether one radial law can generate ellipse, parabola, and hyperbola by changing only one parameter.

4.3 One Polar Formula for the Conic Family

4.3.1 One geometric rule and one shape parameter

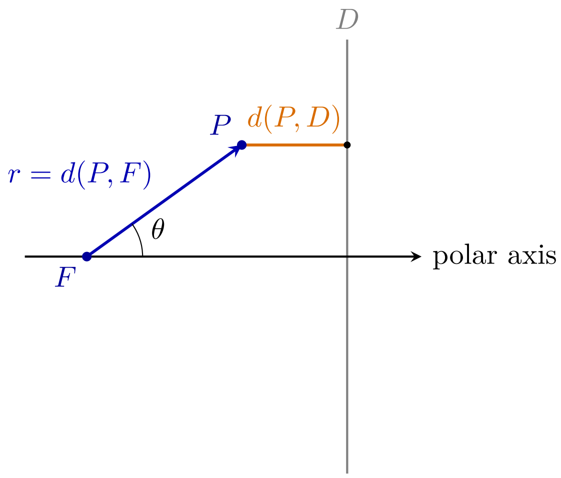
Fix a focus F and a directrix D. A point P belongs to the conic when

$$\boxed{d(P, F) = e\, d(P, D)},$$

where $e > 0$ is the eccentricity.

The value of e compares focal distance with distance from the directrix:

$$\begin{array}{c|c} 0 < e < 1 & \text{ellipse,} \\ e = 1 & \text{parabola,} \\ e > 1 & \text{hyperbola.} \end{array}$$



Why eccentricity measures shape. Small e keeps the focal distance relatively short compared with the distance to the directrix, producing a closed curve. At $e = 1$ the two distances balance. For $e > 1$ the focal distance may dominate, producing an open hyperbolic branch.

4.3.2 Deriving the common polar equation

Place the pole at the focus $F = (0, 0)$ and choose the directrix

$$D : x = d, \qquad d > 0.$$

For a point $P = (r, \theta)$,

$$x = r \cos \theta.$$

Its distance from the directrix is therefore

$$d(P, D) = d - r \cos \theta$$

for the branch lying to the left of the directrix.

The focus–directrix condition becomes

$$r = e(d - r \cos \theta).$$

Collecting the terms containing r gives

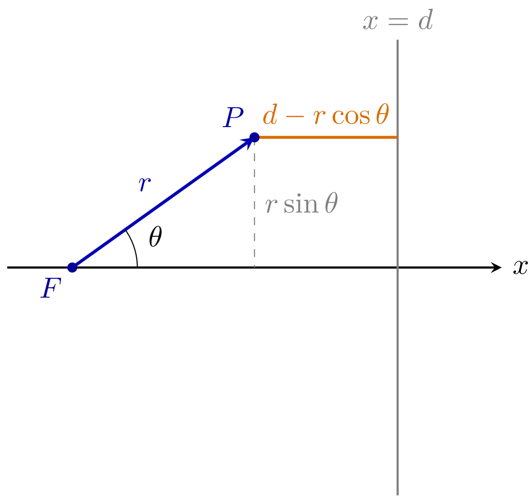
$$r(1 + e \cos \theta) = ed.$$

Define the focal parameter

$$\boxed{p = ed}.$$

Then

$$\boxed{r(\theta) = \frac{p}{1 + e \cos \theta}}.$$



The central idea. The formula is not an algebraic trick. The denominator records how the distance to the directrix changes when the viewing ray turns. Changing e changes the balance between the two distances and therefore changes the type of conic.

4.3.3 Ellipse from the polar formula

Assume

$$0 < e < 1.$$

Then

$$1 - e \leq 1 + e\cos\theta \leq 1 + e,$$

so the denominator is positive for every angle. The radius is finite in every direction and the curve closes around the focus.

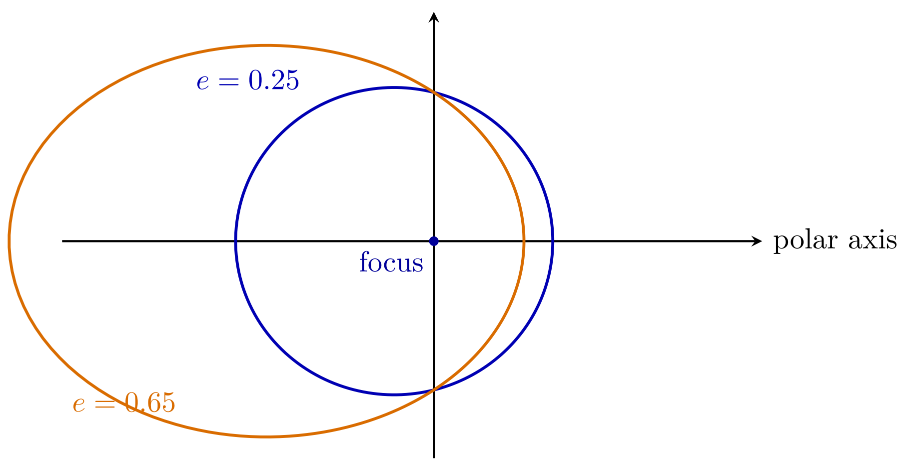
The nearest point occurs at $\theta = 0$:

$$\boxed{r_{\min} = \frac{p}{1+e}}.$$

The farthest point occurs at $\theta = \pi$:

$$\boxed{r_{\max} = \frac{p}{1-e}}.$$

Thus increasing e makes the two extreme focal distances more unequal.



For a Cartesian ellipse with semimajor axis a and semiminor axis b,

$$e = \sqrt{1 - \frac{b^2}{a^2}}, \qquad p = a(1 - e^2) = \frac{b^2}{a}.$$

Therefore the focus-centred form is

$$\boxed{r(\theta) = \frac{a(1 - e^2)}{1 + e\cos\theta}}.$$

Why the focus-centred form matters. The Cartesian equation displays the centre and semiaxes. The polar equation displays the distance from a focus as the direction changes. This second view is the one that later matches central-force motion.

4.3.4 Parabola from the polar formula

Set

$$e = 1.$$

Then

$$\boxed{r(\theta) = \frac{p}{1 + \cos\theta}}.$$

For directions near $\theta = \pi$, the denominator approaches zero and the radius grows without bound. This is the geometric reason the curve is open.

Using

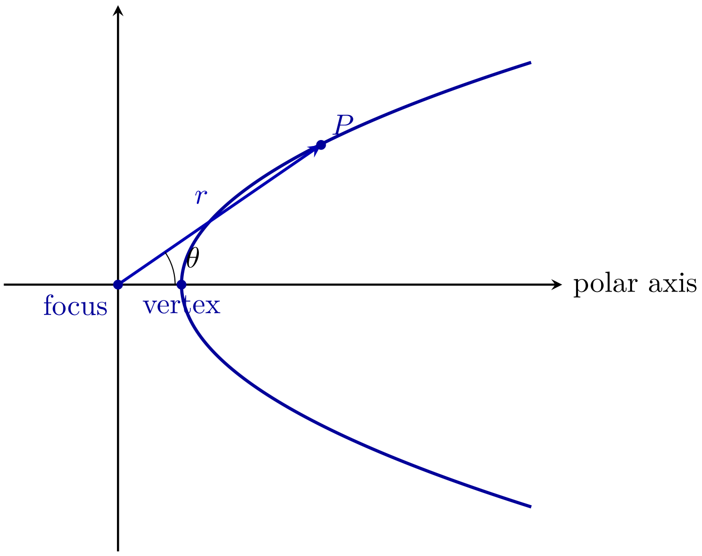
$$1 + \cos\theta = 2\cos^2\frac{\theta}{2},$$

we may also write

$$r = \frac{p}{2\cos^2(\theta/2)}.$$

The formula shows that the vertex lies on the positive polar axis at

$$\boxed{r = \frac{p}{2}}.$$



The exact transition. For an ellipse, the denominator stays away from zero. At $e = 1$ it reaches zero in one limiting direction. The closed orbit has opened into a parabola. Nothing new was inserted into the formula; only one parameter reached its boundary value.

4.3.5 Hyperbola from the polar formula

Assume

$$e > 1.$$

Then the denominator

$$1 + e\cos\theta$$

vanishes at two angles satisfying

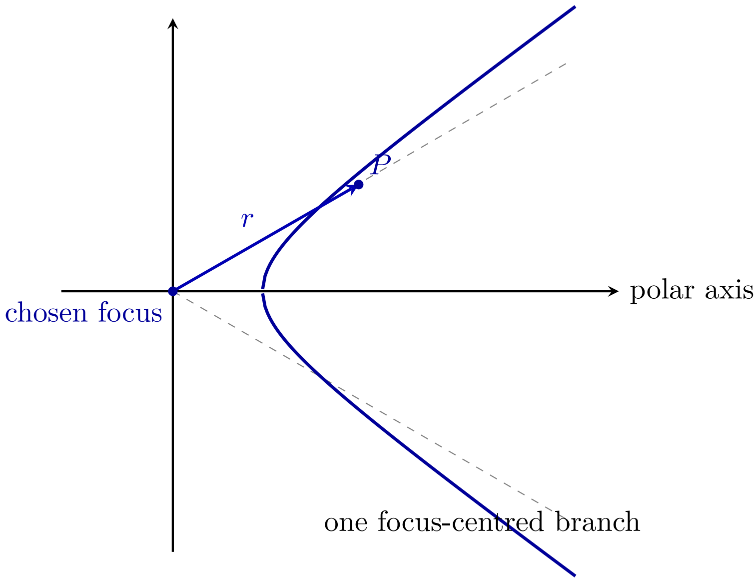
$$\cos\theta = -\frac{1}{e}.$$

These limiting directions are the asymptotic directions of the branch described from the chosen focus.

For $r \geq 0$, the formula

$$\boxed{r = \frac{p}{1 + e\cos\theta}}$$

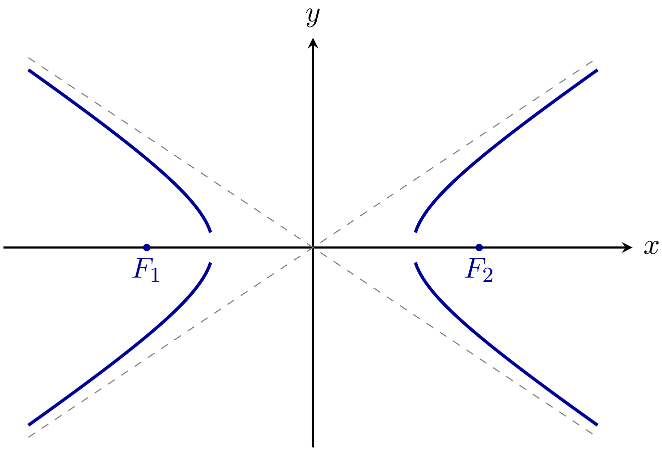
traces the branch lying on the same side of the directrix as the chosen focus. It does not silently represent the entire two-branch hyperbola.



To recover the complete hyperbola, return to the two-focus geometry. The second branch is obtained by choosing the other focus and the corresponding directrix, or by using the Cartesian equation

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1},$$

which displays both branches at once.



the complete hyperbola has two branches

Why the polar and Cartesian pictures differ. The polar equation is adapted to one chosen focus and one directrix. The Cartesian equation is adapted to the centre and displays both branches symmetrically. Neither description is incomplete when its geometric viewpoint is stated correctly.

4.3.6 Orientation, one formula, and the bridge to mechanics

The equation

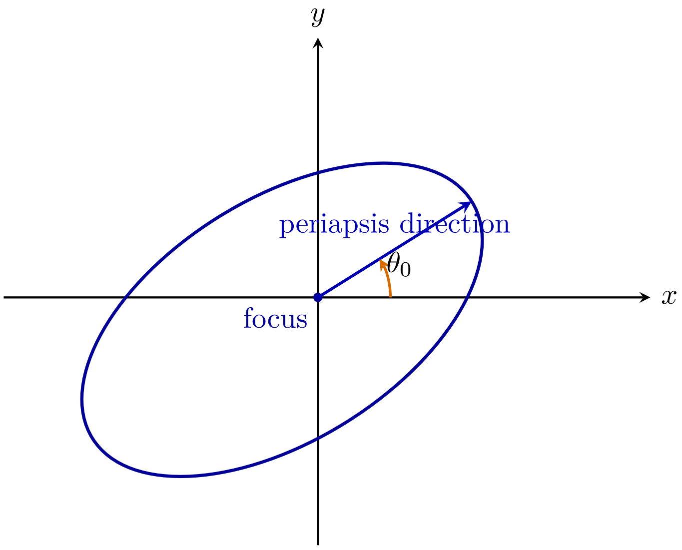
$$r = \frac{p}{1 + e\cos\theta}$$

places the nearest point on the positive polar axis. Rotating the entire conic by an angle θ_0 gives

$$\boxed{r = \frac{p}{1 + e\cos(\theta - \theta_0)}}.$$

The parameters have separate geometric roles:

$$\begin{array}{r|c} p & \text{overall focal scale,} \\ e & \text{shape and conic type,} \\ \theta_0 & \text{orientation in the plane.} \end{array}$$



The same formula now separates the three qualitative regimes:

e	conic	focal motion suggested by the geometry
$0 \leq e < 1$	circle or ellipse	bounded,
$e = 1$	parabola	threshold between bounded and unbounded,
$e > 1$	hyperbola	unbounded passage.

What has been achieved. The chapter did not merely list four equations. It changed the viewpoint until three apparently different curves became outcomes of one radial law. This is the analytic-geometric preparation needed before a later mechanics course derives the same law from a central force.

Choose coordinates adapted to the question. Cartesian coordinates are best for centres, axes, vertices, and complete symmetric pictures. Polar coordinates centred at a focus are best for radial distance, orientation, eccentricity, and orbital interpretation. Analytic geometry works by choosing the representation in which the geometry speaks most clearly.

4.4 Review Exercises

4.4.1 Computational and graphical exercises

1. For $x^2 + y^2 = 16$, write a trigonometric parameterisation and a polar equation centred at the circle centre. Draw all three descriptions on one diagram.

2. For
$$\frac{x^2}{25} + \frac{y^2}{9} = 1,$$
find a, b, c, the foci, and the eccentricity. Mark the semiaxes and both foci on a complete sketch.

3. A parabola has focus $(2, 0)$ and directrix $x = -2$. Derive its Cartesian equation, find its vertex, and draw the equal-distance construction for one point.

4. For
$$\frac{x^2}{16} - \frac{y^2}{9} = 1,$$
find the vertices, foci, eccentricity, and asymptotes. Draw both branches and label which sign of the focal-distance difference corresponds to each branch.

5. Convert $r = 6\cos\theta$ to Cartesian form. Identify the centre and radius, and explain why the curve passes through the pole.

6. For
$$r = \frac{6}{1 + \frac{1}{2}\cos\theta},$$
identify p and e, classify the conic, and compute its nearest and farthest focal distances.

7. Classify and sketch the focal conics
$$r = \frac{3}{1 + 0.4\cos\theta}, \qquad r = \frac{3}{1 + \cos\theta}, \qquad r = \frac{3}{1 + 1.5\cos\theta}.$$
For the hyperbolic case, state clearly which branch is traced when $r \geq 0$.

8. Sketch
$$r = \frac{4}{1 + 0.6\cos(\theta - \pi/4)}.$$
Mark the focus, periapsis direction, nearest point, and farthest point.

4.4.2 Development and reasoning exercises

1. Draw one point in both Cartesian and polar coordinates. Derive

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r = \sqrt{x^2 + y^2},$$

and explain why the polar description is better adapted to a distinguished focus.

2. Starting from

$$d(P, F) = e\, d(P, D),$$

place the pole at F and the directrix at $x = d$. Derive

$$r = \frac{p}{1 + e\cos\theta}, \qquad p = ed,$$

using a fully labelled geometric diagram.

3. Fix $p = 2$ and compare the ellipses obtained for $e = 0.2$, $e = 0.6$, and $e = 0.9$. Explain the changes using $r_{\min}$ and $r_{\max}$ rather than only visual appearance.

4. Explain directly from the denominator why $e = 1$ is the transition between a closed ellipse and an open hyperbola.

5. The polar formula with $e > 1$ and $r \geq 0$ describes one hyperbolic branch relative to one focus and directrix. Explain how the complete two-branch hyperbola is recovered and why the Cartesian equation displays it more naturally.

6. Compare

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

with

$$r = \frac{a(1 - e^2)}{1 + e\cos\theta}.$$

State which geometric information is immediate in each description.

7. Explain why the parameter t in

$$x = a\cos t, \qquad y = b\sin t$$

is not generally the same as the polar angle θ measured from a focus.

8. An orbit has nearest focal distance 4 and farthest focal distance 12. Find a, e, and p, then write its polar equation with periapsis at $\theta = 0$.

9. Explain how p, e, and θ_0 separately control scale, shape, and orientation in

$$r = \frac{p}{1 + e\cos(\theta - \theta_0)}.$$

10. Summarise the chapter as a change of viewpoint rather than a classification table: Cartesian geometry first, polar coordinates second, one focus–directrix law last. Explain why this sequence prepares a later derivation of orbits in mechanics.

Chapter 5

Matrices

5.1 Matrices as Structured Information

The first question is how to read the object. A matrix is not merely a rectangular table of numbers. It is a structured record whose rows, columns, and entries have roles determined by the problem. Before performing any operation, we must identify its shape and interpret what the positions represent.

5.1.1 What a Matrix Is

A matrix is a rectangular arrangement of mathematical objects, usually numbers. That description is correct, but incomplete. A matrix may organise data, record coefficients of equations, describe relations between quantities, or encode a rule that transforms one vector into another. The same array can therefore play very different roles in different problems.

> A matrix with m rows and n columns is called an $m \times n$ matrix. It has the form
>
> $$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$
>
> The entry a_{ij} lies in row i and column j.

How to read the indices. The first index identifies the row and the second identifies the column. Thus a_{ij} is found by moving first to row i and then to column j. This order will matter later when matrices are multiplied and when systems of equations are encoded.

Reading a matrix before calculating. Consider

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}.$$

The matrix has two rows and three columns, so it is a 2×3 matrix. Its entries include

$$m_{11} = 4, \qquad m_{12} = -2, \qquad m_{23} = 5.$$

No operation has been performed. We have only identified the shape and read the positions correctly.

> A matrix is square when it has the same number of rows and columns. An $n \times n$ matrix is called a square matrix of order n.

Shape is part of the mathematical meaning. A row matrix, a column matrix, a square matrix, and a rectangular matrix may contain similar entries but need not represent the same kind of object. A column matrix may represent a vector, a square matrix may represent a transformation from a space to itself, and a rectangular matrix may relate collections of different sizes.

Reading a matrix. Before calculating, determine the number of rows and columns, identify the role of the entries, and decide what the rows and columns represent. A matrix is not understood merely by seeing its numbers; it is understood by reading its structure.

5.2 Operations That Preserve Positions

Entrywise operations. Addition, subtraction, and scalar multiplication preserve the shape of a matrix. They operate on entries occupying the same positions. Their simplicity comes from this positional correspondence.

5.2.1 Addition, Subtraction, and Scalar Multiplication

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size $m \times n$. Their sum and difference are defined by

$$A + B = (a_{ij} + b_{ij}), \qquad A - B = (a_{ij} - b_{ij}).$$

For a scalar λ,

$$\lambda A = (\lambda a_{ij}).$$

Why the sizes must agree. Each entry of A must have a partner in the same position in B. If the shapes differ, there is no complete position-by-position correspondence, so addition and subtraction are not defined.

Three entrywise calculations. Let

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix},$$

$$A - B = \begin{pmatrix} 2-5 & -1-4 \\ 0-(-2) & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix},$$

and

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

For matrices of the same size,

$$A + B = B + A, \qquad (A + B) + C = A + (B + C).$$

More generally, the usual linear rules hold because every statement reduces to the corresponding statement for each individual entry.

Do not transfer this picture to multiplication. These operations are defined entry by entry. Matrix multiplication will combine rows with columns instead, so its definition and properties must be developed separately.

Position-by-position arithmetic. Entrywise operations preserve matrix size and respect fixed positions. Always check compatibility of shapes before calculating.

5.3 Matrix Multiplication

A new kind of operation. Matrix multiplication is not performed entry by entry. It combines each row of the first matrix with each column of the second. The compatibility of dimensions is therefore part of the operation itself.

5.3.1 The Row-by-Column Rule

> Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $n \times p$ matrix. Their product is the $m \times p$ matrix $C = AB$ whose entries are
>
> $$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$
>
> Thus c_{ij} is obtained from row i of A and column j of B.

Why the inner dimensions must match. A row of A contains n entries. It can be paired with a column of B only when that column also contains n entries. Therefore

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions test whether multiplication is possible; the outer dimensions determine the size of the result.

Computing a product. Let

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}.$$

The first entry of AB is

$$1 \cdot 2 + 3 \cdot 5 = 17,$$

while the first row and second column give

$$1 \cdot 0 + 3 \cdot (-1) = -3.$$

Continuing row by column,

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

What each product entry records. Every entry of AB measures one interaction between a row of A and a column of B. Matrix multiplication is therefore a structured combination of whole rows and columns, not a position-by-position product.

5.3.2 Reading a Matrix Through Basis Vectors

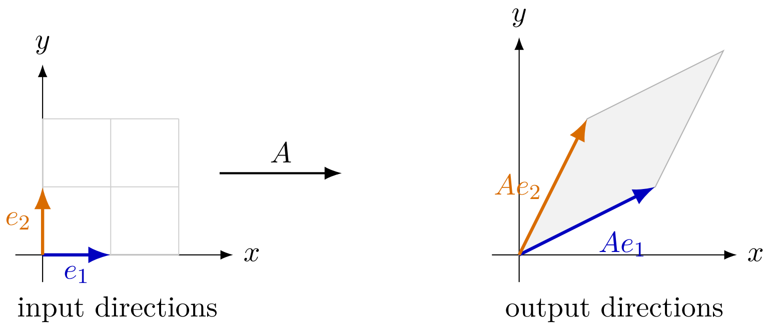
For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

the columns are

$$Ae_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad Ae_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Thus the matrix is determined by where it sends the two coordinate directions.



Columns describe the action. Every vector $x = x_1e_1 + x_2e_2$ is sent to

$$Ax = x_1Ae_1 + x_2Ae_2.$$

Matrix multiplication therefore combines the images of the basis directions with the input coordinates. The row–column calculation and the geometric action are two descriptions of the same operation.

5.3.3 Order, Noncommutativity, and Associativity

Changing the order changes the question. The product AB combines rows of A with columns of B. Reversing the order produces a different interaction, so matrix multiplication is not generally commutative.

The same matrices in the opposite order. For

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix},$$

we already found

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

In the opposite order,

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

Hence $AB \neq BA$.

Commutativity is a property to check. Some pairs of matrices do commute, but this is never automatic. Whenever the order matters in an argument, commutativity must be proved from the structure of the matrices involved.

Whenever all products are defined,

$$(AB)C = A(BC).$$

Associativity allows products in a fixed order to be grouped differently. It does not allow the matrices themselves to be rearranged.

Grouping versus reordering. Parentheses determine which multiplication is carried out first; the written order determines which transformations or interactions occur first. Associativity changes grouping only, while commutativity would change order.

5.4 Special Matrices and Visible Structure

Structure can explain an operation. Certain matrices are important because their pattern is visible before any calculation begins. Zeros, diagonal entries, symmetry, and repeated rows or columns can reveal how the matrix will behave.

5.4.1 The Zero Matrix and the Identity Matrix

The zero matrix is the matrix whose entries are all zero. For every matrix A of the same size,

$$A + 0 = A.$$

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

If $A \in \mathbb{R}^{m \times n}$, then

$$I_m A = A, \qquad A I_n = A.$$

The identity represents no change. The identity matrix plays the role of the number 1 in multiplication. The size must match the side on which it is used: the left identity has size $m \times m$, while the right identity has size $n \times n$.

Identity matrices are size-dependent. There is not one universal identity matrix for every product. The dimensions of the surrounding matrices determine which identity matrix is required.

5.4.2 Transpose and Symmetry

For $A = (a_{ij}) \in \mathbb{R}^{m \times n}$, the transpose is the $n \times m$ matrix

$$A^{\mathsf{T}} = (a_{ji}).$$

Rows of A become columns of A^{T}, and columns become rows.

Transposing a rectangular matrix.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^{\mathsf{T}} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

The original matrix has size 2×3, while its transpose has size 3×2.

A square matrix is symmetric when

$$A^{\mathsf{T}} = A.$$

Equivalently, its entries satisfy $a_{ij} = a_{ji}$, so the matrix is mirrored across the main diagonal.

What symmetry makes visible. In a symmetric matrix, information recorded in position (i, j) agrees with the information in position (j, i). The pattern can therefore be recognised before any calculation is performed.

5.4.3 Diagonal Matrices

A square matrix $D = (d_{ij})$ is diagonal when

$$d_{ij} = 0 \qquad \text{for every } i \neq j.$$

We write

$$D = \mathrm{diag}(d_1, d_2, \ldots, d_n).$$

No mixing between coordinates. A diagonal matrix acts independently on the coordinate positions. Since every off-diagonal entry is zero, one coordinate does not contribute to another. This is why products and powers of diagonal matrices are especially transparent.

Powers of a diagonal matrix. For

$$D = \mathrm{diag}(3, -1, 2),$$

we have

$$D^2 = \mathrm{diag}(9, 1, 4), \qquad D^3 = \mathrm{diag}(27, -1, 8).$$

Each diagonal entry is raised to the corresponding power.

If

$$D_1 = \mathrm{diag}(a_1, \ldots, a_n), \qquad D_2 = \mathrm{diag}(b_1, \ldots, b_n),$$

then

$$D_1 D_2 = D_2 D_1 = \mathrm{diag}(a_1 b_1, \ldots, a_n b_n).$$

The commutativity follows from ordinary multiplication of the diagonal entries.

5.4.4 Patterns Worth Noticing

When a matrix appears, inspect its structure before calculating. Useful questions include:

- Is the matrix diagonal or symmetric?
- Does it contain many zeros?
- Are two rows or columns equal?
- Is one row or column a scalar multiple of another?
- Does the matrix split naturally into simpler blocks?

Seeing structure is part of solving. A visible pattern can explain why an operation simplifies, why two matrices commute, or why some information is redundant. Recognising such a pattern is not a computational trick added after the mathematics; it is part of understanding the mathematical object.

Read before calculating. The zero, identity, transpose, symmetric, and diagonal patterns provide a first vocabulary for structural reasoning. Before applying a general algorithm, check whether the matrix already reveals a simpler route.

5.5 Row Operations and Problem Solving

Controlled transformations reveal structure. Elementary row operations change a matrix in reversible, precisely described ways. Their purpose is not cosmetic simplification but the exposure of relations that were difficult to see in the original arrangement.

5.5.1 Elementary Row Operations

The elementary row operations are:

- swapping two rows;
- multiplying one row by a nonzero scalar;
- adding a scalar multiple of one row to another row.

Why these operations are controlled. Each operation can be reversed: a swap is undone by the same swap, a nonzero scaling is undone by multiplying by the reciprocal, and adding a multiple of one row is undone by subtracting the same multiple. This reversibility is why row operations can transform a problem without losing its essential information.

Eliminating information already carried by another row. Consider

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}.$$

Replace the second row by the second row minus twice the first:

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5).$$

The transformed matrix is

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$

The first two entries of the old second row were already explained by twice the first row. The new row isolates the part that was not.

Three layers of a row operation. A row operation contains arithmetic, a transformation of the matrix, and an interpretation. The arithmetic produces the new row; the transformation changes the representation; the interpretation explains what relation has become visible.

5.5.2 Column Vectors as Matrices

A column vector may be viewed as a matrix with one column. This allows vector and matrix equations to be written in one language.

One vector, two orientations. For

$$u = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix},$$

we have a 3×1 matrix. Its transpose is the 1×3 row matrix

$$u^{\mathsf{T}} = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix}.$$

Orientation changes what products mean. The distinction between a row and a column is structural. The products

$$u^{\mathsf{T}}u \qquad \text{and} \qquad uu^{\mathsf{T}}$$

have different sizes: the first is 1×1, while the second is 3×3. Even when the same entries occur, their arrangement determines which operations are possible and what object is produced.

Read dimensions before multiplying. The notation may look compact, but dimensions carry the compatibility check. Always determine the shapes before applying the row-by-column rule.

5.5.3 How to Read Matrix Problems

Questions to ask before calculating. When working with matrices, determine:

- the size of every matrix;
- which operations are defined;
- whether the operation is entrywise or row-by-column;
- whether the order of multiplication matters;
- whether visible structure simplifies the problem;
- what the result says about the original objects.

A matrix solution is more than its last line. A complete solution should expose the compatibility checks, the chosen operation, the main calculation, and the meaning of the result. The purpose is not merely to produce an array of numbers, but to use the matrix as a mathematical description.

The matrix workflow. Read the shape, identify the operation, calculate according to its definition, and interpret the resulting structure. This workflow prepares the later use of matrices in determinants, inverses, and systems of equations.

Matrices as organised operations. A matrix has a shape, a positional structure, and a role in a problem. Entrywise operations preserve positions; multiplication combines rows with columns; special patterns reveal simplified behaviour; and row operations expose hidden relations. Row operations will return in the next three chapters with new meanings: tracking determinants, constructing inverses, and preserving solution sets.

5.6 Review Exercises

5.6.1 Elementary computational exercises

1. For
$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix},$$
state its size and identify a_{12}, a_{23}, the second row, and the third column.

2. Compute $A + B$, $A - B$, and $3A$ for
$$A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & -2 \\ 5 & 0 \end{pmatrix}.$$

3. Decide which sums are defined and compute those that are:
$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

4. Compute both AB and BA for
$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 4 & 2 \end{pmatrix}.$$

5. For
$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix},$$
compute Ax and display each row–column product.

6. Compute A^T and $(A^\mathsf{T})^\mathsf{T}$ for
$$A = \begin{pmatrix} 1 & -2 & 0 \\ 4 & 3 & 5 \end{pmatrix}.$$

7. For
$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix},$$
compute Dx and describe what happens to each coordinate.

8. Starting from
$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix},$$
perform $R_2 \leftarrow R_2 - 2R_1$ and then undo the operation.

9. Apply in order to
$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$$
the operations $R_2 \leftarrow R_2 - 3R_1$ and $R_1 \leftarrow R_1 - 2R_2$. Record every intermediate matrix.

10. Compute Ax for $x = (4, -1)^\mathsf{T}$ when
$$A = I, \qquad A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$
Identify the action visible in each result.

5.6.2 Development and reasoning exercises

1. Let
$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$
Compute Bx, then $A(Bx)$, then AB and $(AB)x$. Explain how the calculation demonstrates composition and associativity.

2. A linear action sends
$$e_1 \mapsto \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad e_2 \mapsto \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$
Construct its matrix and use it to find the image of $(4, -2)^\mathsf{T}$. Explain why the columns contain the images of the basis vectors.

3. The triangle has vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Apply
$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$
to all three vertices, plot the original and transformed triangles, and describe the visible geometric effect.

4. Compute AB and BA for

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

   Apply both products to e_1 and explain geometrically why changing the order changes the result.

5. Reduce

$$M = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix}$$

   with elementary row operations until the dependence between the rows is visible. State the relation you discover and explain what the arithmetic has exposed.

6. Write the elementary matrix that performs $R_2 \leftarrow R_2 - 3R_1$ on a 2×2 matrix. Find its inverse elementary matrix and verify that their product is the identity.

7. For matrices of sizes 2×3 and 3×4, write a symbolic row–column entry of the product and use it to justify why the result has size 2×4.

8. Explain why matrix addition is positionwise while multiplication combines rows and columns. Include a numerical example showing that entrywise multiplication would not represent composition.

9. Explain the three meanings of a row operation in this course: a calculation on entries, a reversible change of representation, and a tool that reveals a relation. Illustrate all three on one matrix.

10. Given a matrix expression, formulate a decision procedure: inspect dimensions, determine which operations are defined, preserve multiplication order, exploit visible structure, and verify the result on a test vector.

Chapter 6

Determinants

6.1 A Function on Square Matrices

6.1.1 Definition, Scope, and the Order of Investigation

One matrix, one number. For each size n, the determinant is a function

$$\det\colon \mathbb{R}^{n\times n} \longrightarrow \mathbb{R}.$$

It accepts a square matrix and returns one real number. The value is not attached to a rectangular matrix in this course.

The formula is not presented as inevitable. We first define determinants for small matrices, calculate examples, discover stable properties, and only then develop interpretations and general methods.

The order matters. The chapter follows

$$\text{definition} \longrightarrow \text{calculation} \longrightarrow \text{properties} \longrightarrow \text{interpretation}.$$

A definition tells us what object is being studied; subsequent calculations reveal why it is useful.

Square matrices only. Expressions such as

$$\det\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

are not defined because the matrix is not square.

Plan of the chapter. We will define small determinants, derive row and column properties, analyse the condition $\det A = 0$, extend the definition using minors and cofactors, and collect efficient calculation strategies.

6.2 Determinants of Small Matrices

6.2.1 The Determinant of a 1×1 Matrix

For a matrix containing one entry,

$$\det\begin{pmatrix} a \end{pmatrix} = a.$$

The determinant is simply the entry itself. This convention is required when larger determinants are reduced recursively to smaller ones.

6.2.2 The Determinant of a 2×2 Matrix

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is

$$\det A = ad - bc.$$

We also write

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

The products use the two diagonals, but the order is essential: the second product is subtracted from the first.

A nonzero determinant. For

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix},$$

we obtain

$$\det A = 3 \cdot 5 - 1 \cdot 2 = 13.$$

A zero determinant. For

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix},$$

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 0.$$

The cancellation will later be explained by the proportional rows.

Position matters. Exchanging the rows changes

$$\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = -2$$

into

$$\det \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix} = 2.$$

The same entries in different positions can give a different determinant.

6.2.3 The Determinant of a 3×3 Matrix

For

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

define

$$\det A = aei + bfg + cdh - ceg - bdi - afh.$$

Sarrus' rule is a mnemonic for this six-term formula: repeat the first two columns, add the three downward diagonal products, and subtract the three upward diagonal products.

Limited scope of Sarrus' rule. Sarrus' rule applies only to 3×3 determinants. It is not a method for size 4×4 or larger.

Direct six-term calculation. For

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix},$$

$$\begin{aligned}\det A &= (1\cdot 3\cdot 5 + 2\cdot 4\cdot 2 + 0\cdot(-1)\cdot 0)\\ &\quad - (0\cdot 3\cdot 2 + 2\cdot(-1)\cdot 5 + 1\cdot 4\cdot 0)\\ &= (15+16) - (0-10) = 41.\end{aligned}$$

Proportional rows. For
$$C = \begin{pmatrix} 1 & 2 & 0\\ 2 & 4 & 0\\ 3 & 1 & 5\end{pmatrix},$$
the formula gives $\det C = 0$. The first two rows are proportional; later this cancellation becomes a general theorem.

6.2.4 What the Formulas Are Meant to Teach

Two skills. At this stage the student must learn both to apply the formulas accurately and to compare examples for regular behaviour. The second skill is what turns isolated computations into mathematics.

The formulas are the starting point. In the next section numerical experiments suggest patterns; calculations with general entries explain why those patterns always hold.

From calculation to property. Apply the correct formula, keep all signs visible, and then ask what features of the rows or columns caused the resulting value.

6.3 Properties Discovered from the Formulas

6.3.1 Linearity in One Row

Experiment, general calculation, theorem. A numerical example can suggest a property, but only a calculation with general entries explains why it always holds.

Keep the second row fixed and consider two possible first rows (a, b) and (p, q). Then

Addition in one row.

$$\begin{aligned}\det\begin{pmatrix} a+p & b+q\\ c & d\end{pmatrix} &= (a+p)d - (b+q)c\\ &= (ad-bc) + (pd-qc)\\ &= \det\begin{pmatrix} a & b\\ c & d\end{pmatrix} + \det\begin{pmatrix} p & q\\ c & d\end{pmatrix}.\end{aligned}$$

Similarly, multiplying only the first row by k gives

Scaling one row.

$$\begin{aligned}\det\begin{pmatrix} ka & kb\\ c & d\end{pmatrix} &= (ka)d - (kb)c\\ &= k(ad-bc)\\ &= k\det\begin{pmatrix} a & b\\ c & d\end{pmatrix}.\end{aligned}$$

For the determinants already defined, addition in one selected row becomes addition of determinants, and multiplying that row by a scalar multiplies the determinant by the same scalar, provided all other rows remain fixed.

Not linear in the whole matrix. In general $\det(A + B) \neq \det A + \det B$. For example, with $A = B = I_2$, the left side is $\det(2I_2) = 4$, while the right side is 2.

The same mechanism appears in the 3×3 formula because every term contains exactly one entry from each row.

6.3.2 Exchanging Two Rows

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det A = ad - bc.$$

After exchanging the rows,

$$B = \begin{pmatrix} c & d \\ a & b \end{pmatrix}.$$

Why the sign changes.

$$\begin{aligned} \det B &= cb - da \\ &= bc - ad \\ &= -(ad - bc) \\ &= -\det A. \end{aligned}$$

□

Exchanging two rows changes the sign of the determinant.

Order is recorded. The determinant is sensitive not only to the entries but also to the order of the rows and columns.

6.3.3 Equal and Proportional Rows

If the two rows are equal, exchanging them leaves the matrix unchanged, but the row-exchange theorem changes the sign of the determinant. Hence

$$\det A = -\det A,$$

so $\det A = 0$.

Direct check for proportional rows. If

$$A = \begin{pmatrix} a & b \\ ka & kb \end{pmatrix},$$

then

$$\begin{aligned} \det A &= a(kb) - b(ka) \\ &= kab - kab \\ &= 0. \end{aligned}$$

The six-term formula produces the same cancellation for 3×3 matrices with proportional rows.

□

For the small determinants already defined, equal or proportional rows give determinant zero.

A relation removes freedom. When one row repeats or rescales another, the rows do not carry independent information. The determinant records this loss by vanishing.

6.3.4 Adding a Multiple of One Row to Another

Start with

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and replace the second row by

$$(c, d) + k(a, b) = (c + ka, d + kb).$$

The added terms cancel.

$$\begin{aligned} \det \begin{pmatrix} a & b \\ c + ka & d + kb \end{pmatrix} &= a(d + kb) - b(c + ka) \\ &= ad + kab - bc - kab \\ &= ad - bc \\ &= \det A. \end{aligned}$$

Adding a multiple of one row to another row does not change the determinant.

This can also be reconstructed from linearity: the added part creates a determinant with proportional rows, and that determinant is zero.

A determinant-preserving operation. For

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix},$$

replace R_2 by $R_2 - 2R_1 = (1, 1)$. Both the original and transformed matrices have determinant 1.

6.3.5 Rows, Columns, and the Collected Rules

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad A^{\mathsf{T}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix},$$

we have

$$\det(A^{\mathsf{T}}) = ad - cb = ad - bc = \det A.$$

For the determinants considered here,

$$\det(A^{\mathsf{T}}) = \det A.$$

Consequently, every corresponding row property also holds for columns.

Properties discovered from the formulas. The determinant is linear in one selected row, exchanging two rows changes the sign, equal or proportional rows give zero, and adding a multiple of one row to another leaves the determinant unchanged. The same statements hold for columns.

6.4 What a Zero Determinant Reveals

6.4.1 The 2×2 Case

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$\det A = 0 \iff ad = bc.$$

Zero determinant and proportional rows. Assume first that the first row is nonzero. If $a \neq 0$, then

$$d = \frac{bc}{a}, \qquad (c, d) = \frac{c}{a}(a, b).$$

If $a = 0$, then $b \neq 0$ and $ad = bc$ gives $c = 0$, so

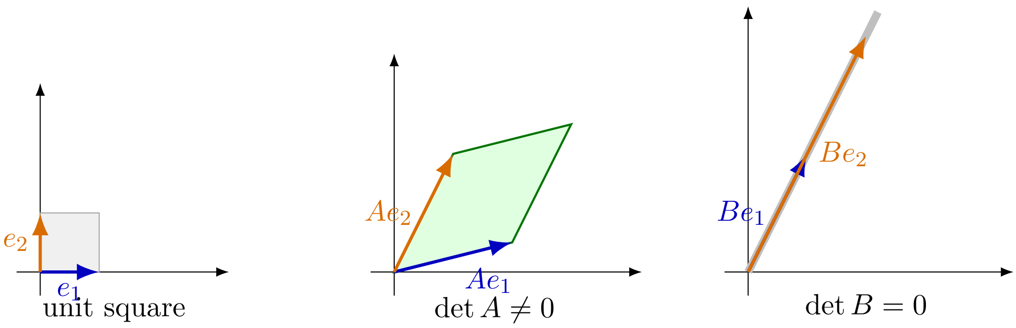
$$(c, d) = \frac{d}{b}(0, b).$$

If the first row is zero, the relation is immediate. $\square$

> For a 2×2 matrix,
> $$\det A = 0$$
> if and only if its rows are proportional. The same statement holds for its columns.

6.4.2 Area Scaling and Collapse in the Plane

For a 2×2 matrix, the absolute determinant measures how the area of the unit square changes. A nonzero determinant produces a genuine parallelogram; a zero determinant makes the two output directions parallel and the area collapses.



What zero means geometrically. When the column images are independent, they span a parallelogram of nonzero area. When one image is a multiple of the other, every combination remains on one line. The determinant is then zero because a two-dimensional set has been compressed into one dimension.

6.4.3 A First Restriction on Matrix Outputs

Suppose the second row is λ times the first:

$$A = \begin{pmatrix} a & b \\ \lambda a & \lambda b \end{pmatrix}.$$

For

$$x = \begin{pmatrix} u \\ v \end{pmatrix},$$

we obtain

$$Ax = \begin{pmatrix} au + bv \\ \lambda au + \lambda bv \end{pmatrix} = \begin{pmatrix} t \\ \lambda t \end{pmatrix}, \qquad t = au + bv.$$

The outputs satisfy a fixed relation. No matter which input is chosen, the second output coordinate is λ times the first. A zero determinant therefore signals that the possible outputs are restricted.

Example. For

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

$\det A = 0$, and every output has the form

$$Ax = \begin{pmatrix} t \\ 2t \end{pmatrix}.$$

The outputs lie in one fixed pattern rather than filling the whole plane.

A similar phenomenon occurs with three rows: if $r_3 = r_1 + r_2$, then the third entry of every output Ax is the sum of the first two.

6.4.4 Parameters and Exceptional Values

When a matrix contains a parameter, the equation

$$\det A(t) = 0$$

identifies values at which a relation among rows or columns appears.

A quadratic parameter condition. Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

Then

$$\det A(t) = t^2 - 4 = (t-2)(t+2).$$

Thus the determinant vanishes for $t = 2$ or $t = -2$. At $t = 2$ the second row is twice the first; at $t = -2$ it is -2 times the first.

A row relation in three dimensions. For

$$B(s) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & s & 2 \end{pmatrix},$$

we obtain

$$\det B(s) = 1 - s.$$

At $s = 1$ the third row satisfies

$$(1,1,2) = (1,0,1) + (0,1,1).$$

How to read a zero determinant. Solve $\det A = 0$, then inspect the exceptional matrices for an explicit relation among rows or columns. The roots are not merely numerical answers; they locate the loss of independence.

6.5 Determinants of Larger Matrices

6.5.1 Minors

Let $A = (a_{ij})$ be an $n \times n$ matrix. Remove row i and column j; the remaining matrix has size $(n-1) \times (n-1)$.

> The determinant of the matrix obtained by deleting row i and column j is the minor of a_{ij}, denoted by
> $$M_{ij}.$$

Example. For

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix},$$

deleting row 1 and column 2 gives

$$M_{12} = \det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = -20.$$

6.5.2 Cofactors

The signs in a Laplace expansion alternate in a checkerboard pattern:

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix}.$$

The sign in position (i, j) is $(-1)^{i+j}$.

> The cofactor of a_{ij} is
> $$C_{ij} = (-1)^{i+j} M_{ij}.$$

For the preceding example,

$$M_{12} = -20, \qquad C_{12} = (-1)^3 M_{12} = 20.$$

Minor versus cofactor. The minor is a determinant of a smaller matrix. The cofactor is that minor together with the sign prescribed by its position.

6.5.3 The Recursive Definition

The base case is

$$\det \begin{pmatrix} a \end{pmatrix} = a.$$

For $n \geq 2$, expand along the first row.

> For $A = (a_{ij}) \in \mathbb{R}^{n \times n}$,
> $$\det A = \sum_{j=1}^{n} a_{1j} C_{1j}.$$

The definition reduces the size by one at each step. Eventually every branch reaches determinants of size 1×1.

Compatibility with the 2×2 *formula.* For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

Laplace expansion gives

$$\det A = a \det \begin{pmatrix} d \end{pmatrix} - b \det \begin{pmatrix} c \end{pmatrix} = ad - bc.$$

□

Compatibility with the 3×3 *formula.* For

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

$$\begin{aligned} \det A &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg), \end{aligned}$$

which expands to the earlier six-term expression. □

6.5.4 Expansion Along Any Row or Column

Although the recursive definition uses the first row, the same determinant is obtained from every row or column.

Expansion along row i gives

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

Expansion along column j gives

$$\det A = \sum_{i=1}^{n} a_{ij} C_{ij}.$$

Proof status. The theorem is verified in small cases and used as the general cofactor expansion rule. A full proof follows from multilinearity and alternation of the determinant; that systematic argument is not developed here.

Freedom of choice. The determinant is independent of the row or column chosen for expansion. The choice changes the amount of work, not the value. The useful choice is therefore the row or column with the simplest pattern, especially many zeros.

6.5.5 Choosing an Expansion Intelligently

Use a sparse column. For

$$B = \begin{pmatrix} 4 & 0 & 1 & 2 \\ 0 & 3 & 0 & -1 \\ 2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 6 \end{pmatrix},$$

the second column contains only one nonzero entry. Therefore

$$\det B = 3C_{22} = 3 \det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix}.$$

The remaining 3×3 determinant is 98, so

$$\det B = 294.$$

Before expanding. Inspect the matrix for zeros, choose a row or column with few nonzero entries, write the cofactor signs explicitly, keep the smaller determinants visible, and combine Laplace expansion with row or column properties when useful.

6.6 General Properties and Efficient Calculation

6.6.1 Multilinearity and Alternating Behaviour

> The determinant is linear in each row separately while all other rows are fixed, and likewise in each column. Exchanging two rows changes the sign. Consequently, a matrix with two equal rows has determinant zero.

If one row is a linear combination of the others, multilinearity reduces the determinant to terms with repeated rows. For example, if

$$r_3 = 2r_1 - r_2,$$

then

$$\begin{aligned} \det(r_1, r_2, r_3) &= 2\det(r_1, r_2, r_1) - \det(r_1, r_2, r_2) \\ &= 0. \end{aligned}$$

Because $\det(A^{\mathsf{T}}) = \det A$, all corresponding statements hold for columns.

6.6.2 Row Operations and Triangular Matrices

Effect of elementary row operations. Exchanging two rows multiplies the determinant by -1; multiplying one row by λ multiplies the determinant by λ; adding a multiple of one row to another leaves the determinant unchanged.

Reduction to triangular form. For

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix},$$

use

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1,$$

then

$$R_3 \leftarrow R_3 + 2R_2.$$

All three operations preserve the determinant and produce

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

Therefore $\det A = 1 \cdot 1 \cdot 5 = 5$.

For an upper or lower triangular matrix,

$$\det A = a_{11}a_{22}\cdots a_{nn}.$$

Hence a triangular matrix has determinant zero exactly when at least one diagonal entry is zero.

6.6.3 Scaling the Matrix and Multiplying Matrices

If every row of an $n \times n$ matrix is multiplied by λ, multilinearity contributes one factor from each row:

$$\boxed{\det(\lambda A) = \lambda^n \det A}.$$

This explains why the determinant is not linear as a function of the whole matrix.

For square matrices of the same size,

$$\det(AB) = \det A \det B.$$

Proof status. This chapter uses the product rule later, but does not give its full proof for arbitrary size. We verify its meaning in simple structured cases and treat the general theorem as an accepted result. A complete proof requires a systematic construction of the determinant from multilinearity and alternation.

For diagonal matrices the rule is immediate because both matrix multiplication and the determinant reduce to products of diagonal entries. It is also consistent with the geometric reading: applying B and then A multiplies the two signed scaling factors.

The product rule concerns ordinary matrix multiplication. It is unrelated to entry-by-entry multiplication and must not be confused with linearity in one row or column.

6.6.4 A Reliable Calculation Strategy

Before computing a determinant. Check whether rows or columns are equal, proportional, or related by a simple linear combination. Look for zeros and triangular structure. Decide whether row or column operations will simplify the matrix. Track sign changes and extracted factors. Use Laplace expansion only after choosing a useful row or column.

The determinant begins as a definition, but its mathematical value lies in the network of properties that allows the definition to be used intelligently.

Determinants. The determinant begins with formulas for small square matrices and extends through minors and cofactors. Its row and column properties explain efficient calculations and show why a zero determinant records a relation among rows, columns, or possible matrix outputs. General identities used later are distinguished from properties that were fully derived in this chapter.

6.7 Review Exercises

6.7.1 Elementary computational exercises

1. Compute

$$\det\begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix}, \qquad \det\begin{pmatrix} 5 & 10 \\ 1 & 2 \end{pmatrix}.$$

Identify the visible row relation in the second matrix.

2. Determine all values of t for which

$$\det \begin{pmatrix} t & 2 \\ 3 & t \end{pmatrix} = 0.$$

3. For

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

compute the determinant after swapping the rows, multiplying the first row by 5, and replacing R_2 by $R_2 - 3R_1$.

4. Compute the determinants of the triangular matrices

$$\begin{pmatrix} 2 & 1 & 4 \\ 0 & -3 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ -1 & 3 & 5 \end{pmatrix}.$$

5. Compute

$$\det \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}$$

by expanding along the first row.

6. Compute

$$\det \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 4 & 0 & -1 \end{pmatrix}$$

by choosing the row or column with the most zeros.

7. Use one row operation to simplify and compute

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 0 & 1 & 2 \end{pmatrix}.$$

Track the determinant correctly.

8. Decide without full expansion which determinants are zero:

$$\det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad \det \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}, \quad \det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 4 \end{pmatrix}.$$

9. For

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & -1 \\ 4 & 1 \end{pmatrix},$$

compute $\det A$, $\det B$, AB, and $\det(AB)$.

10. Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Compute Ae_1 and Ae_2, show that the outputs are parallel, and connect this observation with $\det A = 0$.

6.7.2 Development and reasoning exercises

1. Compute
$$\det\begin{pmatrix}1 & 2 & 0\\ -1 & 3 & 4\\ 2 & 0 & 5\end{pmatrix}$$
first by the direct 3×3 formula and then by Laplace expansion. Compare the work and explain which structure should guide the choice.

2. Use row operations to compute
$$\det\begin{pmatrix}1 & 2 & 1\\ 2 & 5 & 3\\ 3 & 7 & 5\end{pmatrix}.$$
Record every operation and explain its effect on the determinant.

3. Compute
$$\det\begin{pmatrix}2 & 0 & 1 & 0\\ 0 & 3 & 0 & 0\\ 4 & 0 & -1 & 2\\ 0 & 0 & 0 & 5\end{pmatrix}$$
by exploiting zeros. Explain why a full expansion would obscure the structure.

4. Determine all values of t for which
$$\det\begin{pmatrix}1 & t & 0\\ 2 & 2 & 1\\ 0 & 1 & t\end{pmatrix}=0.$$
For every exceptional value, identify a relation among rows, columns, or outputs when possible.

5. Let
$$A=\begin{pmatrix}2 & 1\\ 1 & 1\end{pmatrix}.$$
Map the unit square by A, compute the area of the resulting parallelogram, and compare it with $|\det A|$. Repeat with a singular matrix and describe the collapse.

6. Prove directly from the 2×2 formula that swapping two rows changes the sign of the determinant and that adding a multiple of one row to the other leaves it unchanged.

7. A matrix has two proportional rows. Give a proof of $\det A = 0$ using row operations and a second proof using the interpretation of restricted outputs.

8. Explain the identity
$$\det(AB)=\det A\det B$$
as successive area or volume scaling. Verify the interpretation with one explicit pair of 2×2 matrices.

9. Compare four strategies for a determinant: direct small formula, Laplace expansion, triangular structure, and row reduction. Give one example where each is the natural choice.

10. Design a complete pre-calculation inspection routine for determinants. It must check size, zeros, proportional rows or columns, triangular structure, useful row operations, and the possibility that a parameter creates exceptional cases.

Chapter 7

Matrix Inversion

7.1 The Meaning of an Inverse Matrix

7.1.1 Definition and the Two-Sided Condition

For a nonzero number a, the inverse satisfies $aa^{-1} = 1$. In matrix algebra the identity matrix replaces the number 1.

Let A be an $n \times n$ matrix. A matrix B is an inverse of A if

$$AB = I_n \qquad \text{and} \qquad BA = I_n.$$

If it exists, it is denoted by A^{-1}.

Both sides matter. Matrix multiplication is not commutative. Therefore $AB = I$ and $BA = I$ are not merely two ways of writing the same calculation. The inverse must undo A from both sides.

Remark. The notation A^{-1} does not mean $1/A$. It denotes the matrix that reverses the action of A, provided such a matrix exists.

7.1.2 Undoing a Transformation

If

$$Ax = y$$

and A^{-1} exists, then

$$A^{-1}Ax = A^{-1}y,$$

so

$$\boxed{x = A^{-1}y}.$$

Recovering the original input. An inverse matrix reconstructs the input from the output. If different inputs have been collapsed into the same output, unique recovery is impossible and no inverse can exist.

A diagonal transformation. For

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

the first coordinate is doubled and the second is tripled. Therefore

$$D^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix},$$

and direct multiplication gives $DD^{-1} = I_2$.

7.1.3 When Does an Inverse Exist?

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

If $\det A = 0$, the matrix is singular and has no inverse.

Proof status. The criterion is a central theorem connecting determinants, row reduction, and solvability. This course verifies both directions through the behaviour of elimination and through the adjugate identity, but does not present a completely independent proof from the axiomatic construction of the determinant.

A nonzero determinant means that the matrix has not collapsed independent information. A zero determinant signals structural loss and prevents unique reversal. The practical questions are therefore: what operation does the matrix perform, has information been lost, and if not, which matrix reverses the operation?

7.2 Inverses of 2×2 Matrices

7.2.1 The Formula and Where It Comes From

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}.$$

The condition $AA^{-1} = I_2$ produces two systems:

$$\begin{cases} ap + br = 1, \\ cp + dr = 0, \end{cases} \qquad \begin{cases} aq + bs = 0, \\ cq + ds = 1. \end{cases}$$

Thus finding an inverse already amounts to solving linear systems.

If

$$\det A = ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If $ad - bc = 0$, the inverse does not exist.

The diagonal entries are exchanged, the off-diagonal signs are reversed, and the entire matrix is scaled by the reciprocal determinant.

7.2.2 Invertible and Singular Examples

An invertible matrix. For

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix},$$

$\det A = 1$, so

$$A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

Verification gives

$$AA^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

A singular matrix. For

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix},$$

$\det B = 0$. The formula would require division by zero, but the deeper reason is that the second row is three times the first. Independent information has been lost, so the action cannot be undone uniquely.

Verification is part of the method. The output is not merely a candidate matrix. It must multiply with the original matrix to produce the identity.

7.2.3 How to Read and Use the Formula

The determinant controls reversibility. The factor $1/\det A$ exists only when the transformation is reversible. The rearranged entry matrix is precisely the pattern that cancels the coordinate mixing in A.

Procedure for a 2×2 inverse. Compute $\det A$. If it is zero, stop. Otherwise exchange the diagonal entries, reverse the signs of the off-diagonal entries, divide by the determinant, and verify the product with the original matrix.

7.3 The Gauss–Jordan Method

7.3.1 The Principle and Elementary Matrices

To find A^{-1}, place the identity beside A and perform the same row operations on both blocks:

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

> An elementary matrix is obtained from an identity matrix by one elementary row operation. If E represents that operation, then EA applies it to A.

Why the right block becomes the inverse. If

$$E_k \cdots E_2 E_1 A = I,$$

then

$$E_k \cdots E_2 E_1 = A^{-1}.$$

Applying the same operations to the identity block constructs exactly this product of elementary matrices. □

Gauss–Jordan goal. Create pivots, eliminate entries below and above them, and reduce the entire left block to the identity while preserving every operation on the right block.

7.3.2 A Complete 2×2 Example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

Start from

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right].$$

Reduce the left block. First use

$$R_2 \leftarrow R_2 - 3R_1,$$

which gives

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

Then use

$$R_1 \leftarrow R_1 - 2R_2,$$

obtaining

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

Therefore

$$\boxed{A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}}.$$

Direct multiplication verifies that $AA^{-1} = I_2$.

7.3.3 Why the same method works in every dimension

For an invertible $n \times n$ matrix, the construction is always

$$[A \mid I_n] \longrightarrow [I_n \mid A^{-1}].$$

The number of rows and columns changes, but the idea does not: choose a pivot, use it to eliminate the other entries in its column, and perform every row operation on both blocks.

Dimension changes the workload, not the principle. A larger example contains more pivots and more arithmetic, but no new logical step. Once the complete 2×2 example is understood, a 3×3 calculation is repetition of the same reversible operations.

If the left block can be reduced to I_n, the accumulated operations on the right block form A^{-1}. If reduction cannot create all required pivots, information has been lost and the inverse does not exist.

7.3.4 What Failure of the Reduction Means

A singular left block. For

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

start with

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right].$$

After

$$R_2 \leftarrow R_2 - 2R_1,$$

the left block contains a zero row:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

It cannot be reduced to the identity, so B is not invertible.

The method reveals singularity. A missing pivot is not a defect of Gauss–Jordan elimination. It is evidence that the matrix has no inverse.

Gauss–Jordan procedure. Form $[A \mid I]$, reduce the left block using elementary row operations, and read the inverse from the right block if the left becomes I. If the identity cannot be reached, the matrix is singular.

7.4 The Adjugate Method

7.4.1 The Cofactor Matrix and the Adjugate

For $A = (a_{ij})$, let

$$C_{ij} = (-1)^{i+j} M_{ij}$$

be the cofactor of a_{ij}. Arrange all cofactors in the cofactor matrix C.

> The adjugate of A is the transpose of the cofactor matrix:
>
> $$\operatorname{adj}(A) = C^{\mathsf{T}}.$$

The transpose is essential: cofactors associated with rows become columns in the matrix that reverses the action.

The construction in dimension two. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the cofactor matrix and adjugate are

$$C = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \qquad \operatorname{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Thus the familiar 2×2 inverse formula is already the adjugate formula in its simplest complete form.

7.4.2 The Adjugate Inverse Formula

> For every square matrix,
>
> $$A \operatorname{adj}(A) = (\det A) I.$$
>
> If $\det A \neq 0$, then
>
> $$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A).$$

Proof status. The identity comes from Laplace expansion: diagonal entries of $A \operatorname{adj}(A)$ reproduce $\det A$, while off-diagonal entries cancel because they correspond to determinants with repeated rows or columns. This mechanism is the point retained here; a full index-by-index proof is not developed.

Why the identity gives an inverse. Starting from

$$A \operatorname{adj}(A) = (\det A) I,$$

divide by the nonzero scalar $\det A$:

$$A \left(\frac{1}{\det A} \operatorname{adj}(A) \right) = I.$$

The matrix in parentheses is therefore the inverse.

Determinants and cofactors build the inverse. The same smaller determinants used in Laplace expansion assemble the matrix that reverses A. A zero determinant prevents the final division and records the loss of invertibility.

7.4.3 What the Adjugate Method Is For

For hand calculations, the adjugate method is natural in the 2×2 case because it reproduces the familiar inverse formula immediately. In larger dimensions, computing every cofactor quickly becomes longer than elimination.

Conceptual rather than algorithmic value. Gauss–Jordan is the principal construction method. The adjugate is retained because it explains why cofactors and the determinant control invertibility and why the condition $\det A \neq 0$ appears in every inverse formula.

The student should understand the chain

$$\text{cofactors} \longrightarrow \operatorname{adj}(A) \longrightarrow A\operatorname{adj}(A) = (\det A)I,$$

without treating a long cofactor table as the central skill of the chapter.

7.5 Invertibility and Visible Structure

7.5.1 Diagonal and Triangular Matrices

For

$$D = \operatorname{diag}(d_1, \dots, d_n),$$

we have

$$\det D = d_1 \cdots d_n.$$

Thus D is invertible exactly when every d_i is nonzero, and then

$$\boxed{D^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \dots, \frac{1}{d_n}\right)}.$$

Example. For

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix},$$

$$D^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}.$$

For a triangular matrix, the determinant is the product of the diagonal entries. Therefore it is invertible exactly when all diagonal entries are nonzero.

Pivots and reversibility. A zero diagonal entry in triangular form means a missing pivot. Without one pivot in each coordinate direction, the matrix cannot be reduced to the identity.

7.5.2 Repeated or Dependent Rows

If two rows are equal, one is a multiple of another, or one can be constructed from the remaining rows, then the determinant is zero and the matrix is not invertible.

Example. For

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & 4 \end{pmatrix},$$

the second row is three times the first. Hence $\det A = 0$ and A has no inverse.

Repeated information cannot be reversed. A dependent row contributes no genuinely new information. The matrix therefore lacks enough independent directions to recover arbitrary inputs uniquely.

7.5.3 Products of Invertible Matrices

> If A and B are invertible square matrices of the same size, then AB is invertible and
>
> $$(AB)^{-1} = B^{-1}A^{-1}.$$

The order is reversed because the last action must be undone first. A vector is transformed as

$$x \longmapsto Bx \longmapsto A(Bx),$$

so recovery proceeds by undoing A and then B.

Verification.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I,$$

and similarly

$$(B^{-1}A^{-1})(AB) = I.$$

□

7.5.4 Special Matrix Identities

Sometimes an identity already contains the inverse.

An involution. If

$$A^2 = I,$$

then $AA = I$, so

$$A^{-1} = A.$$

Extracting an inverse from a polynomial identity. Suppose

$$A^2 - 3A + 2I = 0.$$

Then

$$A(A - 3I) = -2I,$$

so

$$A\left(-\frac{1}{2}(A - 3I)\right) = I.$$

Therefore

$$A^{-1} = \frac{1}{2}(3I - A).$$

Recognising invertibility. Inspect diagonal and triangular structure, dependent rows, products of known invertible matrices, and special identities before beginning a long determinant or elimination calculation.

7.6 Matrix Equations

7.6.1 Equations of the Forms $AX = B$ and $XA = B$

If

$$AX = B$$

and A is invertible, multiply on the left:

$$A^{-1}AX = A^{-1}B, \qquad \boxed{X = A^{-1}B}.$$

If instead

$$XA = B,$$

multiply on the right:

$$XAA^{-1} = BA^{-1}, \qquad \boxed{X = BA^{-1}}.$$

The side is determined by the equation. An inverse cancels a matrix only when it is placed next to that matrix on the correct side. Matrix factors cannot be moved through an equation as if they were ordinary numbers.

Different sides give different answers. For

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix},$$

$AX = B$ gives

$$X = A^{-1}B = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix},$$

whereas $XA = B$ gives

$$X = BA^{-1} = \begin{pmatrix} 25 & -7 \\ 29 & -8 \end{pmatrix}.$$

7.6.2 An Unknown Matrix Between Two Known Matrices

Suppose

$$AXB = C$$

and both A and B are invertible. Undo A on the left and B on the right:

$$A^{-1}(AXB)B^{-1} = A^{-1}CB^{-1}.$$

Therefore

$$\boxed{X = A^{-1}CB^{-1}}.$$

Cancellation determines the order. Each inverse must be placed on the side where it becomes adjacent to the matrix it is intended to cancel. The order is a consequence of the original product, not a memorised decoration.

7.6.3 Solving Vector Equations

The equation

$$Ax = b$$

with column vectors x and b is a special case of $AX = B$. If A is invertible, then

$$\boxed{x = A^{-1}b}.$$

Example. Solve

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The coefficient matrix has determinant 1 and inverse

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Hence

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus $x = 2$ and $y = 1$.

This prepares the next chapter, where elimination, inverse matrices, and Cramer's rule will be compared as methods for solving linear systems.

7.6.4 Common Mistakes and a Reliable Procedure

From

$$AX = B$$

it is generally wrong to write $X = BA^{-1}$. Multiplying on the right does not cancel the matrix A standing on the left of X.

Do not move matrix factors like numbers. The correct operation is

$$A^{-1}AX = A^{-1}B,$$

which gives $X = A^{-1}B$. Every cancellation must respect adjacency and order.

Solving matrix equations. Identify the position of the unknown, check that the matrices to be inverted are invertible, multiply on the side where cancellation is valid, preserve factor order, and verify the answer by substitution whenever practical.

Matrix inversion. A square matrix is invertible exactly when its determinant is nonzero. The 2×2 formula gives the first explicit case, Gauss–Jordan is the main general hand method, and the adjugate formula explains the determinant–cofactor connection. Structural identities and matrix equations show how inverses cancel actions on the correct side and in the correct order.

7.7 Review Exercises

7.7.1 Elementary computational exercises

1. Determine whether

   $$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

   is invertible. If so, compute A^{-1} with the 2×2 formula and verify both products.

2. Compute the inverse of
$$A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$$
by Gauss–Jordan elimination, recording every row operation.

3. Compute the inverse of
$$D = \begin{pmatrix} 2 & 0 \\ 0 & -4 \end{pmatrix}$$
and explain the visible coordinate action that is being reversed.

4. Decide which matrices are invertible and compute the inverse when it exists:
$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

5. For
$$A = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 1 \end{pmatrix},$$
compute A^{-1} and solve $Ax = b$.

6. Solve $AX = B$ for
$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}.$$

7. Solve $XA = B$ for the same matrices and compare the multiplication order with the previous exercise.

8. For
$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$
write the cofactor matrix, the adjugate, and the inverse formula. Specialise to $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$.

9. Let
$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}.$$
Compute A^{-1}, B^{-1}, $(AB)^{-1}$, and $B^{-1}A^{-1}$.

10. During Gauss–Jordan elimination the left block reduces to
$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$
Explain the computational conclusion and confirm it for $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ using the determinant.

7.7.2 Development and reasoning exercises

1. Invert

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

by Gauss–Jordan elimination. Before every major step, state the pivot goal; after the calculation, verify one product AA^{-1}.

2. Invert the triangular matrix

$$T = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{pmatrix}$$

by Gauss–Jordan elimination. Explain how the zero pattern reduces the work without changing the method.

3. Apply Gauss–Jordan elimination to

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}.$$

Show where a required pivot fails, identify the row relation, and connect the failure with $\det S = 0$.

4. For

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix},$$

find the inverse directly from structure and again by Gauss–Jordan. Compare the two approaches and explain why structural inspection should come first.

5. Compute the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

with both the formula and Gauss–Jordan. Match the corresponding steps and explain what each method makes most visible.

6. Prove that the Gauss–Jordan construction

$$[A \mid I] \longrightarrow [I \mid A^{-1}]$$

works by interpreting row operations as multiplication by elementary matrices. Use a concrete 2×2 example inside the argument.

7. Explain structurally why

$$A \operatorname{adj}(A) = (\det A)I.$$

Verify the identity for one 2×2 matrix and state why the adjugate method is conceptually useful even when it is not efficient.

8. Prove

$$(AB)^{-1} = B^{-1}A^{-1}$$

and illustrate the reversed order using two simple geometric actions, such as a scaling and a coordinate swap.

9. A student writes $AX = B \Rightarrow X = BA^{-1}$. Diagnose the error, solve one numerical equation $AX = B$, and explain why the correct side of multiplication follows from the order of actions.

10. Give a complete inversion strategy: inspect structure, test existence, choose between a direct formula and Gauss–Jordan, perform and record reversible steps, verify the answer, and interpret a failure. Apply the strategy briefly to one matrix of your choice.

Chapter 8

Systems of Linear Equations

8.1 Systems as Simultaneous Conditions

8.1.1 A Solution Must Satisfy Every Equation

One assignment, all conditions. A linear system is not a list of separate problems. Each equation imposes one condition, and a solution is one assignment of values that satisfies all conditions simultaneously.

Example. For

$$\begin{cases} x + y = 5, \\ 2x - y = 1, \end{cases}$$

the pair $(2, 3)$ satisfies both equations and is therefore a solution. The pair $(1, 4)$ satisfies the first equation but not the second, so it is not a solution of the system.

Passing one test is not enough. Membership in the solution set means passing every equation at the same time.

8.1.2 Unknowns, Coefficients, Constants, and Solution Sets

Throughout this chapter we work with two equations in two unknowns:

$$\begin{cases} a_{11}x + a_{12}y = b_1, \\ a_{21}x + a_{22}y = b_2. \end{cases}$$

The numbers a_{ij} are coefficients, while b_1 and b_2 are the right-hand sides. The same notation extends to larger systems, but the 2×2 case already contains the essential classification and solution methods.

> The solution set is the set of all pairs (x, y) satisfying both equations. It may contain exactly one point, no points, or infinitely many points.

The coefficients determine the directions of the two lines, while the right-hand sides determine their positions. Solving the system means finding their common points.

8.1.3 Matrix Form and the Augmented Matrix

The system

$$\begin{cases} 2x + y = 5, \\ x - y = 1 \end{cases}$$

can be written as

$$\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix},$$

or compactly as

$$\boxed{Ax = b}.$$

The matrix equation. The coefficient matrix acts on the unknown vector and must produce the prescribed right-hand side. Solving means finding which input vector, if any, is sent to b.

The same system is recorded by the augmented matrix

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array}\right].$$

The vertical line separates coefficients from right-hand sides.

Reading a system. Identify unknowns, coefficients, right-hand sides, simultaneous conditions, the matrix equation $Ax = b$, and the information recorded by the augmented matrix before selecting a solution method.

8.2 Geometric Meaning of Solutions

8.2.1 Lines in the Plane and Their Intersections

In two variables, a linear equation

$$ax + by = c$$

usually describes a line. A system of two equations asks for points lying on both lines.

Example. For

$$\begin{cases} x + y = 5, \\ 2x - y = 1, \end{cases}$$

the point $(2, 3)$ lies on both lines. Geometrically, solving the system means finding their common point.

Solution set as intersection. Each equation defines a geometric set. The solution set of the system is the intersection of all those sets.

8.2.2 One Solution, No Solution, or Infinitely Many

One solution. Two nonparallel lines meet in one point. The system

$$\begin{cases} x + y = 5, \\ 2x - y = 1 \end{cases}$$

has the unique solution $(2, 3)$.

No solution. The system

$$\begin{cases} x + y = 1, \\ x + y = 4 \end{cases}$$

describes parallel distinct lines. The same expression cannot equal both constants, so the system is inconsistent.

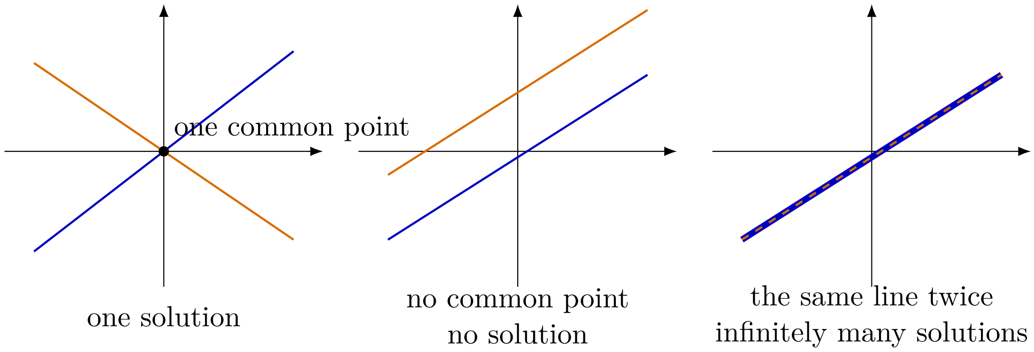
Infinitely many solutions. The system

$$\begin{cases} x + y = 3, \\ 2x + 2y = 6 \end{cases}$$

describes the same line twice. Every point on $x + y = 3$ is a solution.

Independence and compatibility. One solution means the conditions determine one common point. No solution means the conditions conflict. Infinitely many solutions mean at least one condition is redundant and freedom remains.

8.2.3 The Three Cases in One Picture



Algebra must reproduce the geometry. A unique pivot in each variable corresponds to one intersection point. A contradictory row corresponds to parallel distinct lines. A zero row without contradiction means that one equation repeats information already contained in the other, leaving an entire line of solutions.

8.2.4 Algebraic Signals in the Plane

For two variables, each linear equation describes a line. Two lines may intersect once, be parallel and distinct, or coincide.

The three outcomes are already visible in 2×2 systems. Two intersecting lines give exactly one solution. Parallel distinct lines give no solution. Coincident lines give infinitely many solutions.

During elimination,

$$0 = 5$$

signals inconsistency, while

$$0 = 0$$

signals a repeated condition. Proportional left-hand sides describe lines with the same direction; the right-hand sides decide whether the lines coincide or remain distinct.

Algebraic row patterns are therefore geometric information: they record whether the two conditions intersect, conflict, or repeat one another.

8.3 Gaussian Elimination

8.3.1 Equivalent Systems and Allowed Row Operations

Two systems are equivalent when they have the same solution set. Gaussian elimination uses row operations that replace the written equations by equivalent conditions.

> The allowed operations are: exchanging two rows, multiplying a row by a nonzero scalar, and adding a multiple of one row to another.

Why row operations are legitimate. Each operation can be reversed. Therefore it changes the form of the equations without changing which assignments satisfy them.

Remark. The aim is not to make the matrix look different. The aim is to expose the solution set by simplifying the conditions.

8.3.2 A First Elimination Example

Consider

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

Its augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array}\right].$$

Eliminate and back-substitute. Apply

$$R_2 \leftarrow R_2 - 2R_1,$$

obtaining

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \end{array}\right].$$

The second row gives $y = 3$. Substitution into the first row gives $x = 2$.

Therefore

$$\boxed{(x, y) = (2, 3)}.$$

The essential strategy is to simplify until variables can be read from the bottom upward.

8.3.3 Pivots, Back Substitution, and Row Echelon Form

A pivot is a leading nonzero entry used to eliminate other entries in its column. Pivots mark independent information and identify the variables that can be solved.

Back substitution in a 2×2 system. For

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 3 & 6 \end{array}\right],$$

the second row gives $y = 2$, and the first row then gives $x = 1$.

> A matrix is in row echelon form when leading nonzero entries move strictly to the right as one moves downward, and all zero rows are placed at the bottom.

A row such as

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \end{array}\right]$$

represents $0 = 0$ and imposes no new restriction.

8.3.4 Detecting No Solution

A row of the form

$$\left[\begin{array}{cc|c} 0 & 0 & 5 \end{array}\right]$$

represents the impossible equation $0 = 5$.

Parallel inconsistent equations. For

$$\begin{cases} x + y = 1, \\ 2x + 2y = 5, \end{cases}$$

row reduction gives

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 3 \end{array}\right].$$

The second row says $0 = 3$, so the system has no solution.

The contradiction is the algebraic record of two parallel conditions that cannot be satisfied simultaneously.

8.3.5 Free Variables and Infinitely Many Solutions

If a variable column has no pivot, that variable is free and may be used as a parameter.

One equation remains after reduction. The echelon form

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right]$$

represents the single independent condition

$$x + 2y = 3.$$

Set $y = t$. Then

$$x = 3 - 2t,$$

so the complete solution set is

$$(x, y) = (3 - 2t, t), \quad t \in \mathbb{R}.$$

The parameter is not an arbitrary decoration. It records the direction along the line of solutions that remains after one equation has become redundant.

8.3.6 Classification After Row Reduction

Three diagnostic patterns. After reduction:

- a pivot in every variable column and no contradiction gives exactly one solution;
- a row $0 =$ nonzero gives no solution;
- at least one free variable and no contradiction gives infinitely many solutions.

Gaussian elimination is the most flexible method in this chapter. It applies to square and rectangular systems and reveals both the solution and the structure of the solution set.

8.4 Cramer's Rule

8.4.1 The 2×2 Rule

For

$$\begin{cases} ax + by = e, \\ cx + dy = f, \end{cases} \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

assume $\det A \neq 0$. Replace one coefficient column at a time by the right-hand side:

$$A_x = \begin{pmatrix} e & b \\ f & d \end{pmatrix}, \qquad A_y = \begin{pmatrix} a & e \\ c & f \end{pmatrix}.$$

$$x = \frac{\det A_x}{\det A}, \qquad y = \frac{\det A_y}{\det A}.$$

Remark. The denominator condition is structural. If $\det A = 0$, the coefficient matrix is singular and Cramer's rule does not apply.

8.4.2 A Worked Cramer Example

Solve

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det A = -3.$$

Replace one column at a time. For x,

$$A_x = \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det A_x = -6,$$

so

$$x = \frac{-6}{-3} = 2.$$

For y,

$$A_y = \begin{pmatrix} 2 & 5 \\ 1 & 1 \end{pmatrix}, \qquad \det A_y = -3,$$

so

$$y = \frac{-3}{-3} = 1.$$

Thus

$$\boxed{(x, y) = (2, 1)}.$$

Substitution verifies both original equations.

8.4.3 What a Zero Coefficient Determinant Means

If $\det A = 0$, Cramer's rule does not apply. This does not automatically mean that the system is inconsistent; it means only that a unique solution cannot be obtained from an invertible coefficient matrix.

Example. For

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4, \end{cases}$$

the coefficient determinant is zero, but the second equation repeats the first. The system has infinitely many solutions.

Loss of uniqueness is not loss of existence. When $\det A = 0$, further elimination or equation comparison is required to distinguish no solution from infinitely many solutions.

Using Cramer's rule responsibly. Check that the system is square, compute $\det A$, replace the correct column for each unknown, verify the answer, and classify singular cases separately.

8.5 The Inverse Matrix Method

8.5.1 Solving $Ax = b$ by Reversing the Coefficient Matrix

If A is invertible, multiply

$$Ax = b$$

on the left by A^{-1}:

$$A^{-1}Ax = A^{-1}b.$$

Therefore

$$\boxed{x = A^{-1}b}.$$

> If A is an invertible square matrix, then $Ax = b$ has the unique solution $x = A^{-1}b$.

Undoing the coefficient action. The matrix A transforms the unknown vector into b. Applying the inverse to b reconstructs the original input vector.

8.5.2 A Worked Inverse-Matrix Example

Solve

$$\begin{cases} 2x + y = 5, \\ x + y = 3. \end{cases}$$

Write

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Since $\det A = 1$,

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Apply the inverse.

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$\boxed{x = 2, \qquad y = 1}.$$

8.5.3 Determinant Condition and When the Method Is Useful

The inverse method applies exactly when

$$\det A \neq 0.$$

If $\det A = 0$, no inverse exists. The system may then have no solution or infinitely many solutions, but it cannot be solved by multiplying by A^{-1}.

Example. For

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4, \end{cases}$$

the coefficient determinant is zero, yet every point on $x + y = 2$ is a solution. Failure of invertibility means loss of uniqueness, not automatic inconsistency.

When inversion is efficient. The method is especially useful when A^{-1} is already known, when several right-hand sides share the same coefficient matrix, or when the system is being studied explicitly as a matrix transformation. Direct elimination is often cheaper for a single routine system.

Inverse method checklist. Write $Ax = b$, check that A is square and invertible, multiply on the left by A^{-1}, and verify the resulting vector in the original equations.

8.6 Parametric Systems and Classification

8.6.1 The Three Outcomes and a Simple Parameter Example

Every linear system has exactly one solution, no solution, or infinitely many solutions. For a square system $Ax = b$, a nonzero determinant guarantees the first case. A zero determinant removes uniqueness but does not decide existence.

Remark. The condition $\det A = 0$ is a warning sign, not a complete classification.

Consider

$$\begin{cases} x + y = 1, \\ 2x + 2y = a. \end{cases}$$

Twice the first equation gives $2x + 2y = 2$.

Classification. If $a = 2$, the equations describe the same line and there are infinitely many solutions. If $a \neq 2$, they describe parallel distinct lines and there is no solution.

8.6.2 Using the Determinant to Find Exceptional Values

Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

Then

$$\det A(t) = t^2 - 4 = (t - 2)(t + 2).$$

Generic and exceptional parameter values. For $t \neq 2, -2$, the matrix is invertible, so every system $A(t)x = b$ has exactly one solution. The values $t = 2$ and $t = -2$ must be analysed separately.

At an exceptional value, the determinant tells us only that uniqueness is lost. Row reduction or direct equation comparison must decide whether the system is inconsistent or has infinitely many solutions.

8.6.3 Full Classification Examples

A parameter in the coefficient matrix. Classify

$$\begin{cases} x + y = 3, \\ 2x + ay = 6. \end{cases}$$

The coefficient determinant is

$$\det \begin{pmatrix} 1 & 1 \\ 2 & a \end{pmatrix} = a - 2.$$

If $a \neq 2$, there is exactly one solution. If $a = 2$, the second equation is twice the first, so there are infinitely many solutions.

A parameter in the right-hand side. Classify

$$\begin{cases} x + y = 3, \\ 2x + 2y = a. \end{cases}$$

The coefficient determinant is always zero. Comparing with twice the first equation gives:

- if $a = 6$, the equations coincide and there are infinitely many solutions;
- if $a \neq 6$, the equations are inconsistent and there is no solution.

Determinant first, compatibility second. A parameter in the coefficient matrix may create isolated values where uniqueness fails. A parameter in the right-hand side may decide whether dependent equations agree or contradict one another.

8.6.4 Row Reduction with Parameters

For a parameter-dependent 2×2 system, row reduction is often the safest classification method. It exposes contradictions, zero rows, missing pivots, and exceptional parameter values directly.

The diagnostic patterns remain

$$0 = 0$$

for redundancy and

$$0 = 5$$

for inconsistency. A missing pivot produces a free variable.

Do not divide before separating exceptional cases. If a row operation would divide by an expression such as $a - 2$, first consider the case $a - 2 = 0$. Values that make a proposed divisor vanish must be treated separately.

A symbolic elimination step may be reversible for generic parameter values but invalid at exceptional values. Correct classification therefore requires explicit case separation.

8.6.5 A Classification Strategy and Chapter Synthesis

Parameter-system strategy. For a square system $A(t)x = b(t)$:

1. compute $\det A(t)$;
2. solve $\det A(t) = 0$ to find exceptional values;
3. conclude one solution for all non-exceptional values;
4. analyse every exceptional value separately by elimination or equation comparison;
5. classify each case as one solution, no solution, or infinitely many solutions.

Matrices organise coefficients, row operations simplify simultaneous conditions, determinants locate possible loss of uniqueness, inverse matrices solve reversible square systems, and Cramer's rule expresses their unique solutions through determinants.

One object, several methods. These are not isolated tricks. They are different ways of reading the same system of linear conditions: its compatibility, dependence, freedom, and solution set.

Systems of linear equations. A 2×2 system may have one solution, no solution, or infinitely many solutions. Gaussian elimination is the general classification method. When the coefficient determinant is nonzero, Cramer's rule and the inverse formula give two additional descriptions of the same unique solution. Singular and parametric cases return to compatibility, pivots, and free variables.

8.7 Review Exercises

8.7.1 Elementary computational exercises

1. Solve by elimination:
$$\begin{cases} x + y = 7, \\ 2x - y = 2. \end{cases}$$
Write the augmented matrix and verify the solution.

2. Solve by elimination:
$$\begin{cases} 3x + 2y = 4, \\ 5x - 2y = 12. \end{cases}$$
Choose a step that avoids early fractions.

3. Classify and solve:
$$\begin{cases} x + 2y = 3, \\ 2x + 4y = 6. \end{cases}$$
Write the solution set with one parameter.

4. Classify
$$\begin{cases} x - 3y = 2, \\ 2x - 6y = 7. \end{cases}$$
and display the contradictory row.

5. Solve
$$\begin{cases} 2x + y = 5, \\ x - y = 1 \end{cases}$$
with Cramer's rule.

6. Solve
$$\begin{cases} 2x + y = 5, \\ x + y = 3 \end{cases}$$
with the inverse-matrix method.

7. For the parameter a, classify
$$\begin{cases} x + y = 3, \\ 2x + ay = 6. \end{cases}$$

8. For the parameter b, classify
$$\begin{cases} x - 2y = 1, \\ 3x - 6y = b. \end{cases}$$

9. Find and plot the intersection of
$$y = 2x - 3, \qquad y = -x + 6$$
by solving the corresponding system.

10. Reduce
$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 6 & 15 \end{array}\right]$$
to echelon form, identify the pivot and free variable, and write the full solution set.

8.7.2 Development and reasoning exercises

1. Solve by Gaussian elimination:

$$\begin{cases} x + y + z = 6, \\ 2x - y + z = 3, \\ x + 2y - z = 2. \end{cases}$$

   State the pivot goal at each stage and verify the final triple.

2. Classify and solve:

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 6, \\ x - y + z = 1. \end{cases}$$

   Identify a free variable and explain why the same pivot logic as in 2×2 still applies.

3. Classify

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = 3, \\ x - y = 0. \end{cases}$$

   Show the contradictory row and explain what incompatibility means geometrically.

4. Solve the parameter-dependent system

$$\begin{cases} x + y + z = 2, \\ x + ay + z = 2, \\ x + y + az = 2. \end{cases}$$

   Separate exceptional values before division and classify every case.

5. Solve one invertible 2×2 system by elimination, Cramer's rule, and the inverse method. Compare the intermediate objects and explain why all three methods produce the same point.

6. For a singular coefficient matrix, construct one right-hand side giving no solution and another giving infinitely many solutions. Solve both and explain why $\det A = 0$ alone does not decide between the two cases.

7. Explain why elementary row operations preserve the solution set. Demonstrate each of the three operation types on one concrete system and compare the equations before and after.

8. Interpret the rows

$$[\,0 \quad 0 \mid 5\,] \qquad \text{and} \qquad [\,0 \quad 0 \mid 0\,].$$

   Give a system producing each row and describe the corresponding geometry.

9. Compare Gaussian elimination, Cramer's rule, and the inverse-matrix method. Include method scope, classification power, required assumptions, and computational cost.

10. Give a complete strategy for a linear system: translate conditions into an augmented matrix, select pivots, preserve equivalence, detect contradiction or freedom, write the solution set, verify it, and interpret the result geometrically.

Chapter 9

Functions and Sequences

9.1 Functions as Assignments

9.1.1 A function, its domain, and its values

A function is more than a formula. A function is a controlled assignment: every allowed input receives exactly one output. A formula may describe that assignment, but tables, graphs, verbal rules, and recursive rules may describe functions as well.

> Let X and Y be sets. A function
> $$f : X \to Y$$
> assigns to every $x \in X$ exactly one value $f(x) \in Y$. The set X is the domain and Y is the codomain.

The domain is part of the function. The same expression may define different functions when different inputs are allowed. For example,

$$f(x) = \frac{1}{x-2}$$

is meaningful only for $x \neq 2$.

Reading values. For

$$f(x) = x^2 + 1,$$

we have

$$f(3) = 10, \qquad f(-2) = 5.$$

These equations describe two values of the function, not the whole function.

Exactly one output. The phrase "exactly one" excludes two failures: assigning several outputs to one input and leaving an allowed input without an output. This precision makes expressions such as $f(a)$ unambiguous.

Reading a function. Identify the domain, the rule of assignment, the output corresponding to each input, and the way the function is represented.

9.2 Formulas, Tables, and Graphs

9.2.1 Different descriptions of the same assignment

A formula is useful because it gives values for many inputs at once. A table emphasizes selected values, while a graph displays the pairs $(x, f(x))$ geometrically.

A function described by a table. The table

x	1	2	3
$f(x)$	4	4	7

defines a function on the finite domain $\{1, 2, 3\}$. No algebraic formula is required.

The graph of a real function $f : D \to \mathbb{R}$ is the set of points

$$\{(x, f(x)) : x \in D\}.$$

A graph allows us to read approximate values, zeros, signs, intervals of increase or decrease, and visible discontinuities. These observations are geometric statements about the same assignment encoded by the formula or table.

The vertical-line test. A drawn relation represents a function of x only when every vertical line meets it at most once. Two intersections would assign two different outputs to the same input.

Read before calculating. Before manipulating a formula, determine what its graph and domain already tell you. Later, limits and derivatives will refine this visual reading rather than replace it.

9.3 Elementary Real Functions

9.3.1 Basic families and elementary questions

The first examples should remain computationally simple: constants, linear functions, low-degree polynomials, reciprocal functions, absolute value, the exponential function, and the basic sine and cosine functions.

A quadratic function. For

$$f(x) = x^2 - 4,$$

the domain is $\mathbb{R}$, the zeros are $x = \pm 2$, and the function is negative between the zeros and positive outside them.

A simple exponential. For

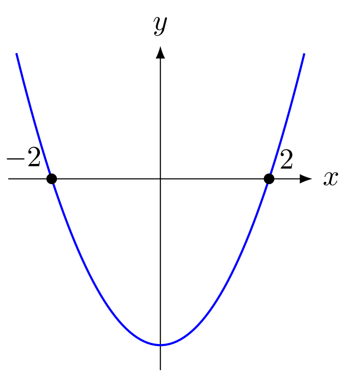
$$g(x) = e^x,$$

the domain is $\mathbb{R}$, every value is positive, and the graph increases as x increases.

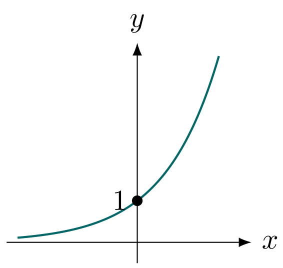
A periodic function. For

$$h(x) = \sin x,$$

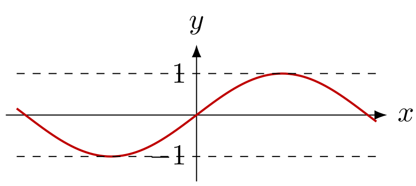
the values repeat every 2π. The function oscillates between -1 and 1.



$f(x) = x^2 - 4$



$g(x) = e^x$



$h(x) = \sin x$

Three graphs, three kinds of behavior. The parabola is organized by its zeros and sign, the exponential by positivity and persistent growth, and the sine function by bounded oscillation and repetition. These visible differences motivate the later questions of limits, derivatives, and integrals.

Questions before calculus. For an elementary function ask about its domain, values, zeros, sign, symmetry, and visible long-term behavior. Calculus will later make statements about change and limits precise.

The purpose of these examples is not to catalogue every possible function. They provide a small vocabulary rich enough to illustrate the central ideas of analysis without hiding them behind difficult algebra.

9.4 Transforming and Combining Functions

9.4.1 Transformations of graphs

A known graph can generate a family of new graphs without starting from zero. If the graph of $y = f(x)$ is known, then

$$y = f(x) + k$$

shifts it vertically by k, while

$$y = f(x - h)$$

shifts it horizontally by h. Multiplication changes scale:

$$y = af(x)$$

stretches or compresses vertical distances, and a negative value of a also reflects the graph across the horizontal axis.

Building graphs from x^2. Starting from $f(x) = x^2$, the function

$$g(x) = -(x-2)^2 + 3$$

is obtained by shifting the parabola two units to the right, reflecting it across the horizontal axis, and shifting it three units upward. Its vertex is therefore $(2, 3)$, and it opens downward.

Why transformations matter. The purpose is not to memorize isolated pictures. Transformations let us read parameters in a formula as geometric instructions. This becomes useful later when parameters change the position of zeros, extrema, or asymptotes.

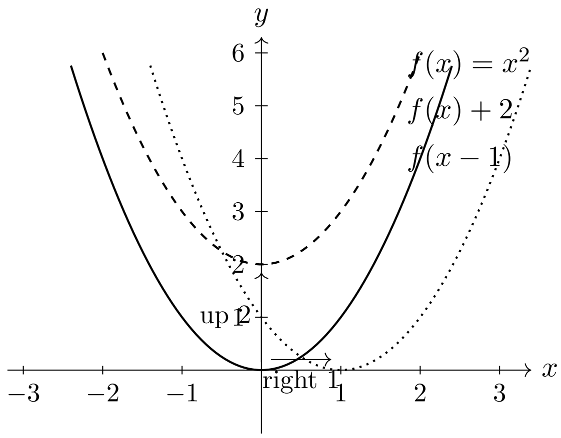
Graph transformations. Separate horizontal changes inside the argument from vertical changes outside the function. State each transformation and then describe its visible effect on the graph.

9.4.2 Seeing transformations on one coordinate plane

The formulas for graph transformations should be connected with a visible motion of the graph. The diagram below starts from

$$f(x) = x^2$$

and compares it with a vertical shift and a horizontal shift.



Read the motion before memorising the formula. Adding outside the function changes output values and therefore moves the graph vertically. Changing the input inside the function moves the graph horizontally, with the direction reversed in the notation: $f(x-1)$ moves the graph one unit to the right.

9.4.3 Composition of functions

Functions can be applied in sequence. If g acts first and f acts second, the resulting function is

$$(f \circ g)(x) = f(g(x)).$$

The order is part of the definition.

Two orders, two results. Let

$$f(x) = x^2, \qquad g(x) = x + 1.$$

Then

$$(f \circ g)(x) = (x+1)^2,$$

whereas

$$(g \circ f)(x) = x^2 + 1.$$

These functions are different. At $x = 1$, the first gives 4, while the second gives 2.

Domain must be checked. For $f(g(x))$ to be defined, the input x must belong to the domain of g, and the output $g(x)$ must belong to the domain of f. Composition is therefore both an algebraic and a domain question.

A model as a chain. Composition describes multi-stage dependence: time may determine temperature, and temperature may determine length; position may determine density, and density may determine mass per unit length. The composed function records the whole chain.

9.5 Inverse Functions and Reversing Assignments

9.5.1 When a function can be reversed

An inverse function reverses an assignment. If f sends x to y, then f^{-1} should send y back to x. This is possible only when different allowed inputs do not produce the same output.

A function is one-to-one on its domain if

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

Only a one-to-one function can have an inverse on its entire range.

Why x^2 needs a restricted domain. On all real numbers, $f(x) = x^2$ is not one-to-one because $f(2) = f(-2) = 4$. On the restricted domain $[0, \infty)$, however, every output has one non-negative input, so the inverse is $f^{-1}(y) = \sqrt{y}$.

Horizontal-line test. A graph represents a one-to-one function precisely when every horizontal line meets it at most once. The test asks whether an output identifies a unique input.

9.5.2 Finding and checking an inverse

For an elementary formula, solve $y = f(x)$ for x, then exchange the roles of input and output.

Inverse of an affine function. Let $f(x) = 3x - 2$. From $y = 3x - 2$ we obtain

$$x = \frac{y+2}{3}.$$

Hence

$$f^{-1}(x) = \frac{x+2}{3}.$$

The check is composition:

$$f^{-1}(f(x)) = \frac{3x-2+2}{3} = x, \qquad f(f^{-1}(x)) = 3\frac{x+2}{3} - 2 = x.$$

Remark. The notation f^{-1} means inverse function, not reciprocal. In general,

$$f^{-1}(x) \neq \frac{1}{f(x)}.$$

9.6 Sequences as Discrete Functions

9.6.1 Explicit rules, recursion, and long-term behavior

A real sequence is a function

$$a : \mathbb{N} \to \mathbb{R}.$$

Its value at the input n is usually written a_n rather than $a(n)$.

A sequence may be given explicitly, for example

$$a_n = \frac{1}{n},$$

or recursively, for example

$$a_1 = 1, \qquad a_{n+1} = \frac{1}{2}a_n.$$

Reading an explicit sequence. For $a_n = 1/n$, the first terms are

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$$

They remain positive and get closer to zero as n grows.

Reading a recursive sequence. If $a_1 = 8$ and $a_{n+1} = a_n/2$, then

$$8, 4, 2, 1, \frac{1}{2}, \ldots$$

Each term is determined from the previous one.

Discrete input. A sequence does not fill an interval of inputs. Its graph consists of separated points indexed by natural numbers. This is why the long-term question is phrased as $n \to \infty$.

9.7 Structured Sequences

9.7.1 Arithmetic and geometric sequences

Two elementary families already show different kinds of repeated change.

> An arithmetic sequence has a constant difference:
>
> $$a_{n+1} - a_n = d.$$
>
> If the first term is a_1, then
>
> $$a_n = a_1 + (n-1)d.$$

Constant additive change. For $a_1 = 4$ and $d = 3$,

$$4, 7, 10, 13, \ldots$$

and

$$a_n = 4 + 3(n-1) = 3n + 1.$$

The index increases by one while the value increases by three.

> A geometric sequence has a constant ratio:
>
> $$\frac{a_{n+1}}{a_n} = q$$
>
> when the quotient is defined. Its explicit form is
>
> $$a_n = a_1 q^{n-1}.$$

Constant multiplicative change. For $a_1 = 8$ and $q = \frac{1}{2}$,

$$8, 4, 2, 1, \frac{1}{2}, \ldots$$

and

$$a_n = 8\left(\frac{1}{2}\right)^{n-1}.$$

This sequence models repeated proportional decay.

Two models of change. Arithmetic sequences describe repeated addition. Geometric sequences describe repeated multiplication. This distinction later becomes the contrast between linear and exponential behavior.

9.7.2 Monotonicity and boundedness

Before asking whether a sequence has a limit, we can ask how its terms are ordered and whether they remain inside a fixed range.

> A sequence is increasing if $a_{n+1} \geq a_n$ for every relevant n, and decreasing if $a_{n+1} \leq a_n$. It is monotone if it is increasing or decreasing.

> A sequence is bounded above if there is a number M such that $a_n \leq M$ for all n. It is bounded below if there is a number m such that $m \leq a_n$ for all n.

A decreasing bounded sequence. For

$$a_n = \frac{1}{n},$$

we have $a_{n+1} < a_n$, so the sequence is decreasing. Also $0 < a_n \leq 1$, so it is bounded below by 0 and above by 1.

An increasing unbounded sequence. For

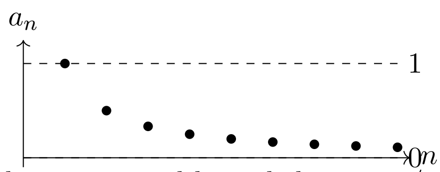
$$b_n = n,$$

we have $b_{n+1} > b_n$, so the sequence is increasing. It is bounded below, but it is not bounded above.

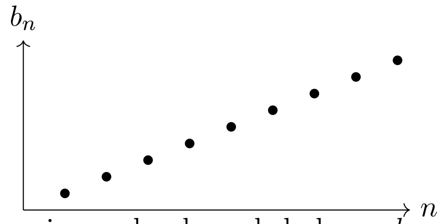
Bounded but not monotone. The sequence

$$c_n = (-1)^n$$

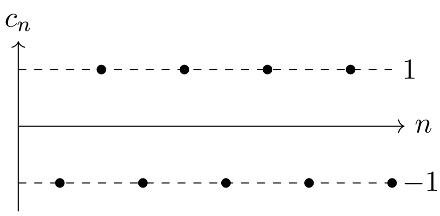
alternates between -1 and 1. It is bounded, but it is neither increasing nor decreasing.



decreasing and bounded: $a_n = 1/n$



increasing and unbounded above: $b_n = n$



bounded but not monotone: $c_n = (-1)^n$

Reading the three pictures. The first sequence moves in one direction and remains trapped between two horizontal levels. The second also moves in one direction, but it escapes every proposed upper bound. The third remains trapped, yet repeatedly changes direction. The pictures show that monotonicity and boundedness answer different questions.

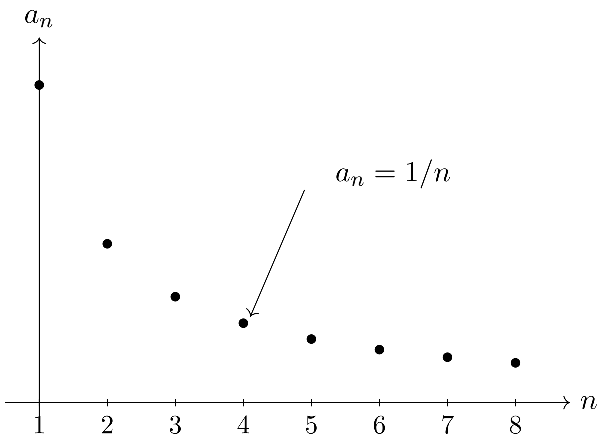
Preparing the limit question. Monotonicity describes direction; boundedness describes confinement. Together they give strong information about long-term behavior, but neither property alone is the same as convergence.

9.7.3 A sequence is drawn as separate points

A sequence has the natural numbers as inputs, so its graph is not a continuous curve. For

$$a_n = \frac{1}{n},$$

the terms form isolated points that move closer to the horizontal axis.



Discrete approach. The points do not fill the spaces between consecutive integers. Nevertheless, their heights approach zero. This is the first visual model for the statement $a_n \to 0$ as $n \to \infty$.

9.8 Questions That Lead to Analysis

9.8.1 From values to behavior

For a real function we may ask where it is positive, where it increases, what happens near a point, and what happens for very large inputs. For a sequence we ask whether it is bounded, monotone, or approaching a limiting value.

The domain shapes the question. Real variables can approach a finite point through nearby real values. Sequence indices move through natural numbers and can only tend toward infinity. The common language is functional, but the manner of approaching is different.

Two approaching questions. The expressions

$$\lim_{x\to 1} f(x) \qquad \text{and} \qquad \lim_{n\to\infty} a_n$$

ask about approached behavior. In the first, a real input approaches a finite point. In the second, a discrete index grows without bound.

Preparation for limits. A function assigns outputs to inputs, its domain is part of the object, its graph records input-output pairs, and a sequence is a function on a discrete domain. These ideas provide the language needed for limits and continuity.

Functions and sequences. Functions organise dependence between quantities. Their domains, representations, transformations, compositions, and possible inverses determine how they can be read. Sequences are functions on discrete domains; arithmetic and geometric patterns, monotonicity, and boundedness prepare the central question of analysis: what behaviour is approached as an input moves toward a point or grows without bound?

9.9 Review Exercises

9.9.1 Elementary computational and graphical exercises

1. For $f(x) = x^2 - 4x + 3$, compute $f(-1)$, $f(0)$, $f(2)$, and $f(5)$. Find the zeros and determine where the function is positive or negative.
2. Determine the natural domain of
$$f(x) = \frac{1}{x-3}, \qquad g(x) = \sqrt{x+2}, \qquad h(x) = \frac{\sqrt{5-x}}{x+1}.$$
3. Prepare a value table for $f(x) = |x-1|$ at $x = -2, -1, 0, 1, 2, 3, 4$ and sketch the graph.
4. Starting from $y = x^2$, sketch
$$y = (x-2)^2 + 1, \qquad y = -x^2, \qquad y = 2x^2,$$
and state each transformation.
5. Let $f(x) = 2x - 1$ and $g(x) = x^2 + 3$. Compute $f \circ g$, $g \circ f$, and both values at $x = 2$.
6. Find the inverse of $f(x) = 3x - 5$ and verify both compositions.
7. An arithmetic sequence has $a_1 = 4$ and $d = -3$. Find a_2, a_3, a_{10} and an explicit formula.
8. A geometric sequence has $b_1 = 2$ and $q = \frac{1}{2}$. Find b_2, b_3, b_6 and an explicit formula.
9. For $a_1 = 1$ and $a_{n+1} = 2a_n + 1$, compute the first six terms and plot (n, a_n).
10. Decide whether each sequence is increasing, decreasing, bounded above, or bounded below:
$$\frac{1}{n}, \qquad n - 3, \qquad (-1)^n, \qquad 2 - \frac{1}{n}.$$
Justify each answer briefly.

9.9.2 Development and reasoning exercises

1. For
$$f(x) = \frac{x+1}{x-2},$$
determine the domain, make a small value table on both sides of $x = 2$, find the zero and axis intercepts, and sketch the main features. Explain which information came from algebra and which from the table.
2. Let
$$f(x) = x^2 - 4, \qquad g(x) = 2x + 1.$$
Find the domains and formulas for $f \circ g$ and $g \circ f$, solve $(f \circ g)(x) = 0$, and explain why the order of composition matters.

3. The function $f(x) = x^2$ is not one-to-one on $\mathbb{R}$. Restrict its domain in two different natural ways, find the corresponding inverse functions, and compare their graphs.

4. A function is described by
$$f(x) = |x - 2| - 1.$$
Produce a table, graph, zeros, sign intervals, minimum value, and a description of the transformations from $|x|$.

5. The arithmetic sequence satisfies $a_4 = 11$ and $a_{10} = 29$. Find a_1, the common difference, and the explicit formula. Explain how two known terms determine the whole sequence.

6. The geometric sequence satisfies $b_2 = 6$ and $b_5 = 162$. Find all real possibilities for the ratio and determine the corresponding explicit formulas.

7. Starting from the recursion
$$a_{n+1} = a_n + 3, \qquad a_1 = 2,$$
derive the explicit formula. Then prove it by induction and explain the roles of the recursive and explicit descriptions.

8. Prove directly that $a_n = 2 - 1/n$ is increasing and bounded above. Identify the inequalities that express direction and confinement.

9. Explain why a sequence is a function on a discrete domain. Compare a graph of isolated sequence points with the graph of a real function and state which operations or questions remain meaningful in both settings.

10. Give a complete analysis of one elementary function and one elementary sequence: specify their domains, representations, transformations or recurrence, monotonicity, boundedness, and the long-term question that leads naturally to limits.

Chapter 10

Limits and Continuity

10.1 Limits of Sequences

10.1.1 The limit of a sequence

A sequence is the simplest setting in which the idea of a limit can be made precise. The input does not move continuously: it takes the values

$$1, 2, 3, \ldots$$

and we ask what happens to the terms far along the sequence.

> We write
> $$\lim_{n\to\infty} a_n = L$$
> when, for every $\varepsilon > 0$, there exists an index $N \in \mathbb{N}$ such that
> $$n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon.$$

The definition says more than "the terms get close to L." It says that after some index they remain inside every prescribed interval

$$(L - \varepsilon, L + \varepsilon).$$

The first terms may behave irregularly; convergence concerns the tail of the sequence.

The sequence $1/n$. Let

$$a_n = \frac{1}{n}.$$

To prove that $a_n \to 0$, let $\varepsilon > 0$. We need

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

It is enough to choose an integer $N > 1/\varepsilon$. Then every $n \geq N$ satisfies $1/n < \varepsilon$. Hence

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Dominant powers. For

$$a_n = \frac{2n+1}{n+3},$$

divide numerator and denominator by n:

$$a_n = \frac{2 + 1/n}{1 + 3/n}.$$

Since $1/n \to 0$, the numerator approaches 2 and the denominator approaches 1. Therefore

$$\lim_{n\to\infty} \frac{2n+1}{n+3} = 2.$$

Oscillation prevents convergence. The sequence

$$b_n = (-1)^n$$

alternates between -1 and 1. Its terms do not eventually remain close to one number, so the sequence has no limit.

Every convergent sequence has exactly one limit and is bounded.

These statements are structural. Uniqueness makes the notation $\lim a_n$ meaningful, while boundedness says that a convergent sequence cannot escape arbitrarily far away.

10.1.2 The Cauchy criterion

The definition of convergence refers to a candidate limit L. Sometimes we want to decide whether a sequence converges before knowing that limit. The Cauchy criterion compares late terms directly with one another.

A sequence (a_n) is a Cauchy sequence if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \quad \Longrightarrow \quad |a_m - a_n| < \varepsilon.$$

Thus, sufficiently late terms must be arbitrarily close to each other. The definition does not name the value they approach.

A real sequence converges if and only if it is a Cauchy sequence.

Why convergence implies the Cauchy property. Suppose $a_n \to L$. Given $\varepsilon > 0$, choose N so that

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \frac{\varepsilon}{2}.$$

Then for $m, n \geq N$, the triangle inequality gives

$$|a_m - a_n| \leq |a_m - L| + |a_n - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence every convergent sequence is Cauchy. $\square$

Proof status. Only the forward implication is proved here. The reverse implication uses completeness of the real numbers: a real Cauchy sequence cannot approach a missing point outside $\mathbb{R}$. We use that completeness result without developing its full construction.

A sequence that is not Cauchy. For $a_n = (-1)^n$, choose arbitrarily large even m and odd n. Then

$$|a_m - a_n| = |1 - (-1)| = 2.$$

The late terms do not become close to one another, so the sequence is not Cauchy and therefore does not converge.

Internal test for convergence. The ordinary definition compares terms with an external target L. The Cauchy criterion tests whether the tail has internally settled down enough that a limit must exist.

10.1.3 Monotone and bounded sequences

The previous chapter introduced monotonicity and boundedness before limits. We can now explain why that combination is so important.

Every increasing sequence that is bounded above converges. Every decreasing sequence that is bounded below converges.

For an increasing bounded sequence, the limit is the least upper bound of the set of its terms. For a decreasing bounded sequence, it is the greatest lower bound.

Proof status. The theorem rests on completeness of $\mathbb{R}$: every nonempty set bounded above has a least upper bound. The text explains how that boundary becomes the limit, but does not develop the completeness axiom itself.

A geometric sequence. Let

$$a_n = 1 - \frac{1}{2^n}.$$

The sequence is increasing because 2^{-n} decreases. It is bounded above by 1. Therefore it converges. Since $2^{-n} \to 0$, its limit is

$$\lim_{n\to\infty} \left(1 - \frac{1}{2^n}\right) = 1.$$

Why one condition is not enough. The sequence $a_n = n$ is increasing but not bounded above, and it does not converge to a real number. The sequence $b_n = (-1)^n$ is bounded but not monotone, and it also does not converge.

Direction plus confinement. Monotonicity prevents repeated reversal of direction. Boundedness prevents escape. Completeness supplies the boundary value toward which the terms are forced.

10.2 Function Limits Through Sequences

10.2.1 From sequence limits to function limits

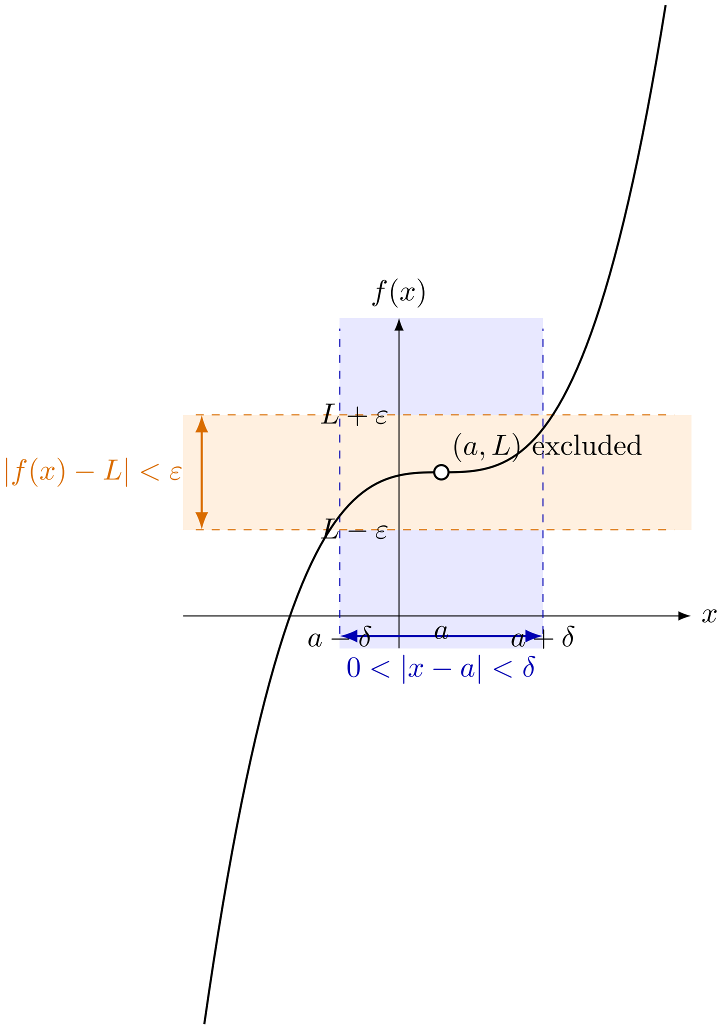
A sequence approaches its limit through its tail. For a function, the input may approach a point through infinitely many different values, so we first state the local definition that every route must satisfy.

Let $f\colon D \to \mathbb{R}$ and let a be an accumulation point of D. We write

$$\lim_{x\to a} f(x) = L$$

when, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every $x \in D$,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$



The ε–δ picture. First choose the permitted output error ε. The definition requires an input distance δ such that every admissible input in the punctured interval $(a-\delta, a+\delta)$ has its output inside the horizontal band $(L-\varepsilon, L+\varepsilon)$. The point $x = a$ itself is removed from the test.

The condition excludes $x = a$: a function limit is determined by nearby inputs, not by the value assigned at the point itself.

A function limit can also be tested by following the function along every sequence of inputs approaching the point. This gives the natural bridge from the discrete theory of sequences to local function behaviour.

Let a be an accumulation point of the domain of f. Then

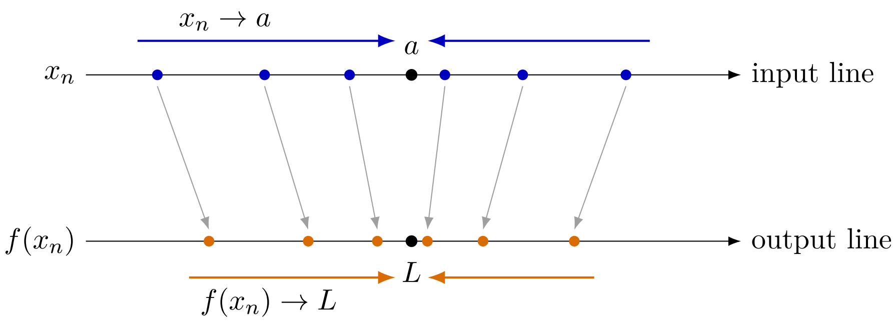
$$\lim_{x\to a} f(x) = L$$

if and only if, for every sequence (x_n) in the domain of f satisfying

$$x_n \neq a \qquad \text{and} \qquad x_n \to a,$$

we have

$$f(x_n) \to L.$$



The sequential picture. A sequence may approach a from the left, from the right, or by alternating between both sides. Heine's theorem requires every such admissible input sequence to produce output values converging to the same number L.

Proof status. The forward implication follows by applying the ε–δ definition to sufficiently late terms x_n. The reverse implication is proved by contradiction: if the local definition failed, one could choose inputs successively closer to a whose outputs remain a fixed distance from L. Those inputs would form a sequence contradicting the sequential condition. The full quantified proof is not expanded here.

Using Heine's theorem. Consider

$$f(x) = 2x + 1.$$

Take any sequence $x_n \to 3$. Then

$$f(x_n) = 2x_n + 1 \to 2 \cdot 3 + 1 = 7.$$

Because the conclusion holds for every sequence approaching 3,

$$\lim_{x \to 3} (2x + 1) = 7.$$

How to disprove a function limit. For

$$f(x) = \frac{x}{|x|}, \qquad x \neq 0,$$

choose

$$x_n = \frac{1}{n} \qquad \text{and} \qquad y_n = -\frac{1}{n}.$$

Both sequences approach 0, but

$$f(x_n) = 1, \qquad f(y_n) = -1.$$

The output sequences have different limits, so $\lim_{x \to 0} f(x)$ does not exist.

A function limit must survive every route. The local definition controls all sufficiently near inputs at once. Heine's theorem expresses the same requirement through every admissible approaching sequence. One chosen sequence can disprove a limit when it behaves differently, but it cannot prove a limit by itself.

10.2.2 The Cauchy criterion for a function limit

Heine's theorem translates a function limit into convergence of output sequences. The corresponding Cauchy criterion removes the need to name the candidate limit in advance.

Let $f\colon D \to \mathbb{R}$, where $D \subseteq \mathbb{R}$, and let a be an accumulation point of D. A finite limit

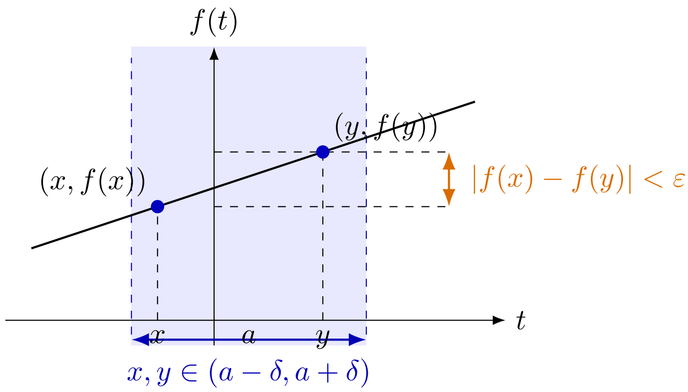
$$\lim_{x\to a} f(x)$$

exists if and only if, for every $\varepsilon > 0$, there is $\delta > 0$ such that whenever $x, y \in D$ satisfy

$$0 < |x-a| < \delta, \qquad 0 < |y-a| < \delta,$$

we have

$$|f(x) - f(y)| < \varepsilon.$$



The Cauchy picture. The criterion does not begin by naming a target value L. It asks whether any two outputs obtained from sufficiently near admissible inputs must be close to each other. Shrinking the input neighbourhood must make the entire collection of nearby outputs fit into an arbitrarily small vertical spread.

Proof status. The criterion is the function analogue of the Cauchy criterion for sequences and ultimately uses completeness of $\mathbb{R}$. We use the equivalence and explain its meaning, but do not repeat the full completeness argument.

The criterion says that all function values sufficiently near a must be close to one another. It does not initially specify the number to which they converge.

A jump fails the Cauchy test. Let

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

No matter how small δ is, we can choose $x < 0 < y$ with

$$|x| < \delta, \qquad |y| < \delta.$$

Then

$$|f(x) - f(y)| = |-1-1| = 2.$$

The nearby values do not become mutually close, so the limit at 0 does not exist.

Heine and Cauchy describe the same settling. Heine follows every approaching input sequence. Cauchy compares nearby output values directly. Both formulations express that all admissible approaches must settle into one common behaviour.

10.2.3 The value at a point and the nearby limit

After the local and sequential descriptions of a function limit have been connected, we can separate two questions that concern a function near a point.

Two different questions. The value

$$f(a)$$

asks what the function assigns at the input a. The limit

$$\lim_{x\to a} f(x)$$

asks what values are approached at nearby inputs, with $x \neq a$.

By Heine's theorem, the nearby limit can be tested along every admissible sequence $x_n \to a$ with $x_n \neq a$. The value $f(a)$ is not part of that test.

A removable hole. For

$$f(x) = \frac{x^2-1}{x-1}, \qquad x \neq 1,$$

we have

$$f(x) = x + 1$$

for every admissible x. Therefore, for every sequence $x_n \to 1$ with $x_n \neq 1$,

$$f(x_n) = x_n + 1 \to 2.$$

Hence

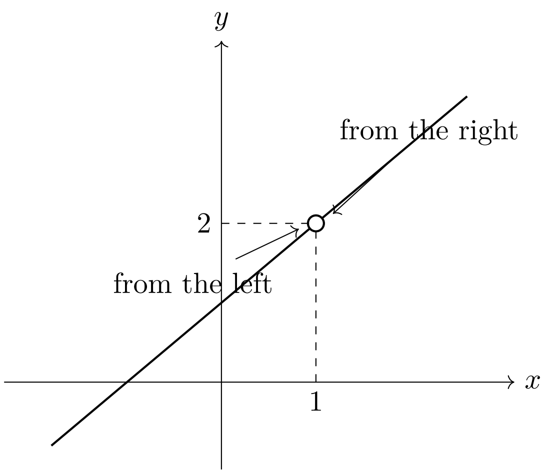
$$\lim_{x\to 1} f(x) = 2,$$

even though $f(1)$ is not defined.

The point itself is excluded from the limit test. Changing, deleting, or redefining the single value $f(a)$ does not change $\lim_{x\to a} f(x)$. The limit is determined by values at nearby inputs.

10.2.4 A hole does not determine the nearby limit

Consider the function that agrees with $y = x + 1$ near $x = 1$ but is not defined at $x = 1$. The graph has a hole at $(1, 2)$, yet every sequence of inputs approaching 1 produces output values approaching 2.



Sequential reading of the graph. Points may approach $x = 1$ from either side or alternate between the two sides. In every case their heights on the line approach 2. This is the geometric content of Heine's theorem in this example.

10.3 Elementary Limit Methods

10.3.1 Substitution, cancellation, and dominant terms

Use the simplest method that reveals the behavior. First try direct substitution. If it produces a meaningful value, the calculation is finished. If it produces an indeterminate form such as $0/0$, transform the expression only as much as needed.

Factoring and cancellation.

$$\lim_{x\to 4}\frac{x^2-2x-8}{x^2-9x+20}=\lim_{x\to 4}\frac{(x-4)(x+2)}{(x-4)(x-5)}=\lim_{x\to 4}\frac{x+2}{x-5}=-6.$$

The cancelled factor records a removable hole, not permission to substitute into the original expression.

A simple radical. For

$$\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x},$$

multiply by the conjugate:

$$\frac{\sqrt{1+x}-1}{x}\cdot\frac{\sqrt{1+x}+1}{\sqrt{1+x}+1}=\frac{1}{\sqrt{1+x}+1}\to\frac{1}{2}.$$

Behavior at infinity.

$$\lim_{x\to\infty}\frac{3x^2+x}{x^2+4}=\lim_{x\to\infty}\frac{3+1/x}{1+4/x^2}=3.$$

Remark. Forms such as $0/0$ are not answers. They indicate that direct substitution has not yet exposed the behavior.

10.4 Limit Laws and Standard Patterns

10.4.1 Algebra of limits

When individual limits exist, sums, differences, products, and quotients can often be handled term by term.

If

$$\lim_{x\to a} f(x)=L,\qquad \lim_{x\to a} g(x)=M,$$

then

$$\lim_{x\to a}(f+g)=L+M,\qquad \lim_{x\to a}(fg)=LM,$$

and, when $M\neq 0$,

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{L}{M}.$$

The same principles hold for sequence limits.

Using laws before using tricks. For

$$\lim_{x\to 2}\frac{x^2+3x}{x+1},$$

the denominator tends to $3 \neq 0$, so direct use of the laws gives

$$\frac{2^2 + 3 \cdot 2}{2 + 1} = \frac{10}{3}.$$

No transformation is needed.

Remark. The expressions $0/0$, ∞/∞, and $\infty - \infty$ are not results. They indicate that the elementary laws cannot be applied directly and that the expression must be reorganized.

10.4.2 The basic trigonometric limit

A central elementary limit is

$$\lim_{x \to 0} \frac{\sin x}{x} = 1,$$

with angles measured in radians.

What the limit says. Near the origin, the functions $\sin x$ and x are approximately equal. Their ratio approaches 1. This is the local fact that later produces the derivative of sine.

A simple transformed limit. Compute

$$\lim_{x \to 0} \frac{\sin(3x)}{x}.$$

Write

$$\frac{\sin(3x)}{x} = 3\frac{\sin(3x)}{3x}.$$

As $x \to 0$, also $3x \to 0$, so

$$\lim_{x \to 0} \frac{\sin(3x)}{x} = 3.$$

Remark. The purpose is not to build a catalogue of difficult trigonometric limits. One standard pattern is enough to show how a local approximation can become a computational tool.

10.5 One-Sided Limits and Infinite Behavior

10.5.1 Approaching from a side and growing without bound

The symbols

$$\lim_{x \to a^-} f(x) \qquad \text{and} \qquad \lim_{x \to a^+} f(x)$$

describe approach from values smaller and larger than a, respectively.

A two-sided finite limit exists only when both one-sided limits exist and are equal.

A jump. For

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0, \end{cases}$$

we have

$$\lim_{x \to 0^-} f(x) = 0, \qquad \lim_{x \to 0^+} f(x) = 1.$$

Therefore the two-sided limit at zero does not exist.

A vertical asymptote. For $f(x) = 1/x$,

$$\lim_{x \to 0^+} \frac{1}{x} = \infty, \qquad \lim_{x \to 0^-} \frac{1}{x} = -\infty.$$

The line $x = 0$ is a vertical asymptote.

Infinite limits. The symbol ∞ does not denote a real output value. It records unbounded growth of the function values.

10.6 Continuity and Discontinuity

10.6.1 Continuity as agreement between value and limit

> A function f is continuous at a when $f(a)$ is defined, the limit $\lim_{x\to a} f(x)$ exists, and
>
> $$\lim_{x\to a} f(x) = f(a).$$

No mismatch at the point. Continuity means that nearby values approach the value assigned at the point. A removable hole, a jump, or unbounded behavior breaks this agreement in a different way.

Repairing a removable discontinuity. Let

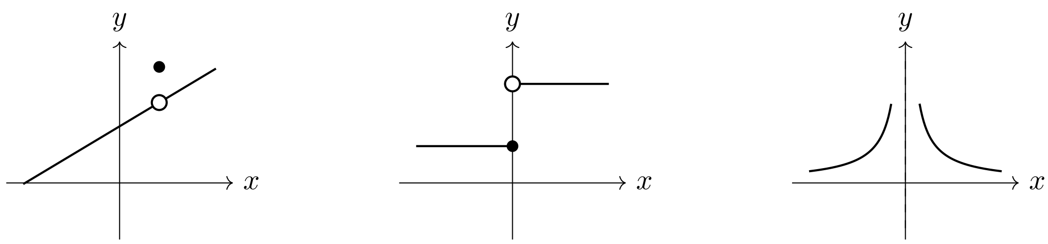
$$f(x) = \frac{x^2 - 1}{x - 1} \quad (x \neq 1).$$

Since the limit at 1 equals 2, defining $f(1) = 2$ fills the hole and produces a continuous function.

Choosing a limit method. Identify the type of approach, try direct substitution, transform only when necessary, compare one-sided limits when formulas differ, and test continuity by matching the limit with the function value.

10.6.2 Three visible ways continuity can fail

The pictures below separate three common phenomena. The algebraic formulas may differ, but the graph reveals the type of failure immediately.



removable   jump   infinite

Classify the failure geometrically. A removable discontinuity has a common nearby height but the wrong or missing value. A jump has different one-sided limits. An infinite discontinuity occurs when values become unbounded near a vertical line.

10.7 Consequences of Continuity

10.7.1 Asymptotes and global behavior

Limits describe not only local holes and jumps but also persistent geometric behavior.

The line $x = a$ is a vertical asymptote when at least one one-sided limit of $f(x)$ as $x \to a$ is infinite. The line $y = L$ is a horizontal asymptote when

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L.$$

Two asymptotes from one simple function. For

$$f(x) = \frac{x+1}{x-1},$$

the denominator vanishes at $x = 1$, while the numerator tends to 2, so $x = 1$ is a vertical asymptote. Dividing numerator and denominator by x gives

$$\frac{1 + 1/x}{1 - 1/x} \to 1,$$

so $y = 1$ is a horizontal asymptote.

An asymptote is a behavior statement. An asymptote does not mean that a graph can never cross the line. It means that a particular limiting relationship holds near a boundary point or far from the origin.

10.7.2 The intermediate value theorem

Continuity has consequences that go beyond evaluating limits.

If f is continuous on $[a, b]$, then every value between $f(a)$ and $f(b)$ is attained at some point of the interval.

Proof status. The geometric mechanism is that a continuous graph cannot jump over an intermediate level. A full proof uses completeness of $\mathbb{R}$, for example through nested intervals or a least-upper-bound argument, and is not developed here.

Proving that a root exists. Let

$$f(x) = x^3 - x - 1.$$

The function is continuous everywhere. Moreover,

$$f(1) = -1, \qquad f(2) = 5.$$

Since 0 lies between -1 and 5, there is at least one $c \in (1, 2)$ such that

$$f(c) = 0.$$

This argument proves existence without finding an exact formula for the root.

Continuity prevents jumps over values. The theorem turns a visible property of an unbroken graph into a precise existence statement. Its purpose is not to compute the solution but to guarantee that a solution must occur.

Remark. The theorem guarantees at least one solution, not uniqueness. Additional information, such as monotonicity, is needed to prove that the solution is unique.

Limits and continuity. Sequence limits provide the basic language of convergence. Heine's theorem transfers this language to functions: every admissible approaching sequence must produce the same output limit. Cauchy criteria and existence theorems are deeper structural results; algebraic techniques, one-sided limits, asymptotes, and continuity are applications of that structure rather than the starting point.

10.8 Review Exercises

10.8.1 Elementary computational exercises

1. Compute the first six terms and determine the limit of

$$a_n = \frac{1}{n}, \qquad b_n = 2 - \frac{1}{n}, \qquad c_n = \left(\frac{1}{2}\right)^n.$$

   State monotonicity and boundedness.

2. For $a_n = 1/n$, find N such that $n \geq N$ implies $|a_n| < 0.01$. Then give a suitable formula for arbitrary $\varepsilon > 0$.

3. Compute

$$\lim_{n\to\infty} \frac{3n+1}{n+4}, \qquad \lim_{n\to\infty} \frac{2n^2-n}{5n^2+3}, \qquad \lim_{n\to\infty} \frac{4}{n^2+1}.$$

4. For $a_n = 1 - 2^{-n}$, show that the sequence is increasing and bounded above, then calculate its limit.

5. For $f(x) = 2x+1$, $a = 3$, and $L = 7$, find δ in terms of ε so that

$$0 < |x-3| < \delta \implies |f(x) - 7| < \varepsilon.$$

6. Let $x_n = 3 + (-1)^n/n$. Compute $f(x_n)$ for $f(x) = 2x+1$ and use the result to illustrate Heine's theorem.

7. Compute by factoring and cancellation:

$$\lim_{x\to 4} \frac{x^2-2x-8}{x^2-9x+20}, \qquad \lim_{x\to 2} \frac{x^2-4}{x-2}.$$

8. Compute

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}$$

   using the conjugate and

$$\lim_{x\to 0} \frac{\sin x}{x}$$

   using the standard limit.

9. For

$$f(x) = \begin{cases} x+1, & x < 0, \\ 2-x, & x > 0, \end{cases}$$

   compute the one-sided limits at 0 and decide whether a value of $f(0)$ can make the function continuous.

10. Find the vertical and horizontal asymptotes of

$$f(x) = \frac{1}{x-2}, \qquad g(x) = \frac{3x^2+x}{x^2+4}$$

    from the relevant limits.

10.8.2 Development and reasoning exercises

1. Prove from the ε–N definition that
$$\frac{3}{2n+1} \longrightarrow 0.$$
Give an explicit $N(\varepsilon)$ and explain every inequality.

2. Prove that
$$\frac{2n+1}{n+3} \longrightarrow 2$$
using the formal definition, not only dominant terms.

3. For $f(x) = x^2$, prove
$$\lim_{x\to 2} x^2 = 4$$
with an ε–δ argument. First control $|x+2|$ by restricting δ, then choose a complete formula for $\delta(\varepsilon)$.

4. Use Heine's theorem to disprove the existence of
$$\lim_{x\to 0} \frac{x}{|x|}.$$
Construct two sequences, compute both output sequences, and state precisely where the common-limit requirement fails.

5. Investigate the Cauchy condition for
$$f(x) = \sin(1/x), \qquad x \neq 0.$$
Choose nearby sequences whose function values remain separated and explain why no finite limit exists at 0.

6. Determine constants a, b so that
$$f(x) = \begin{cases} ax + b, & x < 1, \\ x^2, & x \geq 1 \end{cases}$$
is continuous at 1 and also satisfies $f(0) = 2$. Show the two conditions separately.

7. Analyse
$$f(x) = \frac{x^2 - 1}{x - 1}$$
at $x = 1$: calculate the nearby limit, identify the discontinuity, define a continuous extension, and explain why changing one point does not alter the limit.

8. Use the intermediate value theorem to prove that $x^3 - x - 1 = 0$ has a solution in $(1, 2)$. Then use monotonicity on a suitable interval to discuss uniqueness.

9. Compare the ε–δ, Heine, and Cauchy descriptions of a function limit. Apply all three viewpoints to the same simple linear function and identify what each formulation controls.

10. Give a complete limit-analysis strategy: determine the domain, inspect direct substitution, simplify when necessary, compare one-sided behaviour, use sequences to test failure, identify asymptotes, and decide continuity. Apply the strategy to one rational or piecewise function.

Chapter 11

Derivatives and Function Analysis

11.1 Rates of Change and Tangent Lines

11.1.1 From average change to instantaneous change

If the input changes from a to $a+h$, then the output changes by

$$f(a+h) - f(a).$$

The quotient

$$\frac{f(a+h) - f(a)}{h}$$

is the average rate of change over that interval.

> The derivative of f at a is
> $$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$
> provided the limit exists.

The derivative of x^2. For $f(x) = x^2$,

$$\frac{(a+h)^2 - a^2}{h} = 2a + h,$$

so

$$f'(a) = 2a.$$

Slope and local change. The difference quotient is the slope of a secant line. As $h \to 0$, the secant approaches the tangent, and the derivative gives its slope. In applications the same number represents an instantaneous rate, such as velocity or growth rate.

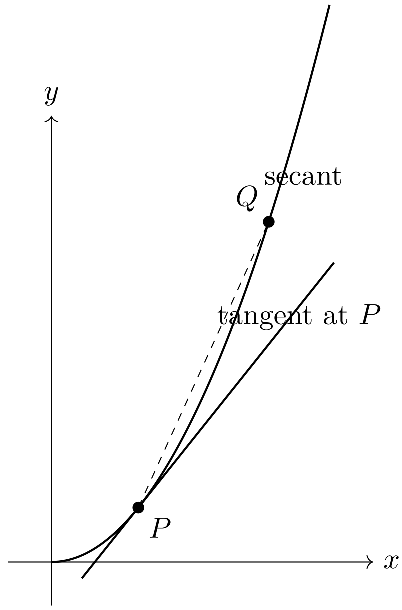
Velocity. If position is $s(t) = t^2$, then

$$v(t) = s'(t) = 2t.$$

At time $t = 3$, the instantaneous velocity is 6 units of distance per unit of time.

11.1.2 From a secant line to a tangent line

For $f(x) = x^2$, the average rate of change between $x = 1$ and $x = 1 + h$ is the slope of a secant line. As h becomes smaller, the second point approaches the first and the secant approaches the tangent.



The derivative is a limiting slope. The secant uses two points and gives an average rate of change. The tangent records the limiting position of those secants and gives the instantaneous rate of change at one point.

11.2 Differentiability and Local Geometry

11.2.1 Continuity and failure of differentiability

A function may be continuous at a point and still fail to have a derivative there.

> If f is differentiable at a, then f is continuous at a.

The converse is false.

A corner. For

$$f(x) = |x|,$$

the function is continuous at 0, but the left-hand slope is -1 and the right-hand slope is 1. Since the one-sided difference quotients do not agree, $f'(0)$ does not exist.

A vertical tangent. For

$$f(x) = x^{1/3},$$

the difference quotient near 0 becomes unbounded. The graph has a vertical tangent rather than a finite tangent slope, so the ordinary derivative at 0 is not a real number.

What can prevent a derivative. Non-differentiability may come from a discontinuity, a corner, a cusp, or an infinite slope. These are geometrically different failures, and the graph should be read before a formula is applied mechanically.

11.2.2 Tangent lines, normal lines, and units

If f is differentiable at $x = a$, the tangent line is

$$y - f(a) = f'(a)(x - a).$$

When $f'(a) \neq 0$, the normal line has slope $-1/f'(a)$.

Tangent and normal to a parabola. For $f(x) = x^2$ at $a = 1$,

$$f(1) = 1, \qquad f'(1) = 2.$$

Hence the tangent is

$$y - 1 = 2(x - 1),$$

and the normal is

$$y - 1 = -\frac{1}{2}(x - 1).$$

Units of a derivative. If x is measured in seconds and $f(x)$ in metres, then $f'(x)$ is measured in metres per second. A derivative always has units of output divided by units of input. This is essential in applications: it distinguishes position, velocity, acceleration, concentration change, and marginal cost.

Remark. A numerical derivative without units may be mathematically correct but incomplete as an applied conclusion.

11.3 Elementary Derivative Rules

11.3.1 Rules sufficient for elementary models

The computational goal is deliberately limited. We need enough rules to differentiate low-degree polynomials, exponentials, sine and cosine, and simple compositions.

For differentiable functions,

$$(c)' = 0, \qquad (x^n)' = nx^{n-1}, \qquad (cf)' = cf', \qquad (f + g)' = f' + g'.$$

We also use

$$(e^x)' = e^x, \qquad (\sin x)' = \cos x, \qquad (\cos x)' = -\sin x.$$

A polynomial. For

$$f(x) = x^3 - 3x,$$

we obtain

$$f'(x) = 3x^2 - 3.$$

A simple oscillation. If

$$y(t) = 2\sin t,$$

then

$$y'(t) = 2\cos t.$$

The derivative describes how rapidly the oscillating quantity changes.

$$(fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2},$$

and

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

An elementary chain rule. For

$$f(x) = (x^2 + 1)^3,$$
$$f'(x) = 3(x^2 + 1)^2 \cdot 2x.$$

The example is chosen to display the nested structure, not to create difficult algebra.

11.4 Monotonicity, Critical Points, and Extrema

11.4.1 Reading change and solving elementary optimisation problems

> Let f be continuous on an interval and differentiable in its interior. If $f'(x) > 0$ throughout the interior, then f is increasing. If $f'(x) < 0$ throughout the interior, then f is decreasing.

Why local slopes control an interval. For two points $a < b$, the mean value theorem gives a point $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

If every derivative value is positive, then the secant slope is positive and therefore $f(b) > f(a)$. This is the bridge from local derivative information to global monotonicity. The full proof of the mean value theorem is not developed here.

A critical point occurs where $f'(x) = 0$ or where the derivative does not exist. A critical point is only a candidate. A change from positive to negative derivative gives a local maximum; a change from negative to positive gives a local minimum.

Reading a cubic. For

$$f(x) = x^3 - 3x, \qquad f'(x) = 3(x - 1)(x + 1).$$

The derivative is positive on $(-\infty, -1)$, negative on $(-1, 1)$, and positive on $(1, \infty)$. Hence f has a local maximum at $x = -1$ and a local minimum at $x = 1$.

Optimisation workflow. Choose the quantity to optimise, express it as a function of one variable, determine the allowed domain, find critical points, check endpoints when relevant, and interpret the result in the original problem.

Maximum area with fixed perimeter. A rectangle has perimeter 20. If one side is x, the other is $10 - x$, so

$$A(x) = x(10 - x) = 10x - x^2, \qquad 0 \le x \le 10.$$

Then

$$A'(x) = 10 - 2x,$$

so the only interior critical point is $x = 5$. The endpoint areas are zero, while $A(5) = 25$. Therefore the maximum occurs for a 5×5 square.

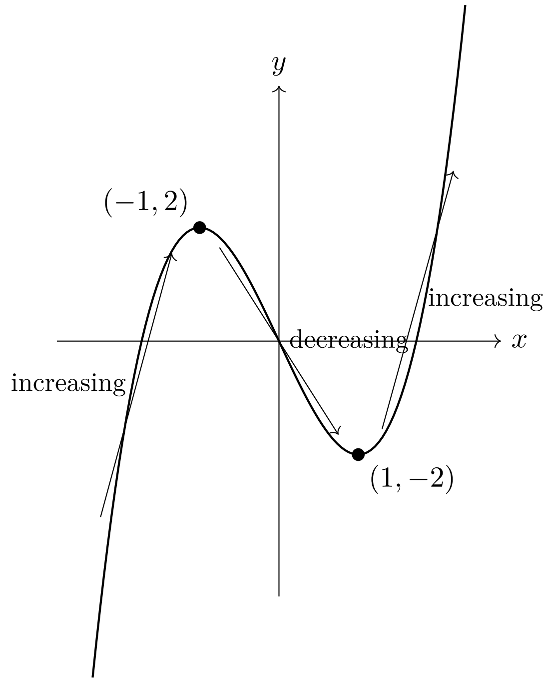
The derivative calculation is short. The substantive work is identifying the correct function, its restrictions, the relevant candidates, and the meaning of the result.

11.4.2 Monotonicity and extrema on the graph

The cubic

$$f(x) = x^3 - 3x$$

has a local maximum at $x = -1$ and a local minimum at $x = 1$. These points divide the graph into increasing and decreasing intervals.



A critical point must be classified. The derivative is zero at both marked points, but the sign change determines the type: positive to negative gives a local maximum, while negative to positive gives a local minimum.

11.5 Second Derivative and Shape

11.5.1 Convexity, inflection, and acceleration

The second derivative measures how the first derivative changes.

> If $f''(x) > 0$ on an interval, the graph is convex there. If $f''(x) < 0$, the graph is concave there. A change of sign of f'' indicates a possible inflection point.

Shape of a cubic. For

$$f(x) = x^3 - 3x,$$

we have

$$f''(x) = 6x.$$

Thus the graph is concave for $x < 0$, convex for $x > 0$, and changes convexity at the origin.

Acceleration. If position is

$$s(t) = t^3 - 3t,$$

then velocity and acceleration are

$$v(t) = s'(t) = 3t^2 - 3, \qquad a(t) = s''(t) = 6t.$$

The sign of $a(t)$ tells us whether velocity is increasing or decreasing.

One calculation, two languages. For a graph, the second derivative describes changing slope and convexity. For motion, it describes changing velocity and acceleration. The mathematics is the same; the interpretation depends on the model.

11.6 The Derivative as Information

11.6.1 Reading the graphs of f and f'

The graph of the derivative records slopes of the original graph. Positive values of f' indicate increase, negative values indicate decrease, and zeros of f' indicate horizontal tangents or possible critical points.

From derivative sign to function shape. Suppose

$$f'(x) = (x + 1)(x - 2).$$

Then $f'(x) > 0$ for $x < -1$, $f'(x) < 0$ for $-1 < x < 2$, and $f'(x) > 0$ for $x > 2$. Therefore f increases, then decreases, then increases. The point $x = -1$ is a local maximum and $x = 2$ is a local minimum.

Reading in the opposite direction. A rising graph of f corresponds to $f' > 0$. A horizontal tangent corresponds to $f' = 0$. If the slopes of f become larger, then f' is increasing and $f'' > 0$. Thus graphs of f, f', and f'' are three views of the same changing object.

Remark. A zero of f' is a candidate for an extremum, not an automatic extremum. The sign of f' on both sides must be checked.

11.6.2 Elementary optimisation models

An optimisation problem has two parts: building the correct function and analyzing it on the allowed domain.

Maximum area with fixed perimeter. A rectangle has perimeter 20. If its width is x, then its length is $10 - x$, so

$$A(x) = x(10 - x) = 10x - x^2, \qquad 0 < x < 10.$$

We compute

$$A'(x) = 10 - 2x.$$

The critical point is $x = 5$. Since

$$A''(x) = -2 < 0,$$

this point gives a maximum. The optimal rectangle is a square of side 5, with area 25.

Minimum material in a simple open box model. An open box with square base has volume 32. If the base side is $x > 0$, then the height is $h = 32/x^2$. Its surface area is

$$S(x) = x^2 + 4xh = x^2 + \frac{128}{x}.$$

Thus

$$S'(x) = 2x - \frac{128}{x^2}.$$

The condition $S'(x) = 0$ gives $2x^3 = 128$, hence $x = 4$. The corresponding height is $h = 2$.

Why the model matters more than the derivative. The differentiation is elementary. The substantial reasoning lies in choosing a variable, expressing all quantities through it, identifying the admissible domain, and interpreting the critical point in the original problem.

11.7 Simple Growth and Decay Models

11.7.1 A rate proportional to the current amount

A simple differential model begins with a law of change rather than a ready-made formula.

Proportional growth law. If a quantity $P(t)$ changes at a rate proportional to its current value, then

$$P'(t) = kP(t).$$

Positive k models growth and negative k models decay.

Checking an exponential model. Let

$$P(t) = P_0 e^{kt}.$$

Then

$$P'(t) = kP_0 e^{kt} = kP(t),$$

so the exponential function satisfies the proportional-rate law.

The derivative states the mechanism. The equation $P' = kP$ says that the present rate depends on the present amount. The exponential formula is a function whose derivative reproduces precisely that mechanism.

The aim here is not a general course in differential equations. The example shows how a derivative can encode a modelling assumption and how a proposed function can be tested against it.

11.8 Complete Function Analysis

11.8.1 A coherent description from simple calculations

A complete analysis should connect domain, limits, derivatives, and conclusions rather than present isolated computations.

An elementary polynomial analysis. For

$$f(x) = x^3 - 3x,$$

the domain is $\mathbb{R}$ and the zeros are $0, \pm\sqrt{3}$. The derivative

$$f'(x) = 3(x-1)(x+1)$$

shows increase on $(-\infty, -1)$ and $(1, \infty)$ and decrease on $(-1, 1)$. Hence there is a local maximum at $(-1, 2)$ and a local minimum at $(1, -2)$. Since

$$f''(x) = 6x,$$

the graph changes convexity at $(0, 0)$. The leading term x^3 determines the opposite behaviors at the two ends.

A simple rational function. For

$$g(x) = \frac{x+1}{x-1},$$

the domain excludes $x = 1$. The limits identify the vertical asymptote $x = 1$ and horizontal asymptote $y = 1$. Since

$$g'(x) = \frac{-2}{(x-1)^2} < 0,$$

the function decreases on each interval of its domain.

Function analysis. Determine the domain, useful zeros and signs, limits at boundaries and infinity, asymptotes, first-derivative sign, extrema, second-derivative sign, and then write one coherent description of the graph or model.

The examples remain algebraically elementary so that assessment can test whether the student understands the conclusions carried by derivatives and limits.

Derivatives in analysis and applications. A derivative connects a local limit with tangent geometry, units, motion, growth, graph shape, and optimal design. The mean value theorem supplies the bridge from local derivative signs to monotonicity on an interval. The central skill is translating derivative information into conclusions while keeping the model, domain, and interpretation visible.

11.9 Review Exercises

11.9.1 Elementary computational exercises

1. For $f(x) = x^2 - 1$, compute the average rate of change on $[1, 3]$ and $[2, 2.5]$. Interpret both as secant slopes.

2. Use the difference quotient to compute the derivative of $f(x) = x^2$ at a general point a. Then find the tangent line at $(3, 9)$.

3. Differentiate
$$f(x) = 4x^3 - 3x^2 + 2x - 7, \qquad g(x) = 3e^x - 2\sin x + \cos x.$$

4. Differentiate
$$f(x) = x^2e^x, \qquad g(x) = \frac{x+1}{x-2}, \qquad h(x) = (x^2+1)^4.$$
Name the rule used in each case.

5. Find the tangent and normal lines to $f(x) = x^3 - x$ at $x = 1$.

6. For $f(x) = x^2 - 4x + 1$, find the critical point and determine where the function increases and decreases.

7. For $f(x) = x^3 - 3x$, find the critical points and classify the local extrema using a sign table for f'.

8. For $f(x) = x^3$, compute f'' and determine the intervals of concavity and the inflection point.

9. A particle has position
$$s(t) = t^2 - 4t + 3.$$
Find velocity and acceleration, determine when the particle is at rest, and state the sign of the velocity before and after that time.

10. Find the absolute maximum and minimum of
$$f(x) = x^2 - 4x + 1$$
on $[0, 5]$ by checking the critical point and endpoints.

11.9.2 Development and reasoning exercises

1. For
$$f(x) = x^4 - 4x^2,$$
perform a complete analysis: compute f' and f'', find critical and inflection points, determine monotonicity and concavity intervals, and sketch the graph. Explain how each derivative contributes new information.

2. A particle has
$$s(t) = t^3 - 6t^2 + 9t, \qquad 0 \le t \le 4.$$
Find velocity and acceleration, intervals of forward and backward motion, rest times, and changes of direction. Interpret every sign change physically.

3. A rectangle has perimeter 20. Build the area function, identify the physical domain, find and classify the critical point, check endpoint behaviour, and interpret the optimal shape.

4. A closed cylindrical can has fixed volume V. Express its surface area using one variable, differentiate, and derive the relation between height and radius at the optimum. State where the modelling assumptions enter.

5. Apply the mean value theorem to $f(x) = x^2$ on $[1, 4]$. Find the point c, then explain how the theorem converts a finite secant slope into a local tangent slope.

6. Prove, using the mean value theorem, that if $f'(x) > 0$ on an interval then f is increasing there. Apply the argument to one concrete function.

7. Give a continuous function with a critical point that is not an extremum. Compute its derivative, analyse the sign, and explain why solving $f'(x) = 0$ is insufficient.

8. Compare the local statements supplied by $f'(a)$ and $f''(a)$ with global conclusions about monotonicity and concavity on intervals. Give examples showing why interval information is required.

9. A model produces a mathematically valid critical point outside its physical domain. Construct or analyse such an example and explain why domain checking is part of the mathematics rather than an external correction.

10. Give a complete function-analysis strategy: domain, zeros, limits or end behaviour, first derivative, critical points, monotonicity, extrema, second derivative, concavity, inflection points, sketch, and interpretation. Apply it to a polynomial of degree at most four.

Chapter 12

Integrals and Accumulation

12.1 Antiderivatives and Accumulation

12.1.1 Undoing differentiation and measuring total change

Two connected meanings. An indefinite integral asks for antiderivatives. A definite integral measures accumulated signed change over an interval. The fundamental theorem will connect these two meanings.

A function F is an antiderivative of f when

$$F'(x) = f(x).$$

We write

$$\int f(x)\,dx = F(x) + C.$$

The constant C is necessary because differentiation removes additive constants.

The expression

$$\int_a^b f(x)\,dx$$

represents accumulated signed contribution of f from a to b.

The same integrand, two questions.

$$\int 2x\,dx = x^2 + C,$$

while

$$\int_0^3 2x\,dx = 9.$$

The first answer is a family of functions; the second is a number.

Accumulation from small pieces. The integral combines many small contributions. For a nonnegative function this can be seen as area; for a rate it represents total change.

12.2 Elementary Antiderivatives

12.2.1 Basic rules and verification

For $n \neq -1$,

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C.$$

We also use

$$\int e^x\,dx = e^x + C, \qquad \int \cos x\,dx = \sin x + C, \qquad \int \sin x\,dx = -\cos x + C.$$

Sums and constant multiples integrate term by term.

A low-degree polynomial.

$$\int (3x^2 - 4x + 2)\, dx = x^3 - 2x^2 + 2x + C.$$

A simple mixed function.

$$\int (e^x + 2\cos x)\, dx = e^x + 2\sin x + C.$$

Check by differentiating. An indefinite integral should be verified by differentiating the proposed answer. This makes integration a controlled reversal of differentiation rather than a memorized lookup procedure.

12.3 Simple Substitution and Integration by Parts

12.3.1 Reversing the chain rule and product rule

Only the elementary cases. Substitution and integration by parts are introduced to reveal the ideas behind them. The examples are chosen so that the method, not algebraic complexity, is the main difficulty.

A direct substitution. Consider

$$\int 2x(x^2+1)^3\, dx.$$

Let

$$u = x^2 + 1, \qquad du = 2x\, dx.$$

Then

$$\int 2x(x^2+1)^3\, dx = \int u^3\, du = \frac{u^4}{4} + C = \frac{(x^2+1)^4}{4} + C.$$

Substitution. Substitution reverses the chain rule: an inner function and its derivative are recognized as one new variable.

$$\int u\, dv = uv - \int v\, du.$$

An elementary example by parts. For

$$\int xe^x\, dx,$$

choose $u = x$ and $dv = e^x\, dx$. Then $du = dx$ and $v = e^x$, so

$$\int xe^x\, dx = xe^x - \int e^x\, dx = xe^x - e^x + C.$$

Integration by parts. The method reverses the product rule. One factor is differentiated while the other is integrated.

12.4 Riemann Sums and Properties of the Definite Integral

12.4.1 From rectangles to an integral

A definite integral can be approached through finite sums. Divide $[a, b]$ into subintervals of width Δx, choose a sample point x_i^* in each subinterval, and form

$$\sum_{i=1}^{n} f(x_i^*)\Delta x.$$

Each term is the area of a thin rectangle when $f \geq 0$, or a signed contribution when f may be negative.

> The definite integral
> $$\int_a^b f(x)\,dx$$
> is the limiting value of such sums as the partition becomes finer, provided the limit exists.

A simple right-endpoint sum. For $f(x) = x$ on $[0, 1]$, divide the interval into n equal parts. With right endpoints $x_i = i/n$,

$$\sum_{i=1}^{n} f(x_i)\frac{1}{n} = \frac{1}{n^2}\sum_{i=1}^{n} i = \frac{n(n+1)}{2n^2} = \frac{n+1}{2n}.$$

As $n \to \infty$, this tends to $1/2$. Hence

$$\int_0^1 x\,dx = \frac{1}{2}.$$

Accumulation from local pieces. The integral is not introduced merely as an antiderivative evaluated at endpoints. It represents a total built from many small contributions. The fundamental theorem later connects this accumulation viewpoint with antidifferentiation.

12.4.2 Properties and average value

The definite integral respects addition and interval decomposition:

$$\int_a^b (f+g) = \int_a^b f + \int_a^b g, \qquad \int_a^c f = \int_a^b f + \int_b^c f.$$

Reversing the limits changes the sign:

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

> For a continuous function on $[a, b]$, its average value is
> $$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Average height of a linear graph. For $f(x) = x$ on $[0, 2]$,

$$f_{\text{avg}} = \frac{1}{2}\int_0^2 x\,dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^2 = 1.$$

This agrees with the geometric average of the endpoint values 0 and 2.

Average value as equal redistribution. The average value is the constant height that would produce the same total accumulated area over the interval.

12.5 Definite Integrals and the Fundamental Theorem

12.5.1 The bridge between accumulation and antiderivatives

Suppose

$$A(x) = \int_a^x f(t)\,dt.$$

Increasing x by a small amount h adds approximately a thin contribution of width h and height $f(x)$. Thus

$$A(x+h) - A(x) \approx f(x)h,$$

so the quotient

$$\frac{A(x+h) - A(x)}{h}$$

should approach $f(x)$. Accumulation therefore creates a function whose derivative recovers the local rate.

> If f is continuous and
>
> $$A(x) = \int_a^x f(t)\,dt,$$
>
> then $A'(x) = f(x)$. Consequently, if $F'(x) = f(x)$ on an interval, then
>
> $$\int_a^b f(x)\,dx = F(b) - F(a).$$

Proof status. The thin-strip argument explains the mechanism of the theorem. A complete proof requires estimates controlling the error in the approximation and is not developed in full here.

This is why the theorem is fundamental: a total defined by a limit of sums can be computed using a function whose derivative is the local contribution.

Area under a simple graph. For $f(x) = x$ on $[0, 2]$,

$$\int_0^2 x\,dx = \left[\frac{x^2}{2}\right]_0^2 = 2.$$

This agrees with the area of a triangle of base 2 and height 2.

Signed accumulation.

$$\int_{-1}^1 x\,dx = 0.$$

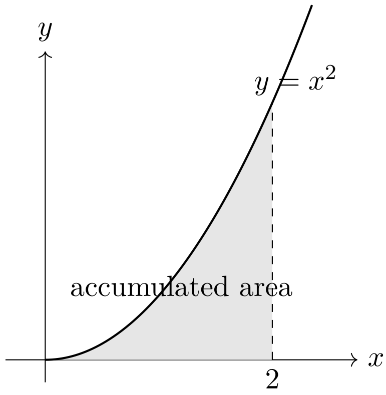
The positive and negative contributions cancel. The integral is signed area, not total geometric area.

For a non-negative integrand, the definite integral is non-negative. This simple sign check is often enough to detect an incorrect calculation.

12.5.2 The definite integral as signed area

For a non-negative function, the definite integral measures the area between the graph and the horizontal axis. The picture below represents

$$\int_0^2 x^2\,dx.$$



The integral adds vertical contributions. Each narrow vertical strip has approximate area $f(x)\,\Delta x$. The definite integral is the limiting total of these strips as their widths become smaller.

12.6 Elementary Applications of Integration

12.6.1 Area, displacement, and work

Area between a graph and the axis. For $f(x) = 2x$ on $[0, 3]$,

$$\int_0^3 2x\,dx = 9.$$

Since the function is non-negative, this is the geometric area under the graph.

Displacement from velocity. If

$$v(t) = 2t,$$

then the displacement from $t = 0$ to $t = 3$ is

$$\int_0^3 v(t)\,dt = 9.$$

Because $v(t) \geq 0$ on this interval, displacement and distance travelled agree.

Remark. When velocity changes sign, the integral gives signed displacement. Total distance requires splitting the interval where the direction of motion changes.

Work done by a simple variable force. Suppose a force depends on position according to

$$F(x) = 3x.$$

The work done from $x = 0$ to $x = 2$ is

$$W = \int_0^2 3x\,dx = 6.$$

Units reveal the meaning. Velocity integrated over time gives distance units. Force integrated over displacement gives work units. The integral combines a rate or density with the variable over which it accumulates.

12.7 Geometric and Physical Accumulation

12.7.1 Area between curves and a simple volume model

If $f(x) \geq g(x)$ on $[a, b]$, then the area between the graphs is

$$\int_a^b (f(x) - g(x))\, dx.$$

Area between a line and a parabola. On $[0, 1]$, the line $y = x$ lies above the parabola $y = x^2$. Therefore

$$A = \int_0^1 (x - x^2)\, dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{6}.$$

A simple volume of revolution about the horizontal axis is obtained by disks:

$$V = \pi \int_a^b r(x)^2\, dx.$$

Rotating a line segment. Rotating $y = x$ on $[0, 1]$ about the horizontal axis produces a cone. The disk formula gives

$$V = \pi \int_0^1 x^2\, dx = \frac{\pi}{3},$$

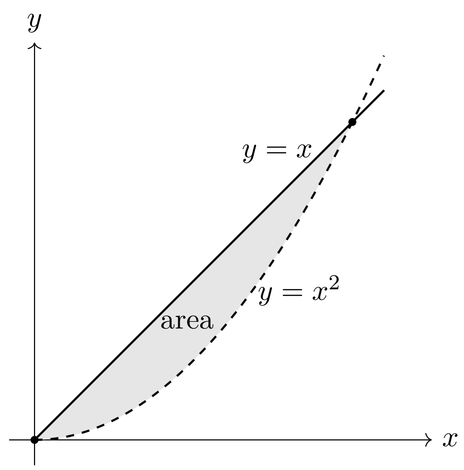
which agrees with the elementary cone formula.

Geometry as accumulation of slices. Area accumulates vertical differences; volume accumulates cross-sectional areas. The integral changes meaning through the quantity being summed, not through a more complicated computational technique.

12.7.2 Area between two graphs

On the interval $[0, 1]$, the graph of $y = x$ lies above the graph of $y = x^2$. The area between them is

$$\int_0^1 (x - x^2)\, dx.$$



Top minus bottom. A vertical strip has height equal to the upper function minus the lower function. Integrating that difference adds the strip areas across the interval.

12.7.3 Motion, work, and total change

If $v(t)$ is velocity, then

$$\int_a^b v(t)\,dt$$

is displacement. Distance requires integrating speed $|v(t)|$, so sign changes must be handled.

Displacement and distance. Let $v(t) = t - 1$ on $[0, 2]$. Then

$$\int_0^2 (t-1)\,dt = 0,$$

so the net displacement is zero. But the particle moves one-half unit before $t = 1$ and one-half unit after $t = 1$, so the total distance is

$$\int_0^2 |t-1|\,dt = 1.$$

If a force $F(x)$ acts along a line, the work done from $x = a$ to $x = b$ is

$$W = \int_a^b F(x)\,dx.$$

Work by a linearly increasing force. For $F(x) = 2x + 1$ from $x = 0$ to $x = 3$,

$$W = \int_0^3 (2x+1)\,dx = \Big[x^2 + x\Big]_0^3 = 12.$$

If force is measured in newtons and distance in metres, the result is 12 joules.

Total change from a rate. Whenever $Q'(t)$ is a rate of change,

$$Q(b) - Q(a) = \int_a^b Q'(t)\,dt.$$

This single principle unifies displacement from velocity, accumulated flow from a flow rate, mass from a linear density, and work from force.

12.8 Choosing an Integration Method

12.8.1 Read the integrand and the modelling question

Method follows structure. First simplify the integrand. Then ask whether it is an elementary antiderivative, a direct substitution pattern, or a simple product suited to integration by parts. For a definite integral, also interpret the interval, sign, and units.

Rewrite before integrating.

$$\int \frac{\sqrt{x}}{x^2}\,dx = \int x^{-3/2}\,dx = -2x^{-1/2} + C.$$

No advanced method is needed after the expression is rewritten clearly.

A deliberately limited toolkit. Use basic antiderivatives for powers, exponentials, sine, and cosine; direct substitution when an inner derivative is visible; and integration by parts only for elementary products such as xe^x. Use definite integrals to express area, accumulated change, displacement, and work. Verify indefinite integrals by differentiation and check signs, scale, units, and restrictions in applications.

The purpose of the chapter is to understand integration as accumulation and reversal of differentiation. Difficult rational decompositions and elaborate trigonometric techniques are outside the intended scope.

Integration as accumulated change. Antiderivatives reverse local differentiation. Riemann sums explain how totals arise from small contributions, and the fundamental theorem is the central bridge connecting those two ideas. Area, average value, volume, displacement, distance, work, and total change are applications of one principle: integrate a local contribution over the interval or region where it acts.

12.9 Review Exercises

12.9.1 Elementary computational exercises

1. Find
$$\int (3x^2 - 4x + 5)\,dx, \qquad \int (2e^x + 3\cos x)\,dx, \qquad \int x^{-1/2}\,dx.$$
Differentiate each answer.

2. Find F satisfying
$$F'(x) = 4x - 3, \qquad F(2) = 5,$$
and compute $F(0)$.

3. Evaluate by substitution:
$$\int 2x(x^2+1)^3\,dx, \qquad \int_0^1 2xe^{x^2}\,dx.$$

4. Evaluate by parts:
$$\int xe^x\,dx, \qquad \int_0^1 xe^x\,dx.$$

5. For $f(x) = x$ on $[0, 2]$, compute right-endpoint Riemann sums for $n = 2$ and $n = 4$, then compare them with the exact integral.

6. Evaluate
$$\int_0^3 (2x+1)\,dx, \qquad \int_{-1}^2 (x^2 - 1)\,dx$$
using the fundamental theorem of calculus.

7. Compute the area under $y = x^2$ on $[0, 2]$ and draw the region.

8. Find the average value of $f(x) = x^2$ on $[0, 3]$ and a point where the function attains that value.

9. A particle has velocity $v(t) = t - 1$ on $[0, 3]$. Compute displacement and total distance, splitting at the sign change.

10. A force $F(x) = 2x + 1$ acts from $x = 0$ to $x = 3$. Compute the work and state its units when force is in newtons and distance in metres.

12.9.2 Development and reasoning exercises

1. Evaluate without being told the method:

$$\int x(x^2+4)^5\,dx, \qquad \int x\cos x\,dx, \qquad \int (3x^2+2e^x)\,dx.$$

   For each integral, justify the method before calculating.

2. Find the intersection points of

$$y = x \qquad \text{and} \qquad y = x^2 - 2x,$$

   determine which graph is above on each interval, and compute the enclosed area.

3. A particle has velocity

$$v(t) = (t-1)(t-3), \qquad 0 \le t \le 4.$$

   Compute displacement and total distance. Explain why the interval must be divided at both zeros.

4. The region under $y = \sqrt{x}$ on $[0,4]$ is rotated about the x-axis. Build the cross-sectional area and compute the volume.

5. A quantity enters a system at rate $r(t) = 3t^2 + 2$ units per minute for $0 \le t \le 2$. Find the accumulated amount and explain the units of the integrand, differential, and integral.

6. Define

$$A(x) = \int_0^x (t^2+1)\,dt.$$

   Find $A(x)$ and $A'(x)$, then explain geometrically why the derivative recovers the current integrand value.

7. Compare right-endpoint Riemann sums for an increasing function with the exact integral. Use $f(x) = x^2$ on $[0,1]$ and explain why the sums overestimate.

8. Construct an example where the definite integral is zero but the total geometric area is positive. Compute both quantities and explain the cancellation.

9. Show explicitly that substitution reverses the chain rule and integration by parts reverses the product rule. Derive both formulas and attach one example to each.

10. Give a complete modelling strategy for accumulated quantities: identify the local contribution and units, choose the interval, determine whether signs or intersections require subdivision, select a method, compute, and interpret the result. Apply it to area, motion, or work.